Representations of Compact Groups

Lecture notes

(based on the *Representations of Finite and Compact Groups*)

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Conventions and prerequisites

Conventions. All vector spaces are complex unless stated otherwise. Inner products $\langle \cdot, \cdot \rangle$ are linear in the first and antilinear in the second argument (Simon uses the opposite convention). Topological groups are Hausdorff, and representations are continuous. "Representation" means finite-dimensional representation unless stated otherwise. We write e or 1 for the neutral element of a group, $\mathbb{T} := \{z \in \mathbb{C} \mid |z| = 1\}$, and $\mathbb{N}_0 := \mathbb{N} \cup \{0\}$.

Prerequisites. Every statement of these notes is proved, with the exception of the following standard results from other courses, which are used freely.

(i) *Analysis and topology:* Heine–Borel; Arzelà–Ascoli; Urysohn's lemma and Tietze's extension theorem for compact Hausdorff spaces; the Riesz–Markov representation theorem (positive linear functionals on $C(X)$, X compact Hausdorff, are given by unique regular Borel measures); density of $C(X)$ in $L^2(X, \mu)$ for regular Borel measures; Fubini–Tonelli and monotone convergence; completeness of $\{\sin(m\theta)\}_{m \in \mathbb{N}}$ in $L^2([0, \pi])$; the lifting of continuous maps $\mathbb{R} \to \mathbb{T}$ along $t \mapsto e^{2\pi i t}$; the spectral theorem for unitary, Hermitian and real orthogonal matrices; Schur triangularization; subgroups of $\mathbb{Z}^m$ are free abelian of rank $\leq m$; a holomorphic function on $\mathbb{C}^N$ vanishing on $\mathbb{R}^N$ vanishes identically.

(ii) *Functional analysis:* the spectral theorem for compact self-adjoint operators (Theorem 4.1.1); Hilbert–Schmidt integral operators are compact; norm limits of finite-rank operators are compact; Riesz' lemma for bounded sesquilinear forms.

(iii) *Differential calculus and geometry:* inverse function theorem and submersion theorem (a smooth map whose differential is surjective at a point maps every neighborhood of that point onto a neighborhood of its image); the change of variables formula for diffeomorphisms; volume forms and the associated measures on manifolds, partitions of unity; the image of a manifold of dimension $< n$ under a smooth map into an n-manifold has measure zero; the surface measure of S^{n-1} is invariant under $\mathrm{O}(n)$; for a closed subgroup H of a Lie group G, the quotient G/H is a manifold such that $G \to G/H$ is a smooth submersion and $G \times G/H \to G/H$ is smooth; every Lie group has an exponential map which is a diffeomorphism from a neighborhood of 0 onto a neighborhood of e (only used in Section 5.5).

1 Topological groups and their representations

1.1 Topological groups

Definition 1.1.1: A *topological group* is a group G equipped with a Hausdorff topology such that the maps $G \times G \to G$, $(g, h) \mapsto gh$, and $G \to G$, $g \mapsto g^{-1}$, are continuous. A topological group is *compact* if it is compact as a topological space.

Remark 1.1.2: In a topological group, left translations $g \mapsto hg$, right translations $g \mapsto gh$ and conjugations $g \mapsto hgh^{-1}$ are homeomorphisms (their inverses are of the same type). Hence if U is a neighborhood of e, then hU is a neighborhood of h.

Example 1.1.3: The following are compact groups.

(i) Every finite group with the discrete topology.

(ii) The circle $\mathbb{T} = \mathrm{U}(1)$ and the n-torus $\mathbb{T}^n$.

(iii) The *unitary group* $\mathrm{U}(n) := \{g \in M_n(\mathbb{C}) \mid g^* g = 1\}$ and $\mathrm{SU}(n) := \{g \in \mathrm{U}(n) \mid \det g = 1\}$.

(iv) The *orthogonal group* $\mathrm{O}(n) := \{g \in M_n(\mathbb{R}) \mid g^T g = 1\}$ and $\mathrm{SO}(n) := \{g \in \mathrm{O}(n) \mid \det g = 1\}$.

(v) The *compact symplectic group* $\mathrm{Sp}(n) := \{g \in \mathrm{U}(2n) \mid g^T J g = J\}$, where $J = \left(\begin{smallmatrix} 0 & 1_n \\ -1_n & 0 \end{smallmatrix}\right)$.

(vi) The infinite product $\prod_{n\in\mathbb{N}} \mathbb{Z}/2\mathbb{Z}$ with the product topology (compact by Tikhonov's theorem).

Lemma 1.1.4: The groups $\mathrm{U}(n)$, $\mathrm{SU}(n)$, $\mathrm{O}(n)$, $\mathrm{SO}(n)$ and $\mathrm{Sp}(n)$ are compact topological groups.

Proof. They are subgroups of $\mathrm{GL}(n,\mathbb{C})$ (resp. $\mathrm{GL}(2n,\mathbb{C})$): e.g. for $\mathrm{Sp}(n)$, if $g^T J g = J = h^T J h$ then $(gh)^T J(gh) = h^T J h = J$ and $(g^{-1})^T J g^{-1} = (g^{-1})^T (g^T J g) g^{-1} = J$. They are closed in $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$, being defined by polynomial equations in the entries and their conjugates, and bounded, since the columns of a unitary matrix are unit vectors, so $|g_{jk}| \le 1$. By Heine–Borel they are compact. Multiplication is polynomial and inversion $g \mapsto g^{-1} = g^*$ is continuous. ☐

Lemma 1.1.5 (Uniform continuity): Let G be a compact group and $f \in C(G)$. For every $\varepsilon > 0$ there is a neighborhood U of e such that

$$|f(hg) - f(g)| < \varepsilon \quad \text{and} \quad |f(gh) - f(g)| < \varepsilon \qquad \text{for all } g \in G,\ h \in U.$$

Proof. We prove the first inequality; the second follows by applying the first to $\tilde{f}(g) := f(g^{-1})$ and replacing U by $U \cap U^{-1}$. For each $g \in G$, continuity of $(h, g') \mapsto f(hg')$ at (e, g) gives open neighborhoods U_g of e and V_g of g with $|f(hg') - f(g)| < \varepsilon/2$ for $h \in U_g$, $g' \in V_g$. Cover G by finitely many $V_{g_1}, \dots, V_{g_m}$ and set $U := \bigcap_j U_{g_j}$. If $g \in V_{g_j}$ and $h \in U$, then $|f(hg) - f(g)| \le |f(hg) - f(g_j)| + |f(g_j) - f(g)| < \varepsilon$. ☐

1.2 Representations

Definition 1.2.1: Let G be a topological group. A *representation* of G is a pair (π, V) of a finite-dimensional complex vector space V and a continuous homomorphism $\pi : G \to \mathrm{GL}(V)$; $d_\pi := \dim V$ is its *dimension*. If V carries an inner product for which all $\pi(g)$ are unitary, π is a *unitary representation*. For a Hilbert space $\mathcal{H}$ of arbitrary dimension, a *unitary representation* is a homomorphism $\pi : G \to \mathrm{U}(\mathcal{H})$ which is *strongly continuous*, i.e., $g \mapsto \pi(g)v$ is continuous for every $v \in \mathcal{H}$.

Definition 1.2.2: Let (π, V) and (σ, W) be representations of G.

(i) A subspace $V' \subset V$ is *invariant* if $\pi(g)V' \subset V'$ for all $g \in G$; then $\pi|_{V'}$ is a *subrepresentation*. (For Hilbert space representations we require invariant subspaces to be closed.)

(ii) π is *irreducible* if $V \neq \{0\}$ and $\{0\}$ and V are the only invariant subspaces.

(iii) A linear map $A : V \to W$ is an *intertwiner* if $A\pi(g) = \sigma(g)A$ for all $g \in G$. The intertwiners form a vector space $\mathrm{Hom}_G(V, W)$.

(iv) π and σ are *equivalent*, $\pi \cong \sigma$, if there is an invertible intertwiner.

We denote by $\widehat{G}$ the set of equivalence classes of irreducible representations of G.

Definition 1.2.3 (Constructions): For representations (π, V), (σ, W) of G we have the *direct sum* $(\pi \oplus \sigma)(g) := \pi(g) \oplus \sigma(g)$, the *tensor product* $(\pi \otimes \sigma)(g) := \pi(g) \otimes \sigma(g)$, the *contragredient representation* π^* on V^* given by $(\pi^*(g)\ell)(v) := \ell(\pi(g)^{-1}v)$, and the representation $g \cdot A := \sigma(g)A\pi(g)^{-1}$ on $\mathrm{Hom}(V, W)$, whose fixed vectors are exactly the intertwiners.

Example 1.2.4: (i) The *trivial representation* $\pi(g) = 1$ on $\mathbb{C}$.

(ii) For $k \in \mathbb{Z}$, $\chi_k(z) := z^k$ is a one-dimensional representation of $\mathbb{T}$.

(iii) The *defining representation* of a closed subgroup $G \subset \mathrm{GL}(n, \mathbb{C})$ on $\mathbb{C}^n$.

(iv) Let V_n be the space of homogeneous polynomials of degree n in $z = (z_1, z_2)$, so $\dim V_n = n+1$. For $g \in \mathrm{SU}(2)$ put $(\pi_n(g)P)(z) := P(g^{-1}z)$. Then

$$(\pi_n(g)\pi_n(h)P)(z) = (\pi_n(h)P)(g^{-1}z) = P(h^{-1}g^{-1}z) = (\pi_n(gh)P)(z),$$

and continuity is clear since the coefficients of $\pi_n(g)P$ are polynomials in the entries of $g^{-1} = g^*$. So π_n is a representation of $\mathrm{SU}(2)$.

1.3 Schur's lemma

Lemma 1.3.1 (Schur's lemma): Let (π, V) and (σ, W) be irreducible representations of G.

(i) Every $A \in \mathrm{Hom}_G(V, W)$ is either zero or invertible.

(ii) $\mathrm{Hom}_G(V, V) = \mathbb{C} \cdot 1$.

Consequently, $\dim \mathrm{Hom}_G(V, W) = 1$ if $\pi \cong \sigma$, and $\mathrm{Hom}_G(V, W) = 0$ otherwise.

Proof. (i) $\ker A \subset V$ and $\operatorname{im} A \subset W$ are invariant. If $A \neq 0$, then $\ker A \neq V$, so $\ker A = 0$, and $\operatorname{im} A \neq 0$, so $\operatorname{im} A = W$. (ii) $A \in \mathrm{Hom}_G(V, V)$ has an eigenvalue $\lambda \in \mathbb{C}$, and $A - \lambda$ is a non-invertible intertwiner, hence 0 by (i). For the last claim, if $A_0 : V \to W$ is an invertible intertwiner, then $A \mapsto A_0^{-1}A$ identifies $\mathrm{Hom}_G(V, W)$ with $\mathrm{Hom}_G(V, V) = \mathbb{C}$. ☐

Corollary 1.3.2: Every irreducible representation of an abelian group is one-dimensional.

Proof. Every $\pi(h)$ is an intertwiner, hence a scalar, so every subspace is invariant, and irreducibility forces $\dim V = 1$. ☐

Proposition 1.3.3: The irreducible representations of $\mathbb{T}^n$ are exactly the characters

$$\chi_k(z_1, \ldots, z_n) = z_1^{k_1} \cdots z_n^{k_n}, \qquad k \in \mathbb{Z}^n,$$

and they are pairwise inequivalent.

Proof. By Corollary 1.3.2, an irreducible representation is a continuous homomorphism $\chi : \mathbb{T}^n \to \mathbb{C}^\times$; its image is a compact subgroup of $\mathbb{C}^\times$, hence lies in $\mathbb{T}$ (if $|\chi(z)| \neq 1$, the powers $\chi(z)^{\pm m}$ would be unbounded). Since $\chi(z) = \prod_j \chi(1, \ldots, z_j, \ldots, 1)$, it suffices to treat $n = 1$. The map $\psi(t) := \chi(e^{2\pi i t})$ is a continuous homomorphism $\mathbb{R} \to \mathbb{T}$; let $\tilde{\psi} : \mathbb{R} \to \mathbb{R}$ be the continuous lift with $\psi(t) = e^{2\pi i \tilde{\psi}(t)}$, $\tilde{\psi}(0) = 0$. The function $(s, t) \mapsto \tilde{\psi}(s+t) - \tilde{\psi}(s) - \tilde{\psi}(t)$ is continuous, integer-valued on the connected set $\mathbb{R}^2$, and vanishes at $(0,0)$; hence $\tilde{\psi}$ is additive. A continuous additive function is linear, $\tilde{\psi}(t) = kt$, and $1 = \psi(1) = e^{2\pi i k}$ gives $k \in \mathbb{Z}$. Characters with different k are different functions, hence inequivalent. ☐

2 Haar measure and complete reducibility

2.1 Existence and uniqueness of Haar measure

Throughout this subsection, G is a compact group, and $C(G)$ carries the sup norm $\|\cdot\|_\infty$. For $f \in C(G)$ and $h \in G$ let

$$(\lambda_h f)(g) := f(hg), \qquad (\rho_h f)(g) := f(gh).$$

Note $\lambda_k\lambda_h = \lambda_{hk}$ and $\rho_k\rho_h = \rho_{kh}$, so the translates of a translate of f are translates of f. For real-valued $f \in C(G)$ let K_f^λ (resp. K_f^ρ) be the closure in $C(G)$ of the convex hull of $\{\lambda_h f \mid h \in G\}$ (resp. of $\{\rho_h f \mid h \in G\}$). Since λ_h is a linear isometry of $C(G)$, $\lambda_h(K_f^\lambda) \subset K_f^\lambda$, and $K_\xi^\lambda \subset K_f^\lambda$ for every $\xi \in K_f^\lambda$; similarly for ρ.

Lemma 2.1.1: K_f^λ and K_f^ρ are compact, and every element of them takes values in $[\min f, \max f]$.

Proof. The set $\{\lambda_h f\}_{h\in G}$ is bounded by $\|f\|_\infty$ and equicontinuous: given $\varepsilon > 0$, choose U as in Lemma 1.1.5; for $g' = gu$ with $u \in U$, $|\lambda_h f(g') - \lambda_h f(g)| = |f(hgu) - f(hg)| < \varepsilon$ for all h. Convex combinations and uniform limits preserve the bound, the estimate (with $\leq \varepsilon$) and the range $[\min f, \max f]$. By Arzelà–Ascoli, the closed set K_f^λ is compact. For K_f^ρ argue in the same way with $g' = ug$, using $|\rho_h f(ug) - \rho_h f(g)| = |f(ugh) - f(gh)| < \varepsilon$ (first inequality of Lemma 1.1.5). ☐

Lemma 2.1.2: K_f^λ and K_f^ρ contain constant functions.

Proof. The oscillation $\omega(\varphi) := \max\varphi - \min\varphi$ is continuous on $C(G)$, so it attains its minimum on the compact set K_f^λ at some φ_0. Suppose $M := \max\varphi_0 > m := \min\varphi_0$. The set $U := \{g \mid \varphi_0(g) < \frac{M+m}{2}\}$ is open and nonempty. The translates kU, $k \in G$, cover G (as $x \in xu^{-1}U$ for $u \in U$), so $G = k_1U \cup \cdots \cup k_nU$. Let $\psi := \frac{1}{n}\sum_i \lambda_{k_i^{-1}}\varphi_0 \in K_f^\lambda$. For every g there is i with $k_i^{-1}g \in U$, so

$$\psi(g) \leq \frac{1}{n}\Big((n-1)M + \frac{M+m}{2}\Big) < M.$$

Since ψ is continuous on the compact space G, $\max\psi < M$, while $\min\psi \geq m$. Thus $\omega(\psi) < \omega(\varphi_0)$, a contradiction. Hence φ_0 is constant. The same proof works for K_f^ρ. ☐

Lemma 2.1.3: If $c \in K_f^\lambda$ and $c' \in K_f^\rho$ are constants, then $c = c'$. Consequently, K_f^λ and K_f^ρ contain exactly one constant, and it is the same; we denote it by $I(f)$.

Proof. Let $\varepsilon > 0$ and choose convex combinations with $|\sum_i a_i f(h_i g) - c| \leq \varepsilon$ and $|\sum_j b_j f(gk_j) - c'| \leq \varepsilon$ for all g. Then the number $S := \sum_{i,j} a_i b_j f(h_i k_j)$ satisfies $|S - c| \leq \varepsilon$ (put $g = k_j$ in the first estimate and average with weights b_j) and $|S - c'| \leq \varepsilon$ (put $g = h_i$ in the second and average with weights a_i). So $|c - c'| \leq 2\varepsilon$ for all ε. ☐

Lemma 2.1.4: For real $f, f_1, f_2 \in C(G)$, $a \in \mathbb{R}$ and $h \in G$:

(i) $I(1) = 1$, $I(af) = aI(f)$ and $\min f \leq I(f) \leq \max f$;

(ii) $I(\lambda_h f) = I(\rho_h f) = I(f)$, and $I(f^\vee) = I(f)$ for $f^\vee(g) := f(g^{-1})$;

(iii) if $\xi \in K_f^\lambda$ and $\|\xi - c\|_\infty \leq \delta$ for a constant c, then $|I(f) - c| \leq \delta$;

(iv) $I(f_1 + f_2) = I(f_1) + I(f_2)$;

(v) if $f \geq 0$ and $f \neq 0$, then $I(f) > 0$.

Proof. (i) $K_{af}^\lambda = aK_f^\lambda$, and Lemma 2.1.1. (ii) The left translates of $\lambda_h f$ are exactly the left translates of f, so $K_{\lambda_h f}^\lambda = K_f^\lambda$ and hence $I(\lambda_h f) = I(f)$. Similarly $K_{\rho_h f}^\rho = K_f^\rho$; since $I(\rho_h f)$ is the constant in $K_{\rho_h f}^\rho$ and $I(f)$ the constant in K_f^ρ (Lemma 2.1.3), $I(\rho_h f) = I(f)$. For the inversion, $\lambda_h(f^\vee) = (\rho_{h^{-1}} f)^\vee$, so the isometry $\varphi \mapsto \varphi^\vee$ maps K_f^ρ onto $K_{f^\vee}^\lambda$ and fixes constants. (iii) $K_\xi^\lambda \subset K_f^\lambda$, so the unique constant of K_ξ^λ is $I(f)$. All translates of ξ, and hence all elements of K_ξ^λ, are within δ of c. (iv) Let $\varepsilon > 0$ and choose $\varphi = \sum_i a_i \lambda_{h_i} f_1$ with $\|\varphi - I(f_1)\|_\infty \leq \varepsilon$.

Put $f_2' := \sum_i a_i\lambda_{h_i} f_2 \in K^\lambda_{f_2}$, and since $K^\lambda_{f_2'} \subset K^\lambda_{f_2}$, the unique constant of $K^\lambda_{f_2'}$ is $I(f_2)$, i.e., $I(f_2') = I(f_2)$. Choose $\psi = \sum_j b_j\lambda_{k_j} f_2'$ with $\|\psi - I(f_2)\|_\infty \le \varepsilon$. Then

$$\xi := \sum_j b_j\lambda_{k_j} \sum_i a_i\lambda_{h_i}(f_1 + f_2) = \sum_j b_j\lambda_{k_j}\varphi + \psi \in K^\lambda_{f_1+f_2}$$

satisfies $\|\xi - I(f_1) - I(f_2)\|_\infty \le 2\varepsilon$, so $|I(f_1 + f_2) - I(f_1) - I(f_2)| \le 2\varepsilon$ by (iii). (v) Let $c := \frac{1}{2}\max f > 0$ and $U := \{f > c\}$. As in the proof of Lemma 2.1.2, $G = k_1U \cup \cdots \cup k_nU$, and then $\sum_i \lambda_{k_i^{-1}} f \ge c$ on G. By (i), (ii), (iv), $nI(f) = I(\sum_i \lambda_{k_i^{-1}} f) \ge c$. ☐

We extend I to complex-valued $f = u + iv$ by $I(f) := I(u) + iI(v)$. By Lemma 2.1.4, I is a positive normalized linear functional on $C(G)$ which is invariant under left and right translations and under inversion.

Theorem 2.1.1: Haar measure

Let G be a compact group. There is a unique regular Borel probability measure μ on G which is left invariant, i.e., $\mu(hA) = \mu(A)$ for all Borel sets A and $h \in G$. It is also right invariant and invariant under $g \mapsto g^{-1}$, and $\mu(U) > 0$ for every nonempty open $U \subset G$.

Proof. By Riesz–Markov there is a unique regular Borel measure μ with $\int f\,d\mu = I(f)$ for $f \in C(G)$, and $\mu(G) = I(1) = 1$. For $h \in G$, $\mu_h(A) := \mu(hA)$ is again a regular Borel measure with $\int f\,d\mu_h = \int f(h^{-1}g)\,d\mu(g) = I(\lambda_{h^{-1}}f) = I(f)$, so $\mu_h = \mu$ by the uniqueness part of Riesz–Markov. Right invariance and inversion invariance follow in the same way. If $U \ne \emptyset$ is open, Urysohn gives $f \in C(G)$, $0 \le f \le 1_U$, $f \ne 0$, so $\mu(U) \ge I(f) > 0$.

Uniqueness: let ν be a left-invariant regular Borel probability measure and $J(f) := \int f\,d\nu$; then $J(\lambda_h f) = J(f)$. For real f and $\varepsilon > 0$ choose a convex combination $\varphi = \sum_i a_i\lambda_{h_i} f$ with $\|\varphi - I(f)\|_\infty \le \varepsilon$. Then $J(\varphi) = J(f)$ and $|J(\varphi) - I(f)| \le \varepsilon$ since J is positive with $J(1) = 1$. Hence $J = I$ and $\nu = \mu$ by Riesz–Markov. ☐

We write $\int_G f(g)\,dg := \int f\,d\mu$. Thus for $f \in L^1(G)$ and $h \in G$,

$$\int_G f(hg)\,dg = \int_G f(gh)\,dg = \int_G f(g^{-1})\,dg = \int_G f(g)\,dg.$$

Example 2.1.5: (i) For finite G, $\int_G f = \frac{1}{|G|}\sum_{g\in G} f(g)$, since this measure is left invariant.

(ii) For $\mathbb{T}^n$, $\int f = (2\pi)^{-n}\int_{[0,2\pi]^n} f(e^{i\theta_1}, \dots, e^{i\theta_n})\,d\theta$, by translation invariance of Lebesgue measure and periodicity.

(iii) Writing $g = \left(\begin{smallmatrix} a & -\bar b \\ b & \bar a \end{smallmatrix}\right)$ with $|a|^2 + |b|^2 = 1$ identifies $\mathrm{SU}(2)$ with $S^3 \subset \mathbb{C}^2 \cong \mathbb{R}^4$. (Indeed, if $g = \left(\begin{smallmatrix} a & c \\ b & d \end{smallmatrix}\right) \in \mathrm{SU}(2)$, comparing $g^{-1} = g^*$ with $g^{-1} = \left(\begin{smallmatrix} d & -c \\ -b & a \end{smallmatrix}\right)$ gives $d = \bar a$, $c = -\bar b$; conversely such matrices are unitary of determinant 1.) Left multiplication $x \mapsto gx$ by $g \in \mathrm{SU}(2)$ is a linear isometry of $\mathbb{C}^2 = \mathbb{R}^4$ preserving S^3, so the normalized surface measure of S^3, which is $\mathrm{O}(4)$-invariant, is left invariant, hence it is the Haar measure of $\mathrm{SU}(2)$.

2.2 Unitarity and complete reducibility

Theorem 2.2.1: Weyl's unitarian trick

Let (π, V) be a representation of a compact group G. Then V has an inner product for which all $\pi(g)$ are unitary.

Proof. Let $(\cdot,\cdot)$ be any inner product on V and $\langle v, w\rangle := \int_G (\pi(g)v, \pi(g)w)\, dg$. This is a Hermitian form. For $v \neq 0$ the integrand $g \mapsto (\pi(g)v, \pi(g)v)$ is continuous and strictly positive, hence bounded below by a positive constant on the compact G, so $\langle v, v\rangle > 0$. By right invariance, $\langle \pi(h)v, \pi(h)w\rangle = \int_G (\pi(gh)v, \pi(gh)w)\, dg = \langle v, w\rangle$. ☐

Corollary 2.2.1 (Complete reducibility): Every representation of a compact group is a direct sum of irreducible representations.

Proof. By Theorem 2.2.1 we may assume π unitary. If $V' \subset V$ is invariant, so is $V'^{\perp}$: for $w \in V'^{\perp}$, $v \in V'$, $\langle \pi(g)w, v\rangle = \langle w, \pi(g)^{-1}v\rangle = 0$. Now induct on $\dim V$. ☐

Remark 2.2.2: Compactness is essential: the representation $t \mapsto \left(\begin{smallmatrix} 1 & t \\ 0 & 1 \end{smallmatrix}\right)$ of $(\mathbb{R}, +)$ on $\mathbb{C}^2$ has the invariant line $\mathbb{C}e_1$, and no other invariant line (an eigenvector of $\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)$ lies in $\mathbb{C}e_1$), so it has no invariant complement.

Proposition 2.2.3 (Multiplicities): Let $V = V_1 \oplus \cdots \oplus V_m$ be a decomposition of a representation of a compact group into irreducibles, and (σ, W) irreducible. Then the number of V_j equivalent to σ equals $\dim \mathrm{Hom}_G(W, V)$. In particular this number, the *multiplicity* $m_\sigma(V)$, does not depend on the decomposition, and two representations with the same multiplicities are equivalent.

Proof. $\mathrm{Hom}_G(W, V) = \bigoplus_j \mathrm{Hom}_G(W, V_j)$, and $\dim \mathrm{Hom}_G(W, V_j) \in \{0, 1\}$ according to whether $V_j \cong W$ (Lemma 1.3.1). The last statement follows since both representations are then equivalent to $\bigoplus_\sigma m_\sigma\, \sigma$. ☐

3 Schur orthogonality and characters

3.1 Matrix coefficients and the orthogonality relations

Definition 3.1.1: Let (π, V) be a unitary representation. For $u, v \in V$, the function $f^\pi_{u,v}(g) := \langle \pi(g)u, v\rangle$ is a *matrix coefficient* of π. If $e_1, \ldots, e_d$ is an orthonormal basis, $\pi_{ij}(g) := \langle \pi(g)e_j, e_i\rangle$ are the entries of the matrix of $\pi(g)$.

On $L^2(G) := L^2(G, dg)$ we use $\langle f_1, f_2\rangle_{L^2} := \int_G f_1(g)\overline{f_2(g)}\, dg$.

Theorem 3.1.1: Schur orthogonality relations

Let (π, V), (σ, W) be irreducible unitary representations of a compact group G.

(i) If $\pi \not\cong \sigma$, then $\langle f^\pi_{u,v}, f^\sigma_{u',v'}\rangle_{L^2} = 0$ for all $u, v \in V$, $u', v' \in W$.

(ii) For $u_1, v_1, u_2, v_2 \in V$: $\displaystyle\int_G \langle \pi(g)u_1, v_1\rangle\overline{\langle \pi(g)u_2, v_2\rangle}\, dg = \frac{1}{d_\pi}\langle u_1, u_2\rangle\overline{\langle v_1, v_2\rangle}$.

In matrix form: $\displaystyle\int_G \pi_{ij}(g)\overline{\sigma_{kl}(g)}\, dg = \frac{\delta_{\pi\sigma}\delta_{ik}\delta_{jl}}{d_\pi}$, where $\delta_{\pi\sigma} = 1$ if $\pi = \sigma$ and 0 if $\pi \not\cong \sigma$.

Proof. For a linear map $A : W \to V$ let $\tilde{A} := \int_G \pi(g)A\sigma(g)^{-1}\, dg$ (entrywise integral). By left invariance, $\pi(h)\tilde{A}\sigma(h)^{-1} = \int_G \pi(hg)A\sigma(hg)^{-1}\, dg = \tilde{A}$, so $\tilde{A} \in \mathrm{Hom}_G(W, V)$. By Schur's lemma, $\tilde{A} = 0$ if $\pi \not\cong \sigma$, while for $\sigma = \pi$, $\tilde{A} = c \cdot 1$ with $cd_\pi = \operatorname{tr}\tilde{A} = \operatorname{tr} A$.

Take $Ax := \langle x, u'\rangle u$ with $u \in V$, $u' \in W$. For $w \in V$, $w' \in W$, using unitarity of σ,

$$\langle \tilde{A}w', w\rangle = \int_G \langle \sigma(g)^{-1}w', u'\rangle\langle \pi(g)u, w\rangle\, dg = \int_G \langle \pi(g)u, w\rangle\overline{\langle \sigma(g)u', w'\rangle}\, dg.$$

If $\pi \ncong \sigma$ this vanishes, which is (i). If $\sigma = \pi$, then $\operatorname{tr} A = \langle u, u' \rangle$ and $\langle \tilde{A} w', w \rangle = \frac{1}{d_\pi} \langle u, u' \rangle \langle w', w \rangle$; renaming $u = u_1$, $w = v_1$, $u' = u_2$, $w' = v_2$ gives (ii). The matrix form follows with $u_1 = e_j$, $v_1 = e_i$, $u_2 = e_l$, $v_2 = e_k$. ☐

3.2 Characters

Definition 3.2.1: The *character* of a representation (π, V) is $\chi_\pi(g) := \operatorname{tr} \pi(g)$. A function f on G is a *class function* if $f(hgh^{-1}) = f(g)$ for all g, h.

Lemma 3.2.2: For representations π, σ of a compact group:

(i) $\chi_\pi(e) = d_\pi$, and χ_π is a continuous class function;

(ii) $\pi \cong \sigma$ implies $\chi_\pi = \chi_\sigma$;

(iii) $\chi_{\pi \oplus \sigma} = \chi_\pi + \chi_\sigma$ and $\chi_{\pi \otimes \sigma} = \chi_\pi \chi_\sigma$;

(iv) $\chi_{\pi^*}(g) = \chi_\pi(g^{-1}) = \overline{\chi_\pi(g)}$.

Proof. (i)–(iii) follow from $\operatorname{tr}(ABA^{-1}) = \operatorname{tr} B$, $\operatorname{tr}(A \oplus B) = \operatorname{tr} A + \operatorname{tr} B$ and $\operatorname{tr}(A \otimes B) = \operatorname{tr} A \operatorname{tr} B$. (iv): the matrix of $\pi^*(g)$ in the dual basis is the transpose of that of $\pi(g^{-1})$; with an invariant inner product (Theorem 2.2.1), $\pi(g^{-1}) = \pi(g)^*$ has trace $\overline{\operatorname{tr} \pi(g)}$. ☐

Theorem 3.2.1: Orthogonality of characters

For irreducible representations π, σ of a compact group, $\langle \chi_\pi, \chi_\sigma \rangle_{L^2} = 1$ if $\pi \cong \sigma$ and $= 0$ otherwise.

Proof. By Lemma 3.2.2(ii) and Theorem 2.2.1 we may assume both unitary and $\sigma = \pi$ in the equivalent case. Then $\chi_\pi = \sum_i \pi_{ii}$, and Theorem 3.1.1 gives $\langle \chi_\pi, \chi_\sigma \rangle = \sum_{i,k} \delta_{\pi\sigma} \delta_{ik} / d_\pi = \delta_{\pi\sigma}$. ☐

Corollary 3.2.3: Let π be a representation of a compact group G.

(i) $m_\sigma(\pi) = \langle \chi_\pi, \chi_\sigma \rangle_{L^2}$ for every irreducible σ.

(ii) π is determined up to equivalence by χ_π.

(iii) $\|\chi_\pi\|_{L^2}^2 = \sum_\sigma m_\sigma(\pi)^2$; in particular π is irreducible if and only if $\|\chi_\pi\|_{L^2} = 1$.

(iv) $\dim V^G = \int_G \chi_\pi(g)\, dg$, where V^G is the space of fixed vectors.

Proof. Write $\pi \cong \bigoplus_\sigma m_\sigma(\pi) \sigma$ (Corollary 2.2.1), so $\chi_\pi = \sum_\sigma m_\sigma(\pi) \chi_\sigma$, and apply Theorem 3.2.1; (ii) then follows from Proposition 2.2.3. For (iv), V^G is the isotypic component of the trivial representation, whose character is 1, so $\dim V^G = m_1(\pi) = \langle \chi_\pi, 1 \rangle$. ☐

3.3 First example: the representations of SU(2)

For $\theta \in \mathbb{R}$ put $t_\theta := \operatorname{diag}(e^{i\theta}, e^{-i\theta}) \in \mathrm{SU}(2)$.

Lemma 3.3.1: Every $g \in \mathrm{SU}(2)$ is conjugate in $\mathrm{SU}(2)$ to t_θ for a unique $\theta \in [0, \pi]$, determined by $\operatorname{tr} g = 2\cos\theta$.

Proof. By the spectral theorem there is $k \in \mathrm{U}(2)$ with $k^{-1} g k$ diagonal; its entries λ, λ^{-1} have modulus 1. Replacing k by $k \cdot (\det k)^{-1/2}$ (any square root) gives $k \in \mathrm{SU}(2)$. So g is conjugate to some t_θ, and $t_{-\theta} = w t_\theta w^{-1}$ with $w = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) \in \mathrm{SU}(2)$. Uniqueness: $\operatorname{tr} t_\theta = 2\cos\theta$ is injective on $[0, \pi]$. ☐

Lemma 3.3.2 (Integration formula for SU(2)): For every continuous class function f on SU(2),

$$\int_{\mathrm{SU}(2)} f(g)\,dg = \frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta.$$

Proof. By Example 2.1.5(iii), Haar measure is the normalized surface measure of S^3. In the coordinates $x = (x_0, \dots, x_3) = (\operatorname{Re} a, \operatorname{Im} a, \operatorname{Re} b, \operatorname{Im} b)$, $\operatorname{tr} g = 2x_0$, so by Lemma 3.3.1, $f(g) = F(x_0)$ with $F(\cos\theta) := f(t_\theta)$. The map $\Phi(\theta, \omega) := (\cos\theta, \sin\theta\,\omega)$, $(\theta, \omega) \in (0, \pi) \times S^2$, is a diffeomorphism onto S^3 minus two points. Its derivative maps ∂_θ to a unit vector orthogonal to $\Phi_*(T_\omega S^2)$, and scales $T_\omega S^2$ by $\sin\theta$; hence the surface measure is $\sin^2\theta\,d\theta\,d\sigma_{S^2}(\omega)$. Thus $\operatorname{vol}(S^3) = 4\pi\int_0^\pi \sin^2\theta\,d\theta = 2\pi^2$ and

$$\int_{\mathrm{SU}(2)} f = \frac{4\pi}{2\pi^2}\int_0^\pi F(\cos\theta)\sin^2\theta\,d\theta = \frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta. \qquad \square$$

Theorem 3.3.1: Irreducible representations of SU(2)

The representations π_n, $n \in \mathbb{N}_0$, of Example 1.2.4 are irreducible and pairwise inequivalent, with

$$\chi_n(t_\theta) = \sum_{k=0}^n e^{i(n-2k)\theta} = \frac{\sin((n+1)\theta)}{\sin\theta} \qquad (\theta \notin \pi\mathbb{Z}).$$

Every irreducible representation of SU(2) is equivalent to some π_n.

Proof. Since $t_\theta^{-1} z = (e^{-i\theta} z_1, e^{i\theta} z_2)$, the monomial $z_1^k z_2^{n-k}$ is an eigenvector of $\pi_n(t_\theta)$ with eigenvalue $e^{i(n-2k)\theta}$; summing the geometric series gives χ_n. By Lemma 3.3.2,

$$\langle \chi_n, \chi_m \rangle_{L^2} = \frac{2}{\pi}\int_0^\pi \sin((n+1)\theta)\sin((m+1)\theta)\,d\theta = \delta_{nm},$$

so by Corollary 3.2.3 the π_n are irreducible and pairwise inequivalent. Let σ be irreducible and not equivalent to any π_n. Then for all n, using that χ_n is real,

$$0 = \langle \chi_\sigma, \chi_n \rangle_{L^2} = \frac{2}{\pi}\int_0^\pi \big(\chi_\sigma(t_\theta)\sin\theta\big)\sin((n+1)\theta)\,d\theta.$$

By completeness of $\{\sin(m\theta)\}_{m\geq 1}$ in $L^2([0,\pi])$, the continuous function $\chi_\sigma(t_\theta)\sin\theta$ vanishes on $[0,\pi]$, so $\chi_\sigma(t_\theta) = 0$ for $\theta \in (0,\pi)$ and by continuity for $\theta = 0$. This contradicts $\chi_\sigma(e) = d_\sigma \geq 1$. $\square$

Corollary 3.3.3 (Clebsch–Gordan): For $m \geq n \geq 0$: $\pi_m \otimes \pi_n \cong \bigoplus_{k=0}^n \pi_{m+n-2k}$.

Proof. By Corollary 3.2.3(ii) and Lemma 3.2.2(iii), it suffices to compare characters on the conjugacy class representatives t_θ, $\theta \notin \pi\mathbb{Z}$ (and then everywhere by continuity). With $x = e^{i\theta}$,

$$\chi_m\chi_n = \sum_{j=0}^n \chi_m(t_\theta)x^{n-2j} = \sum_{j=0}^n \frac{x^{m+n-2j+1} - x^{-(m-n+2j+1)}}{x - x^{-1}}.$$

Substituting $j \mapsto n - j$ in the second part, this becomes $\sum_{k=0}^n \frac{x^{m+n-2k+1} - x^{-(m+n-2k+1)}}{x - x^{-1}} = \sum_{k=0}^n \chi_{m+n-2k}$. $\square$

Remark 3.3.4: In physics one writes $j = n/2$ (*spin*) and $D^{(j)} := \pi_{2j}$; then $D^{(j_1)} \otimes D^{(j_2)} \cong \bigoplus_{J=|j_1-j_2|}^{j_1+j_2} D^{(J)}$.

4 The Peter–Weyl theorem

4.1 Compact operators and the regular representation

Theorem 4.1.1: Spectral theorem for compact self-adjoint operators

Let K be a compact self-adjoint operator on a Hilbert space $\mathcal{H}$. Then $\mathcal{H} = \ker K \oplus \widehat{\bigoplus}_\lambda \mathcal{H}_\lambda$ (orthogonal Hilbert sum), where λ runs through the nonzero eigenvalues of K, which are real, and each $\mathcal{H}_\lambda := \ker(K - \lambda)$ is finite-dimensional. If $K \neq 0$, there is at least one nonzero eigenvalue.

This is a prerequisite from functional analysis.

Definition 4.1.1: The *left* and *right regular representations* of a compact group G on $L^2(G)$ are $(L(h)f)(g) := f(h^{-1}g)$ and $(R(h)f)(g) := f(gh)$.

Lemma 4.1.2: L and R are unitary representations, and they commute.

Proof. Unitarity and the homomorphism property follow from invariance of Haar measure, and $L(h)R(k)f(g) = f(h^{-1}gk) = R(k)L(h)f(g)$. For $f \in C(G)$, Lemma 1.1.5 gives $\|L(h)f - f\|_2 \leq \|L(h)f - f\|_\infty \to 0$ as $h \to e$. For $f \in L^2(G)$ and $\varepsilon > 0$ choose $f_0 \in C(G)$ with $\|f - f_0\|_2 < \varepsilon$; then $\|L(h)f - f\|_2 \leq 2\varepsilon + \|L(h)f_0 - f_0\|_2$. Continuity at h_0 follows from $\|L(h)f - L(h_0)f\|_2 = \|L(h_0^{-1}h)f - f\|_2$. The same applies to R. ☐

Definition 4.1.3: For $\varphi \in C(G)$ and $f \in L^2(G)$ let $(\varphi * f)(g) := \int_G \varphi(h)f(h^{-1}g)\,dh = \int_G \varphi(gh^{-1})f(h)\,dh$ (the second form follows by substituting $h \mapsto gh^{-1}$, using invariance under right translation and inversion).

Lemma 4.1.4: Let $\varphi \in C(G)$ and $T_\varphi f := \varphi * f$.

(i) T_φ is a Hilbert–Schmidt operator with kernel $K(g,h) = \varphi(gh^{-1})$, hence compact. If $\varphi(g^{-1}) = \overline{\varphi(g)}$, then T_φ is self-adjoint.

(ii) $T_\varphi R(k) = R(k)T_\varphi$ for all $k \in G$.

(iii) $\varphi * f$ is continuous for $f \in L^2(G)$, and $\|\varphi * f\|_\infty \leq \|\varphi\|_2\|f\|_2$.

(iv) $\varphi * f = \int_G \varphi(h)L(h)f\,dh$ in the sense that $\langle \varphi * f, u\rangle = \int_G \varphi(h)\langle L(h)f, u\rangle\,dh$ for $u \in L^2(G)$.

Proof. (i) The kernel is continuous on $G \times G$, hence square integrable; $\overline{K(h,g)} = \overline{\varphi(hg^{-1})} = \varphi(gh^{-1}) = K(g,h)$. (ii) $(\varphi * R(k)f)(g) = \int \varphi(h)f(h^{-1}gk)\,dh = (\varphi * f)(gk)$. (iii) By Cauchy–Schwarz and invariance, $|(\varphi * f)(g) - (\varphi * f)(g')| \leq \left(\int_G |\varphi(gh^{-1}) - \varphi(g'h^{-1})|^2 dh\right)^{1/2}\|f\|_2$, which tends to 0 as $g' \to g$ by Lemma 1.1.5; and $|(\varphi * f)(g)| \leq \|\varphi\|_2\|f\|_2$. (iv) Fubini. ☐

Lemma 4.1.5 (Approximate identities): For every neighborhood U of e there is $\varphi_U \in C(G)$ with $\varphi_U \geq 0$, $\operatorname{supp}\varphi_U \subset U$, $\int_G \varphi_U = 1$ and $\varphi_U(g^{-1}) = \varphi_U(g)$. For $f \in L^2(G)$, $\|\varphi_U * f - f\|_2 \leq \sup_{h\in U}\|L(h)f - f\|_2$, and for $f \in C(G)$, $\|\varphi_U * f - f\|_\infty \leq \sup_{h\in U,\, g\in G}|f(h^{-1}g) - f(g)|$. In particular $\varphi_U * f \to f$ in L^2 (resp. uniformly for continuous f) as U shrinks.

Proof. By Urysohn choose $\psi \in C(G)$, $0 \leq \psi \leq 1$, $\psi(e) = 1$, $\operatorname{supp}\psi \subset U \cap U^{-1}$. Then $\psi + \psi^\vee \geq 0$ is nonzero, so its integral is positive (Theorem 2.1.1), and $\varphi_U := (\psi + \psi^\vee)/\int(\psi + \psi^\vee)$ works. Since $\int \varphi_U = 1$, $\varphi_U * f - f = \int \varphi_U(h)(L(h)f - f)\,dh$ (weakly, Lemma 4.1.4(iv)), so $|\langle \varphi_U * f - f, u\rangle| \leq \sup_{h\in U}\|L(h)f - f\|_2\|u\|_2$. The uniform estimate is immediate, and convergence follows from Lemmas 9.3.2 and 1.1.5. ☐

4.2 The Peter–Weyl theorem

Let $\mathcal{E}(G) \subset C(G)$ be the span of all matrix coefficients of (finite-dimensional) representations. By Theorem 2.2.1 and Corollary 2.2.1, $\mathcal{E}(G)$ is spanned by the functions π_{ij} for unitary irreducible π and orthonormal bases. For a representation π, the space $\mathcal{E}_\pi := \operatorname{span}\{f^\pi_{u,v}\}$ is finite-dimensional and invariant under L and R, since $L(h)f^\pi_{u,v} = f^\pi_{u,\pi(h)v}$ and $R(h)f^\pi_{u,v} = f^\pi_{\pi(h)u,v}$.

Theorem 4.2.1: Peter–Weyl

Let G be a compact group, and fix a unitary representative π of each class in $\widehat{G}$ together with an orthonormal basis.

(i) $\mathcal{E}(G)$ is dense in $C(G)$ with respect to $\|\cdot\|_\infty$.

(ii) $\{\sqrt{d_\pi}\,\pi_{ij} \mid \pi \in \widehat{G},\ 1 \le i,j \le d_\pi\}$ is an orthonormal basis of $L^2(G)$.

(iii) The maps $A \mapsto f_A$, $f_A(x) := \sqrt{d_\pi}\operatorname{tr}(A\pi(x)^{-1})$, define a unitary equivalence of $G \times G$-representations

$$\widehat{\bigoplus_{\pi\in\widehat{G}}} \operatorname{End}(V_\pi) \xrightarrow{\cong} L^2(G),$$

where $(g,h)\cdot A := \pi(g)A\pi(h)^{-1}$ and $(g,h)\cdot f := L(g)R(h)f$, and $\operatorname{End}(V_\pi)$ carries the Hilbert–Schmidt inner product $\langle A, B\rangle = \operatorname{tr}(AB^*)$.

(iv) $\{\chi_\pi\}_{\pi\in\widehat{G}}$ is an orthonormal basis of the closed subspace $L^2(G)^{\mathrm{class}}$ of class functions.

Proof. Step 1: $\mathcal{E}(G)$ is dense in $L^2(G)$. Suppose $0 \neq f \perp \mathcal{E}(G)$. By Lemma 4.1.5 there is a symmetric approximate identity φ with $\varphi * f \neq 0$. By Lemma 4.1.4, T_φ is compact, self-adjoint and commutes with R. Let $\lambda \neq 0$ be an eigenvalue and $\mathcal{H}_\lambda$ its finite-dimensional eigenspace. It is R-invariant, and its elements are continuous since $\psi = \lambda^{-1}\varphi * \psi$. Choose a basis $\psi_1, \dots, \psi_m$ of $\mathcal{H}_\lambda$ and write $R(h)\psi_j = \sum_i \rho_{ij}(h)\psi_i$; then $\rho = R|_{\mathcal{H}_\lambda}$ is a representation of G (continuity follows from Lemma 9.3.2). Both sides are continuous functions, so evaluating at e gives

$$\psi_j(h) = \sum_i \rho_{ij}(h)\psi_i(e),$$

so $\mathcal{H}_\lambda \subset \mathcal{E}(G)$ and $f \perp \mathcal{H}_\lambda$. This holds for every $\lambda \neq 0$, so $f \in \ker T_\varphi$ by Theorem 4.1.1, contradicting $\varphi * f \neq 0$.

Step 2: (i). Let $f \in C(G)$, $\varepsilon > 0$. Choose φ with $\|\varphi * f - f\|_\infty < \varepsilon$ (Lemma 4.1.5) and, by Step 1, $e' \in \mathcal{E}(G)$ with $\|f - e'\|_2 < \varepsilon/(1 + \|\varphi\|_2)$. If $e' \in \mathcal{E}_{\pi_1} + \cdots + \mathcal{E}_{\pi_r}$, a finite-dimensional L-invariant space E, then $\varphi * e' = \int \varphi(h)L(h)e'\,dh \in E$ (a weak integral of a continuous E-valued function lies in the closed subspace E). By Lemma 4.1.4(iii), $\|f - \varphi * e'\|_\infty \le \|f - \varphi * f\|_\infty + \|\varphi * (f - e')\|_\infty < 2\varepsilon$.

Step 3: (ii) and (iii). Orthonormality is Theorem 3.1.1, completeness is Step 1. For (iii), $L(g)R(h)f_A(x) = \sqrt{d_\pi}\operatorname{tr}(A\pi(h)^{-1}\pi(x)^{-1}\pi(g)) = f_{\pi(g)A\pi(h)^{-1}}(x)$. Since $\operatorname{tr}(A\pi(x)^*) = \sum_{ij} A_{ij}\overline{\pi_{ij}(x)}$, the map sends the orthonormal basis $\{E_{ij}\}$ of $\operatorname{End}(V_\pi)$ to $\{\sqrt{d_\pi}\overline{\pi_{ij}}\}$. Now $\overline{\pi_{ij}}$ are the matrix coefficients of π^* in the dual basis (Lemma 3.2.2(iv) and its proof), and $\pi \mapsto \pi^*$ permutes $\widehat{G}$ (since $\pi^{**} \cong \pi$ and π^* is irreducible as $\chi_{\pi^*} = \overline{\chi_\pi}$ has norm 1). So (iii) follows from (ii).

Step 4: (iv). The characters are orthonormal class functions. Let P_π be the orthogonal projection onto $\mathcal{E}_\pi$. As $\mathcal{E}_\pi$ and $\mathcal{E}_\pi^\perp$ are invariant under the unitary operators $C(h) := L(h)R(h)$, P_π commutes with $C(h)$; class functions are the fixed vectors of all $C(h)$, so P_π maps class functions to class functions. A class function in $\mathcal{E}_\pi$ has the form $x \mapsto \operatorname{tr}(A\pi(x))$ with $\operatorname{tr}(A\pi(hxh^{-1})) = \operatorname{tr}(A\pi(x))$; averaging over h gives $\operatorname{tr}(A\pi(x)) = \operatorname{tr}(\tilde{A}\pi(x))$ with $\tilde{A} = \int_G \pi(h)^{-1}A\pi(h)\,dh$, an intertwiner, hence $\tilde{A} = \frac{\operatorname{tr} A}{d_\pi} 1$ by Schur. So $P_\pi f \in \mathbb{C}\chi_\pi$, and by (ii), $f = \sum_\pi P_\pi f = \sum_\pi \langle f, \chi_\pi \rangle \chi_\pi$ for every class function $f \in L^2(G)$. ☐

Example 4.2.1 (Fourier series): For $G = \mathbb{T}$, $\widehat{G} = \{\chi_k\}_{k \in \mathbb{Z}}$ (Proposition 1.3.3), and Theorem 4.2.1 states completeness of $\{e^{ik\theta}\}$ in $L^2(\mathbb{T})$ and uniform density of trigonometric polynomials in $C(\mathbb{T})$. In general, $f = \sum_\pi d_\pi \operatorname{tr}(\hat{f}(\pi)\pi(\cdot))$ in $L^2(G)$, with $\hat{f}(\pi) := \int_G f(g)\pi(g)^{-1}\,dg$, and $\|f\|_2^2 = \sum_\pi d_\pi \|\hat{f}(\pi)\|_{\mathrm{HS}}^2$ (Plancherel). Indeed, $\langle f, \pi_{ij} \rangle = \hat{f}(\pi)_{ji}$, so the component of f in $\mathcal{E}_\pi$ is $\sum_{ij} d_\pi \hat{f}(\pi)_{ji} \pi_{ij} = d_\pi \operatorname{tr}(\hat{f}(\pi)\pi(\cdot))$, of norm squared $d_\pi^2 \sum_{ij} |\hat{f}(\pi)_{ji}|^2 / d_\pi$.

4.3 Consequences

Corollary 4.3.1: Let G be a compact group.

(i) If G is finite, then $\sum_{\pi \in \widehat{G}} d_\pi^2 = |G|$, and $|\widehat{G}|$ equals the number of conjugacy classes.

(ii) For $g \neq e$ there is $\pi \in \widehat{G}$ with $\pi(g) \neq 1$.

Proof. (i) Compare the dimension $|G|$ of $L^2(G)$, resp. the dimension of the space of class functions (the number of conjugacy classes), with Theorem 4.2.1(ii) and (iv). (ii) By Urysohn there is $f \in C(G)$ with $f(e) = 0$, $f(g) = 1$; by Theorem 4.2.1(i), some $e' \in \mathcal{E}(G)$ satisfies $e'(g) \neq e'(e)$. If all $\pi(g) = 1$, all matrix coefficients would take the same value at g and e. ☐

Theorem 4.3.1: Unitary representations of compact groups

Let $(\pi, \mathcal{H})$ be a unitary representation of a compact group G on a Hilbert space. Then $\mathcal{H}$ is an orthogonal Hilbert sum of finite-dimensional irreducible invariant subspaces. In particular every irreducible unitary representation is finite-dimensional.

Proof. Claim: every nonzero closed invariant subspace $\mathcal{H}'$ contains a finite-dimensional irreducible invariant subspace. Replacing $\mathcal{H}$ by $\mathcal{H}'$, we prove this for $\mathcal{H}$. Let $v \neq 0$ and let $R_g := \langle \cdot, \pi(g)v \rangle \pi(g)v$ (rank one). The sesquilinear form $(x, y) \mapsto \int_G \langle R_g x, y \rangle\,dg$ is bounded by $\|v\|^2 \|x\| \|y\|$, so it defines a bounded operator K with $\langle Kx, y \rangle = \int_G \langle x, \pi(g)v \rangle \langle \pi(g)v, y \rangle\,dg$. It is self-adjoint and positive, and by left invariance of Haar measure $\pi(h)K\pi(h)^{-1} = K$. Moreover $\langle Kv, v \rangle = \int_G |\langle v, \pi(g)v \rangle|^2\,dg > 0$, since the integrand is continuous, nonnegative and positive at e (Theorem 2.1.1). K is compact: $\|R_g - R_{g'}\| \leq 2\|v\| \|\pi(g)v - \pi(g')v\|$, so $g \mapsto R_g$ is norm-continuous. Given $\varepsilon > 0$, cover G by finitely many open sets $U_i \ni g_i$ with $\|R_g - R_{g_i}\| < \varepsilon$ on U_i, refine to a Borel partition $B_i \subset U_i$, and let $K_\varepsilon := \sum_i \mu(B_i) R_{g_i}$, a finite-rank operator with $|\langle (K - K_\varepsilon)x, y \rangle| \leq \varepsilon \|x\| \|y\|$. So K is a norm limit of finite-rank operators. By Theorem 4.1.1, K has a nonzero eigenvalue with finite-dimensional, nonzero, π-invariant eigenspace, and an invariant subspace of minimal positive dimension in it is irreducible.

Decomposition. By Zorn's lemma choose a maximal family $\{\mathcal{H}_\alpha\}$ of mutually orthogonal finite-dimensional irreducible invariant subspaces. The orthogonal complement of $\widehat{\bigoplus}_\alpha \mathcal{H}_\alpha$ is closed and invariant (π is unitary); if it were nonzero, the claim would contradict maximality. Finally, if π is irreducible, the claim shows that $\mathcal{H}$ itself is finite-dimensional. ☐

Proposition 4.3.2 (Homogeneous spaces and Frobenius reciprocity): Let $K \subset G$ be a closed subgroup with Haar measure dk, and define $L^2(G/K) := \{f \in L^2(G) \mid R(k)f = f \text{ for all } k \in K\}$, a closed L-invariant subspace. Then, as representations of G (acting by L),

$$L^2(G/K) \cong \widehat{\bigoplus_{\pi \in \widehat{G}}} (\dim V_\pi^K) \cdot \pi,$$

i.e., π occurs in $L^2(G/K)$ with multiplicity $\dim V_\pi^K$, where V_π^K is the space of K-fixed vectors.

Proof. Under the isomorphism of Theorem 4.2.1(iii), $R(k)$ acts on $\mathrm{End}(V_\pi)$ by $A \mapsto A\pi(k)^{-1}$, componentwise, so $L^2(G/K)$ corresponds to $\widehat{\bigoplus}_\pi \{A \mid A\pi(k) = A\ \forall k \in K\}$. Let $P := \int_K \pi(k)\,dk$. By invariance of dk, $\pi(k)P = P\pi(k) = P$, $P^2 = P$ and $P^* = P$; so P is the orthogonal projection onto V_π^K (its image is fixed by K, and $Pv = v$ for $v \in V_\pi^K$). If $A\pi(k) = A$ for all k, then $A = AP$; conversely $AP\pi(k) = AP$. Hence the K-invariant part of $\mathrm{End}(V_\pi)$ is $\{A \mid A = AP\} \cong \mathrm{Hom}(V_\pi^K, V_\pi)$, on which $L(g)$ acts by $A \mapsto \pi(g)A$; choosing a basis of V_π^K identifies this with $\dim V_\pi^K$ copies of π. ☐

5 Lie groups and Lie algebras

We mostly work with *matrix Lie groups*, i.e., closed subgroups of $\mathrm{GL}(n, \mathbb{C})$. By Theorem 5.5.1, every compact Lie group is of this form. On $M_n(\mathbb{C})$ we use the operator norm.

5.1 The matrix exponential

Lemma 5.1.1: For $X, Y \in M_n(\mathbb{C})$ and $g \in \mathrm{GL}(n, \mathbb{C})$:

(i) $\exp X := e^X := \sum_{k \geq 0} X^k/k!$ converges absolutely and locally uniformly, and exp is smooth (indeed real analytic).

(ii) If $XY = YX$, then $e^{X+Y} = e^X e^Y$. In particular $(e^X)^{-1} = e^{-X}$ and $t \mapsto e^{tX}$ is a homomorphism $\mathbb{R} \to \mathrm{GL}(n, \mathbb{C})$.

(iii) $ge^X g^{-1} = e^{gXg^{-1}}$, $(e^X)^* = e^{X^*}$, $(e^X)^T = e^{X^T}$ and $\det e^X = e^{\mathrm{tr}\, X}$.

(iv) $\frac{d}{dt} e^{tX} = Xe^{tX} = e^{tX}X$.

(v) There are $r > 0$ and an open neighborhood V of 1 such that $\exp : B_{2r}(0) \to V$ is a diffeomorphism. We denote its inverse by $\log : V \to B_{2r}(0)$.

Proof. (i) $\|X^k/k!\| \leq \|X\|^k/k!$; the entries of e^X are power series in the $2n^2$ real coordinates converging everywhere, hence real analytic. (ii) By absolute convergence we may form the Cauchy product: $e^X e^Y = \sum_m \sum_{k+l=m} \frac{X^k Y^l}{k!l!} = \sum_m \frac{(X+Y)^m}{m!}$ by the binomial theorem, which needs $XY = YX$. (iii) Apply $A \mapsto gAg^{-1}$, $A \mapsto A^*$, $A \mapsto A^T$ termwise. For the determinant, write $X = UTU^{-1}$ with T upper triangular (Schur); e^T is upper triangular with diagonal entries $e^{T_{jj}}$, so $\det e^X = \det e^T = \prod_j e^{T_{jj}} = e^{\mathrm{tr}\, T} = e^{\mathrm{tr}\, X}$. (iv) Differentiate the power series in t termwise. (v) By (iv), $d\exp_0 = \mathrm{id}$; apply the inverse function theorem. ☐

Lemma 5.1.2 (Lie product formula): $e^{X+Y} = \lim_{k\to\infty}(e^{X/k}e^{Y/k})^k$.

Proof. By the power series, $A_k := e^{X/k}e^{Y/k} - 1 = \frac{X+Y}{k} + R_k$ with $\|R_k\| \leq C/k^2$. Since log is smooth near 1 with $d\log_1 = \mathrm{id}$, Taylor's theorem gives $\|\log(1 + A) - A\| \leq C'\|A\|^2$ for small A. Hence for large k, $Z_k := \log(e^{X/k}e^{Y/k}) = \frac{X+Y}{k} + O(k^{-2})$, and $(e^{X/k}e^{Y/k})^k = e^{kZ_k} \to e^{X+Y}$ by continuity of exp, as $kZ_k \to X + Y$. ☐

Proposition 5.1.3 (One-parameter subgroups): Every continuous homomorphism $\gamma : \mathbb{R} \to \mathrm{GL}(n, \mathbb{C})$ is of the form $\gamma(t) = e^{tX}$ for a unique $X \in M_n(\mathbb{C})$.

Proof. Uniqueness: $X = \frac{d}{dt}|_{t=0} e^{tX}$. Existence: let r, V be as in Lemma 5.1.1(v), and choose $\varepsilon > 0$ with $\gamma([-\varepsilon, \varepsilon]) \subset \exp(B_r(0))$. Let $Y := \log \gamma(\varepsilon)$ and $Z := \log \gamma(\varepsilon/2)$, both in $B_r(0)$. Then $e^{2Z} = \gamma(\varepsilon/2)^2 = \gamma(\varepsilon) = e^Y$ with $2Z, Y \in B_{2r}(0)$, so $2Z = Y$. Inductively $\gamma(\varepsilon/2^m) = e^{Y/2^m}$, hence $\gamma(q\varepsilon/2^m) = e^{qY/2^m}$ for $q \in \mathbb{Z}$, $m \in \mathbb{N}$. Both sides of $\gamma(t) = e^{tY/\varepsilon}$ are continuous in t and agree on a dense set. ☐

5.2 The Lie algebra of a matrix Lie group

Definition 5.2.1: A real or complex *Lie algebra* is a vector space $\mathfrak{g}$ with a bilinear, antisymmetric map $[\cdot,\cdot]$ satisfying the Jacobi identity $[X,[Y,Z]]+[Y,[Z,X]]+[Z,[X,Y]]=0$. Homomorphisms preserve brackets, and $\operatorname{ad} X(Y) := [X,Y]$. A *representation* of $\mathfrak{g}$ on V is a homomorphism $\mathfrak{g} \to \mathfrak{gl}(V) = \operatorname{End}(V)$ (with the commutator).

Definition 5.2.2: For a closed subgroup $G \subset \mathrm{GL}(n,\mathbb{C})$, its *Lie algebra* is $\mathfrak{g} := \{X \in M_n(\mathbb{C}) \mid e^{tX} \in G\ \forall t \in \mathbb{R}\}$, and $\operatorname{Ad}(g)X := gXg^{-1}$ for $g \in G$.

If $X \in \mathfrak{g}$ and $g \in G$, then $e^{t\operatorname{Ad}(g)X} = ge^{tX}g^{-1} \in G$, so $\operatorname{Ad}(g)\mathfrak{g} \subset \mathfrak{g}$.

Theorem 5.2.1: Lie algebra of a closed subgroup

$\mathfrak{g}$ is a real Lie subalgebra of $M_n(\mathbb{C})$ with $[X,Y] = XY - YX$.

Proof. $\mathbb{R}\mathfrak{g} \subset \mathfrak{g}$ is clear. For $X, Y \in \mathfrak{g}$, $e^{t(X+Y)} = \lim_k (e^{tX/k}e^{tY/k})^k \in G$ by Lemma 5.1.2 and closedness, so $X+Y \in \mathfrak{g}$. Next, $t \mapsto \operatorname{Ad}(e^{tX})Y$ is a smooth curve in the closed subspace $\mathfrak{g}$, so its derivative at 0, which is $XY - YX$, lies in $\mathfrak{g}$. The Jacobi identity holds for commutators of matrices. ☐

Example 5.2.3:

G	$\mathfrak{g}$	$\dim_{\mathbb{R}} \mathfrak{g}$
$\mathrm{SL}(n,\mathbb{C}) = \{\det = 1\}$	$\mathfrak{sl}(n,\mathbb{C}) = \{X \mid \operatorname{tr} X = 0\}$	$2n^2 - 2$
$\mathrm{U}(n)$	$\mathfrak{u}(n) = \{X \mid X^* = -X\}$	n^2
$\mathrm{SU}(n)$	$\mathfrak{su}(n) = \{X \in \mathfrak{u}(n) \mid \operatorname{tr} X = 0\}$	$n^2 - 1$
$\mathrm{SO}(n)$	$\mathfrak{so}(n) = \{X \in M_n(\mathbb{R}) \mid X^T = -X\}$	$n(n-1)/2$
$\mathrm{Sp}(n)$	$\mathfrak{sp}(n) = \{X \in \mathfrak{u}(2n) \mid X^TJ + JX = 0\}$	$n(2n+1)$
$\mathbb{T}^n$ (diagonal)	$\{\operatorname{diag}(i\theta_1, \dots, i\theta_n) \mid \theta \in \mathbb{R}^n\}$	n

Proof. $\mathrm{U}(n)$: if $e^{tX} \in \mathrm{U}(n)$ for all t, differentiating $e^{tX^*}e^{tX} = 1$ at 0 gives $X^* + X = 0$; conversely if $X^* = -X$ then $(e^{tX})^* = e^{-tX} = (e^{tX})^{-1}$. The real dimension of skew-Hermitian matrices is n (imaginary diagonal) $+2\binom{n}{2}$ (upper triangle) $= n^2$. SL, SU: $\det e^{tX} = e^{t\operatorname{tr} X} = 1$ for all t iff $\operatorname{tr} X = 0$. $\mathrm{SO}(n)$: e^{tX} real for all t iff X real (derivative at 0), then argue as for $\mathrm{U}(n)$; $\det e^{tX} = e^{t\operatorname{tr} X} = 1$ automatically. $\mathrm{Sp}(n)$: if $(e^{tX})^TJe^{tX} = J$ for all t, differentiation gives $X^TJ + JX = 0$; conversely $X^TJ = -JX$ implies $(X^T)^kJ = J(-X)^k$, hence $(e^{tX})^TJ = Je^{-tX}$. Writing $X = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, one computes that $X^TJ + JX = 0$ means $D = -A^T$, $B = B^T$, $C = C^T$, and $X^* = -X$ then means $A^* = -A$ and $C = -B^*$. So $\mathfrak{sp}(n) \cong \mathfrak{u}(n) \oplus \operatorname{Sym}_n(\mathbb{C})$, of real dimension $n^2 + n(n+1)$. The torus case is clear. ☐

Theorem 5.2.2: Closed subgroup theorem

Let $G \subset \mathrm{GL}(n,\mathbb{C})$ be closed with Lie algebra $\mathfrak{g}$. There are a neighborhood U of 0 in $\mathfrak{g}$ and an open neighborhood V of 1 in G such that $\exp : U \to V$ is a homeomorphism. Consequently G is an embedded submanifold of $\mathrm{GL}(n,\mathbb{C})$ of dimension $\dim_{\mathbb{R}} \mathfrak{g}$ with $T_1G = \mathfrak{g}$, multiplication and inversion are smooth, and smooth maps into $\mathrm{GL}(n,\mathbb{C})$ with values in G are smooth into G.

Proof. Choose a real complement $\mathfrak{m}$ of $\mathfrak{g}$ in $M_n(\mathbb{C})$. The map $\Phi(X+Y) := e^Xe^Y$ ($X \in \mathfrak{g}$, $Y \in \mathfrak{m}$) has $d\Phi_0 = \mathrm{id}$, so it restricts to a diffeomorphism from a neighborhood $U_1 \times W_1$ of 0 onto an open neighborhood O of 1. We claim that there is a neighborhood $W \subset W_1$ of 0 in $\mathfrak{m}$ such that $Y \in W$, $e^Y \in G$ imply $Y = 0$. Otherwise there are $0 \neq Y_k \to 0$ in $\mathfrak{m}$ with $e^{Y_k} \in G$; passing to a subsequence, $Y_k/\|Y_k\| \to Y \in \mathfrak{m}$ with $\|Y\| = 1$. For $t \in \mathbb{R}$ let $n_k := \lfloor t/\|Y_k\| \rfloor$; then $n_kY_k \to tY$, so $e^{tY} = \lim (e^{Y_k})^{n_k} \in G$. Thus $Y \in \mathfrak{g} \cap \mathfrak{m} = \{0\}$, a contradiction.

Now shrink U_1, W_1 so that $W_1 \subset W$ and set $U := U_1$, $V := \Phi(U \times W) \cap G$, which is open in G. For $X \in U$, $Y \in W$: $e^X e^Y \in G \iff e^Y \in G \iff Y = 0$. Hence $V = \exp(U)$, and Φ is a chart of $\mathrm{GL}(n,\mathbb{C})$ near 1 in which G corresponds to the linear subspace $\mathfrak{g}$. Left translation by $g \in G$ transports this to a neighborhood of g. The remaining statements follow since G is an embedded submanifold. ☐

Proposition 5.2.4: Let G be a matrix Lie group. The subgroup G_0 generated by $\exp(\mathfrak{g})$ is open and closed in G, and it is the connected component of 1. If G is compact, G/G_0 is finite. If G is connected, G is generated by $\exp(\mathfrak{g})$.

Proof. G_0 contains the neighborhood V of 1 from Theorem 5.2.2, hence $hV \subset G_0$ for $h \in G_0$, so G_0 is open. Its complement is a union of cosets gG_0, which are open, so G_0 is closed. G_0 is path-connected: $t \mapsto e^{tX_1} \cdots e^{tX_k}$ connects 1 to any product of exponentials. Hence G_0 is the component of 1. If G is compact, the open cosets gG_0 cover G, so there are finitely many. ☐

5.3 Homomorphisms and derived representations

Theorem 5.3.1: Lie correspondence for homomorphisms

Let G, H be matrix Lie groups and $\Phi : G \to H$ a continuous homomorphism. There is a unique map $\phi = d\Phi : \mathfrak{g} \to \mathfrak{h}$ with $\Phi(e^X) = e^{\phi(X)}$ for all $X \in \mathfrak{g}$. It is a real-linear Lie algebra homomorphism, $\phi(X) = \frac{d}{dt}|_{t=0}\Phi(e^{tX})$, and $\phi(\mathrm{Ad}(g)X) = \mathrm{Ad}(\Phi(g))\phi(X)$.

Proof. $t \mapsto \Phi(e^{tX})$ is a continuous one-parameter subgroup, so $\Phi(e^{tX}) = e^{t\phi(X)}$ for a unique $\phi(X)$ (Proposition 5.1.3), which lies in $\mathfrak{h}$, and $\phi(X) = \frac{d}{dt}|_0 \Phi(e^{tX})$. Clearly $\phi(sX) = s\phi(X)$. By Lemma 5.1.2, $\Phi(e^{t(X+Y)}) = \lim_k (e^{t\phi(X)/k} e^{t\phi(Y)/k})^k = e^{t(\phi(X)+\phi(Y))}$, so ϕ is additive. From $\Phi(g e^{tX} g^{-1}) = \Phi(g) e^{t\phi(X)} \Phi(g)^{-1}$ we get the equivariance. Applying it with $g = e^{sX}$ and differentiating at $s = 0$ (using linearity of ϕ) gives $\phi([X,Y]) = [\phi(X), \phi(Y)]$. ☐

Corollary 5.3.1: $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$ is a continuous homomorphism with $d\,\mathrm{Ad} = \mathrm{ad}$, i.e., $\mathrm{Ad}(e^X) = e^{\mathrm{ad}\,X}$.

Proof. $\frac{d}{dt}|_0 e^{tX} Y e^{-tX} = XY - YX$, and Theorem 5.3.1. ☐

Definition 5.3.2: For a representation (π, V) of a matrix Lie group G (viewing $\mathrm{GL}(V) \cong \mathrm{GL}(d,\mathbb{C})$), $d\pi : \mathfrak{g} \to \mathfrak{gl}(V)$ is the *derived representation.* Its complex-linear extension to the *complexification* $\mathfrak{g}_{\mathbb{C}} := \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C} = \mathfrak{g} \oplus i\mathfrak{g}$ (with the $\mathbb{C}$-bilinear extension of the bracket) is again denoted $d\pi$.

Proposition 5.3.3: Let G be a connected matrix Lie group and (π, V), (σ, W) representations. A subspace of V is G-invariant if and only if it is invariant under $d\pi(\mathfrak{g})$ (equivalently, under $d\pi(\mathfrak{g}_{\mathbb{C}})$), and $\mathrm{Hom}_G(V,W) = \{A \mid A\, d\pi(X) = d\sigma(X) A \ \forall X \in \mathfrak{g}\}$. In particular, π is irreducible iff $d\pi$ is.

Proof. If V' is $d\pi(\mathfrak{g})$-invariant, it is invariant under $\pi(e^X) = e^{d\pi(X)}$, hence under the group generated by $\exp(\mathfrak{g})$, which is G (Proposition 5.2.4). Conversely, if V' is G-invariant, it contains $d\pi(X)v = \lim_{t\to 0}(\pi(e^{tX})v - v)/t$ for $v \in V'$. The statement on intertwiners is proved in the same way. ☐

5.4 The representations of sl(2,C); SU(2) and SO(3)

We have $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$, since $M_n(\mathbb{C}) = \mathfrak{u}(n) \oplus i\mathfrak{u}(n)$ and the trace condition is preserved. Put

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \qquad [H,E] = 2E,\ [H,F] = -2F,\ [E,F] = H.$$

Note that iH, $E-F$ and $i(E+F)$ form a basis of $\mathfrak{su}(2)$.

Theorem 5.4.1: Representations of sl(2,C)

(i) For every $n \in \mathbb{N}_0$ there is, up to equivalence, exactly one irreducible representation $V^{(n)}$ of $\mathfrak{sl}(2,\mathbb{C})$ of dimension $n+1$. It has a basis $v_0, \dots, v_n$ with

$$Hv_k = (n-2k)v_k, \qquad Fv_k = v_{k+1}\ (v_{n+1} := 0), \qquad Ev_k = k(n-k+1)v_{k-1}.$$

(ii) Let V be a finite-dimensional representation of $\mathfrak{sl}(2,\mathbb{C})$ with an inner product for which $\mathfrak{su}(2)$ acts by skew-Hermitian operators. Then V is a direct sum of irreducible representations; in particular H is diagonalizable with integer eigenvalues.

Proof. (i) Let V be irreducible. From the relations, $Hv = \mu v$ implies $H(Ev) = (\mu+2)Ev$ and $H(Fv) = (\mu-2)Fv$. Starting from an eigenvector of H and applying E repeatedly, since H has finitely many eigenvalues we find $v_0 \neq 0$ with $Hv_0 = \mu v_0$, $Ev_0 = 0$. Let $v_k := F^k v_0$. Then $Hv_k = (\mu - 2k)v_k$, and by induction, using $EF = FE + H$,

$$Ev_k = EFv_{k-1} = F(k-1)(\mu-k+2)v_{k-2} + (\mu-2k+2)v_{k-1} = k(\mu-k+1)v_{k-1}.$$

The nonzero v_k are linearly independent (distinct eigenvalues of H); let n be maximal with $v_n \neq 0$. Then $0 = Ev_{n+1} = (n+1)(\mu-n)v_n$ gives $\mu = n$. The span of $v_0, \dots, v_n$ is invariant, hence equals V, and the formulas show that V is determined by $n = \dim V - 1$ up to equivalence. Conversely, the formulas define a representation: on v_k,

$$[E,F]v_k = (k+1)(n-k)v_k - k(n-k+1)v_k = (n-2k)v_k, \qquad [H,E]v_k = 2k(n-k+1)v_{k-1} = 2Ev_k,$$

and $[H,F]v_k = \big((n-2k-2)-(n-2k)\big)v_{k+1} = -2Fv_k$. It is irreducible: an invariant subspace $\neq 0$ contains an H-eigenvector, i.e., some v_k, and applying E and F yields all v_j.

(ii) The operators iH, $E-F$, $i(E+F)$ are skew-Hermitian, so $H^* = H$, and $E^* - F^* = F - E$, $E^* + F^* = E + F$, i.e., $E^* = F$. If $W \subset V$ is invariant, so is $W^\perp$: e.g. $\langle Ew', w\rangle = \langle w', Fw\rangle = 0$ for $w' \in W^\perp$, $w \in W$. Induction on $\dim V$ and (i) finish the proof. ☐

Lemma 5.4.1: U(n), SU(n) and SO(n) are connected.

Proof. If $g \in \mathrm{U}(n)$, then $g = k\,\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})k^{-1}$ with k unitary, and $s \mapsto k\,\mathrm{diag}(e^{is\theta_j})k^{-1}$ connects 1 to g in U(n). If $\det g = 1$ we may choose the θ_j with $\sum_j \theta_j = 0$, and the path stays in SU(n). If $g \in \mathrm{SO}(n)$, the real normal form of orthogonal matrices gives $k \in \mathrm{O}(n)$ with $k^{-1}gk$ block diagonal with blocks $R(\theta_j) = \left(\begin{smallmatrix} \cos\theta_j & -\sin\theta_j \\ \sin\theta_j & \cos\theta_j \end{smallmatrix}\right)$ and ± 1; as $\det g = 1$, the -1's come in pairs, which we write as $R(\pi)$. Then $s \mapsto k\,\mathrm{diag}(R(s\theta_j), 1, \dots, 1)k^{-1}$ is a path in SO(n) from 1 to g. ☐

Proposition 5.4.2 (SU(2) covers SO(3)): With the inner product $\langle X, Y\rangle := -2\,\mathrm{tr}(XY)$ on $\mathfrak{su}(2)$, $\mathrm{Ad} : \mathrm{SU}(2) \to \mathrm{SO}(\mathfrak{su}(2)) \cong \mathrm{SO}(3)$ is surjective with kernel $\{\pm 1\}$, and it induces an isomorphism of topological groups $\mathrm{SU}(2)/\{\pm 1\} \cong \mathrm{SO}(3)$. The irreducible representations of SO(3) correspond to the π_n with n even: writing $n = 2\ell$, they are the representations $D^{(\ell)}$, $\ell \in \mathbb{N}_0$, of dimension $2\ell + 1$.

Proof. $X_a := -\frac{i}{2}\sigma_a$ (σ_a the Pauli matrices) is an orthonormal basis, and $\mathrm{Ad}(g)$ preserves $\mathrm{tr}(XY)$, so $\mathrm{Ad}(g)$ is orthogonal; $\det \mathrm{Ad}(g) \in \{\pm 1\}$ depends continuously on $g \in \mathrm{SU}(2) \cong S^3$, which is connected, so $\mathrm{Ad}(g) \in \mathrm{SO}(3)$. If $\mathrm{Ad}(g) = 1$, then g commutes with $\mathfrak{su}(2)$ and hence with its complex span $M_2(\mathbb{C})$, so $g = \pm 1$. The derivative $\mathrm{ad} : \mathfrak{su}(2) \to \mathfrak{so}(3)$ is injective by the same argument and therefore bijective (both have dimension 3). By Corollary 5.3.1, $\mathrm{Ad}(\mathrm{SU}(2)) \supset \exp(\mathfrak{so}(3))$, which contains a neighborhood of 1 in $\mathrm{SO}(3)$ (Theorem 5.2.2); so the image is an open subgroup, hence closed, hence all of the connected group $\mathrm{SO}(3)$ (Lemma 5.4.1). The induced continuous bijection $\mathrm{SU}(2)/\{\pm 1\} \to \mathrm{SO}(3)$ from a compact to a Hausdorff space is a homeomorphism. Thus representations of $\mathrm{SO}(3)$ are the same as representations of $\mathrm{SU}(2)$ trivial on -1, with the same invariant subspaces; and $\pi_n(-1)P(z) = P(-z) = (-1)^n P(z)$. ☐

5.5 Compact Lie groups are matrix groups

Lemma 5.5.1 (No small subgroups): A Lie group G has a neighborhood V of e containing no subgroup other than $\{e\}$.

Proof. Choose $r > 0$ such that exp is injective on $B_{2r}(0) \subset \mathfrak{g}$ (for some norm on $\mathfrak{g}$) with open image, and put $V := \exp(B_r(0))$. Let $H \subset V$ be a subgroup and $e \neq h = \exp X \in H$ with $0 < \|X\| < r$. Choose $k \in \mathbb{N}$ with $r \leq k\|X\| < 2r$. Then $h^k = \exp(kX) \in H \subset V = \exp(B_r(0))$, which contradicts injectivity of exp on $B_{2r}(0)$. ☐

Remark 5.5.2: $\prod_{n\in\mathbb{N}} \mathbb{Z}/2\mathbb{Z}$ is a compact group which is not a Lie group: every neighborhood of e contains a subgroup $\{x \mid x_1 = \cdots = x_N = 0\} \neq \{e\}$.

Theorem 5.5.1: Compact Lie groups are linear

Every compact Lie group G has a faithful unitary representation, and is therefore isomorphic as a topological group to a closed subgroup of some $\mathrm{U}(N)$.

Proof. For $\pi \in \widehat{G}$, $K_\pi := \ker \pi$ is a closed subgroup, and $\bigcap_\pi K_\pi = \{e\}$ by Corollary 4.3.1(ii). Let V be an open neighborhood of e as in Lemma 5.5.1. The compact sets $K_\pi \setminus V$ have empty intersection, so finitely many of them, say for $\pi_1, \dots, \pi_k$, already do. Then $\bigcap_j K_{\pi_j}$ is a subgroup contained in V, hence trivial, and $\pi := \pi_1 \oplus \cdots \oplus \pi_k$ (unitary by Theorem 2.2.1) is injective. A continuous injection of a compact space into a Hausdorff space is a homeomorphism onto its (compact, hence closed) image. ☐

6 Compact Lie algebras and the exponential map

6.1 Invariant inner products

Proposition 6.1.1: Let G be a compact matrix Lie group. There is an inner product on $\mathfrak{g}$ which is Ad-invariant. For it, every $\mathrm{ad}\, X$ is skew-symmetric, $\langle [X,Y], Z\rangle = -\langle Y, [X,Z]\rangle$, and hence diagonalizable over $\mathbb{C}$ with purely imaginary eigenvalues.

Proof. Average any inner product as in Theorem 2.2.1 (the argument works verbatim for real representations). Differentiating $\langle \mathrm{Ad}(e^{tX})Y, \mathrm{Ad}(e^{tX})Z\rangle = \langle Y, Z\rangle$ at $t = 0$ and using Corollary 5.3.1 gives skew-symmetry. A real skew-symmetric matrix is normal, hence unitarily diagonalizable, with eigenvalues λ satisfying $\bar{\lambda} = -\lambda$. ☐

Example 6.1.2: For $G \subset \mathrm{U}(n)$, $\langle X, Y\rangle := -\mathrm{tr}(XY) = \mathrm{tr}(XY^*)$ is Ad-invariant and positive definite on $\mathfrak{g}$.

Proposition 6.1.3: Let $\mathfrak{z}$ be the center of $\mathfrak{g}$ and fix an Ad-invariant inner product. Then $\mathfrak{g} = \mathfrak{z} \oplus [\mathfrak{g}, \mathfrak{g}]$ orthogonally, $[\mathfrak{g}, \mathfrak{g}]$ has trivial center, and the Killing form $B(X, Y) := \operatorname{tr}(\operatorname{ad} X \operatorname{ad} Y)$ is negative semidefinite with kernel $\mathfrak{z}$.

Proof. By skew-symmetry, $\langle Z, [X, Y]\rangle = -\langle [X, Z], Y\rangle$, which vanishes for all X, Y iff $Z \in \mathfrak{z}$; thus $[\mathfrak{g}, \mathfrak{g}]^{\perp} = \mathfrak{z}$. An element of the center of $[\mathfrak{g}, \mathfrak{g}]$ commutes with $[\mathfrak{g}, \mathfrak{g}]$ and with $\mathfrak{z}$, hence lies in $\mathfrak{z} \cap [\mathfrak{g}, \mathfrak{g}] = 0$. Finally $B(X, X) = \operatorname{tr}((\operatorname{ad} X)^2) = -\operatorname{tr}((\operatorname{ad} X)^T \operatorname{ad} X) \leq 0$ with equality iff $\operatorname{ad} X = 0$, i.e., $X \in \mathfrak{z}$; since B is symmetric and semidefinite, its kernel is $\{X \mid B(X, X) = 0\} = \mathfrak{z}$. ☐

6.2 The differential of the exponential map

Proposition 6.2.1: For $X, Y \in M_n(\mathbb{C})$,

$$e^{-X} \frac{d}{ds}\Big|_{s=0} e^{X+sY} = \frac{1 - e^{-\operatorname{ad} X}}{\operatorname{ad} X} Y := \sum_{k \geq 0} \frac{(-1)^k}{(k+1)!} (\operatorname{ad} X)^k Y.$$

In particular, $d\exp_X$ is invertible if and only if $\operatorname{ad} X$ has no eigenvalue in $2\pi i\mathbb{Z} \setminus \{0\}$.

Proof. Let $Z(s) := X + sY$ and $\Gamma(s, t) := e^{-tZ(s)} \partial_s e^{tZ(s)}$. Then $\partial_t \Gamma = e^{-tZ}(-Z\partial_s e^{tZ} + \partial_s(Ze^{tZ})) = e^{-tZ} Y e^{tZ}$, which at $s = 0$ equals $e^{-t \operatorname{ad} X} Y$ by Corollary 5.3.1 (applied in $\mathrm{GL}(n, \mathbb{C})$). Since $\Gamma(s, 0) = 0$, integrating over $t \in [0, 1]$ gives the formula. The operator $\frac{1-e^{-\operatorname{ad} X}}{\operatorname{ad} X}$ has eigenvalues $\frac{1-e^{-z}}{z}$ ($:= 1$ for $z = 0$) for the eigenvalues z of $\operatorname{ad} X$ (triangularize $\operatorname{ad} X$), and these vanish exactly for $z \in 2\pi i\mathbb{Z} \setminus \{0\}$. ☐

Remark 6.2.2: The exponential map of a connected matrix Lie group need not be surjective. For $A = \left(\begin{smallmatrix} -1 & 1 \\ 0 & -1 \end{smallmatrix}\right) \in \mathrm{SL}(2, \mathbb{R})$ there is no $X \in \mathfrak{sl}(2, \mathbb{R})$ with $e^X = A$: the eigenvalues of X are $\pm\lambda$ with $\lambda^2 = -\det X \in \mathbb{R}$. If $\lambda \in \mathbb{R} \setminus \{0\}$, e^X has the positive eigenvalues $e^{\pm\lambda}$; if $\lambda = 0$, X is nilpotent and $e^X = 1 + X$ has eigenvalue 1; if $\lambda \in i\mathbb{R} \setminus \{0\}$, X is diagonalizable (distinct eigenvalues), so e^X is diagonalizable, and having eigenvalues $-1, -1$ it would equal $-1 \neq A$. For compact connected groups, exp is surjective (Corollary 7.3.2).

7 Maximal tori

From now on, G denotes a compact connected matrix Lie group with Lie algebra $\mathfrak{g}$, equipped with a fixed Ad-invariant inner product $\langle \cdot, \cdot \rangle$ (Proposition 6.1.1).

7.1 Tori

Definition 7.1.1: A *torus* is a compact connected abelian matrix Lie group.

Lemma 7.1.2: Let Λ be a discrete subgroup of a finite-dimensional real vector space E. Then $\Lambda = \mathbb{Z}v_1 \oplus \cdots \oplus \mathbb{Z}v_m$ for linearly independent $v_1, \ldots, v_m$.

Proof. Let $m := \dim \operatorname{span} \Lambda$ and choose $w_1, \ldots, w_m \in \Lambda$ forming a basis of $\operatorname{span} \Lambda$. The bounded set $\{\sum c_j w_j \mid 0 \leq c_j < 1\}$ contains only finitely many points of the discrete closed set Λ, and every coset of $\Lambda_0 := \bigoplus \mathbb{Z}w_j$ in Λ meets it; so $N := [\Lambda : \Lambda_0] < \infty$ and $\Lambda \subset \frac{1}{N}\Lambda_0 \cong \mathbb{Z}^m$. Subgroups of $\mathbb{Z}^m$ are free of rank $\leq m$, and Λ spans an m-dimensional space, so a $\mathbb{Z}$-basis of Λ consists of m linearly independent vectors. ☐

Proposition 7.1.3: Let T be a torus of dimension k with Lie algebra $\mathfrak{t}$. Then $\exp : \mathfrak{t} \to T$ is a surjective homomorphism whose kernel Λ is a lattice of rank k, and $\mathfrak{t}/\Lambda \cong T$ as topological groups. In particular $T \cong \mathbb{R}^k/\mathbb{Z}^k \cong \mathbb{T}^k$.

Proof. T is abelian, so Ad is trivial and $\mathfrak{t}$ is abelian (Corollary 5.3.1); hence exp is a homomorphism (Lemma 5.1.1(ii)). Its image is a subgroup containing $\exp(\mathfrak{t})$, so it is T by Proposition 5.2.4. Since exp is injective near 0, Λ is discrete, so $\Lambda = \bigoplus_{j=1}^{m} \mathbb{Z}v_j$ by Lemma 7.1.2. The induced bijection $\mathfrak{t}/\Lambda \to T$ is a homomorphism and a local homeomorphism (Theorem 5.2.2), hence a homeomorphism. So $\mathfrak{t}/\Lambda \cong \mathbb{T}^m \times \mathbb{R}^{k-m}$ is compact, forcing $m = k$. ☐

Definition 7.1.4: For a torus $T = \exp(\mathfrak{t})$ with $\Lambda = \ker \exp$, the *weight lattice* is $P(T) := \{\lambda \in \mathfrak{t}^* \mid \lambda(\Lambda) \subset 2\pi\mathbb{Z}\}$. For $\lambda \in P(T)$ and $t = \exp H$ put $t^\lambda := e^{i\lambda(H)}$.

Proposition 7.1.5: $\lambda \mapsto (t \mapsto t^\lambda)$ is a bijection from $P(T)$ onto the set of irreducible representations (characters) of T, and $t^{\lambda+\mu} = t^\lambda t^\mu$. The characters t^λ, $\lambda \in P(T)$, form an orthonormal basis of $L^2(T)$.

Proof. Choose a $\mathbb{Z}$-basis $v_1, \ldots, v_k$ of Λ; then $\mathbb{T}^k \to T$, $(e^{2\pi i s_j})_j \mapsto \exp(\sum_j s_j v_j)$ is an isomorphism. Under it, t^λ corresponds to $\prod_j z_j^{m_j}$ with $m_j = \lambda(v_j)/2\pi \in \mathbb{Z}$, and every $m \in \mathbb{Z}^k$ arises from a unique λ. Now apply Proposition 1.3.3 and Theorem 4.2.1(ii). ☐

Definition 7.1.6: $t \in T$ is a *generator* of T if $\{t^m \mid m \in \mathbb{Z}\}$ is dense in T.

Theorem 7.1.1: Kronecker

The element $(e^{2\pi i\theta_1}, \ldots, e^{2\pi i\theta_k}) \in \mathbb{T}^k$ is a generator if and only if $1, \theta_1, \ldots, \theta_k$ are linearly independent over $\mathbb{Q}$. Every torus has a generator.

Proof. Let t be the given element and S the closure of $\{t^m\}$, a closed subgroup. If $S \neq \mathbb{T}^k$, then $\mathbb{T}^k/S$ is a nontrivial compact abelian group, so by Corollary 4.3.1(ii) and Corollary 1.3.2 it has a nontrivial character; composing with $\mathbb{T}^k \to \mathbb{T}^k/S$ gives a character z^m, $m \neq 0$, with t^m-value $e^{2\pi i \sum m_j\theta_j} = 1$, i.e., $\sum_j m_j\theta_j \in \mathbb{Z}$, a rational relation. Conversely, a relation $\sum_j m_j\theta_j \in \mathbb{Z}$ with $m \neq 0$ gives a nontrivial character equal to 1 on all t^m, whose kernel is a proper closed subgroup containing $\{t^m\}$. Existence: since the $\mathbb{Q}$-span of finitely many reals is countable, we can choose inductively $\theta_j \notin \operatorname{span}_{\mathbb{Q}}\{1, \theta_1, \ldots, \theta_{j-1}\}$. For a general torus use $T \cong \mathbb{T}^k$. ☐

7.2 Maximal tori and the Weyl group

Definition 7.2.1: A *maximal torus* of G is a torus $T \subset G$ not properly contained in another torus. Its dimension is the *rank* of G.

Lemma 7.2.2: (i) If $\mathfrak{a} \subset \mathfrak{g}$ is an abelian subalgebra, then the closure $\overline{\exp(\mathfrak{a})}$ is a torus with Lie algebra containing $\mathfrak{a}$.

(ii) If $T \subset T'$ are tori of the same dimension, then $T = T'$. In particular, maximal tori exist.

Proof. (i) $\exp(\mathfrak{a})$ is a connected abelian subgroup (Lemma 5.1.1(ii)), so its closure is a compact connected abelian closed subgroup of G. (ii) T is an open subgroup of T' by Theorem 5.2.2, hence closed, hence equal to T' as T' is connected. Thus any increasing chain of tori stabilizes, as dimensions are bounded by $\dim G$. ☐

Lemma 7.2.3: Let $T \subset G$ be a torus with Lie algebra $\mathfrak{t}$. Then T is a maximal torus if and only if

$$\mathfrak{g}^T := \{X \in \mathfrak{g} \mid \mathrm{Ad}(t)X = X \ \forall t \in T\} = \{X \in \mathfrak{g} \mid [H, X] = 0 \ \forall H \in \mathfrak{t}\}$$

equals $\mathfrak{t}$, i.e., iff $\mathfrak{t}$ is a maximal abelian subalgebra of $\mathfrak{g}$.

Proof. The two descriptions of $\mathfrak{g}^T$ agree: since $T = \exp(\mathfrak{t})$ and $\mathrm{Ad}(e^H) = e^{\operatorname{ad} H}$, X is fixed by $\mathrm{Ad}(T)$ iff $e^{s \operatorname{ad} H} X = X$ for all s, H, iff $[H, X] = 0$ for all $H \in \mathfrak{t}$. If T is maximal and $X \in \mathfrak{g}^T$, then $\mathfrak{t} + \mathbb{R}X$ is abelian, so by Lemma 7.2.2(i) $T \subset \overline{\exp(\mathfrak{t} + \mathbb{R}X)}$, a torus; by maximality they are equal, and $e^{sX} \in T$ for all s, i.e., $X \in \mathfrak{t}$. Conversely, if $\mathfrak{g}^T = \mathfrak{t}$ and $T \subset T'$ with T' a torus, then $\mathrm{Lie}(T')$ is abelian and contains $\mathfrak{t}$, so it lies in $\mathfrak{g}^T = \mathfrak{t}$; by Lemma 7.2.2(ii), $T' = T$. ☐

Example 7.2.4: The following are maximal tori.

(i) $\mathrm{U}(n)$: $T = \{\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n})\}$; rank n.

(ii) $\mathrm{SU}(n)$: $T \cap \mathrm{SU}(n)$; rank $n - 1$.

(iii) $\mathrm{SO}(2n)$: $T = \{\mathrm{diag}(R(\theta_1), \dots, R(\theta_n))\}$; $\mathrm{SO}(2n+1)$: $T = \{\mathrm{diag}(R(\theta_1), \dots, R(\theta_n), 1)\}$; rank n.

(iv) $\mathrm{Sp}(n)$: $T = \{\mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_n}, e^{-i\theta_1}, \dots, e^{-i\theta_n})\}$; rank n.

Proof. Each T is a compact connected abelian closed subgroup, hence a torus; we verify $\mathfrak{g}^T = \mathfrak{t}$ (Lemma 7.2.3). (i), (ii): $\mathfrak{t}$ contains a diagonal matrix H with distinct entries (with trace 0 for $\mathrm{SU}(n)$, $n \geq 2$); a matrix commuting with H preserves its eigenspaces, the coordinate axes, so it is diagonal. Diagonal skew-Hermitian (traceless) matrices form $\mathfrak{t}$. (iii) Take $H = \mathrm{diag}(\theta_1 J_2, \dots, \theta_n J_2)$ (plus a 0 for $\mathrm{SO}(2n+1)$), $J_2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$, with $0 < \theta_1 < \cdots < \theta_n$. Its eigenspaces in $\mathbb{C}^{2n}$ (resp. $\mathbb{C}^{2n+1}$) for the distinct eigenvalues $\pm i\theta_j$ (and 0) are lines in the coordinate planes (resp. the last axis). A real matrix X commuting with H preserves these, hence preserves each coordinate plane, so X is block diagonal with 2×2 blocks commuting with J_2, i.e., of the form $a + bJ_2$ (and a 1×1 block). If $X^T = -X$, then $a = 0$ and the 1×1 block vanishes, so $X \in \mathfrak{t}$. (iv) A generic element of $\mathfrak{t}$ is diagonal with distinct entries $\pm i\theta_j$, so its centralizer in $\mathfrak{u}(2n)$ consists of diagonal matrices $\mathrm{diag}(a, b)$, and the condition $X^T J + JX = 0$ gives $b = -a$ (cf. the proof of Example 5.2.3); hence $\mathfrak{g}^T = \mathfrak{t}$. ☐

Definition 7.2.5: Let $T \subset G$ be a maximal torus and $N(T) := \{g \in G \mid gTg^{-1} = T\}$. The *Weyl group* is $W := N(T)/T$. It acts on T by $w \cdot t := ntn^{-1}$ ($w = nT$), on $\mathfrak{t}$ by $\mathrm{Ad}(n)$, and on $\mathfrak{t}^*$ by $(w\lambda)(H) := \lambda(\mathrm{Ad}(n)^{-1} H)$. These actions are well defined since T is abelian, and w preserves Λ and $P(T)$, with $(w \cdot t)^{w\lambda} = t^\lambda$.

Proposition 7.2.6: *W is finite.*

Proof. $N(T)$ is closed in G, hence compact, so $N(T)/N(T)_0$ is finite (Proposition 5.2.4). For $n \in N(T)$, $\mathrm{Ad}(n)|_{\mathfrak{t}}$ preserves Λ, so it lies in the discrete set $\{A \in \mathrm{GL}(\mathfrak{t}) \mid A\Lambda = \Lambda\}$. The continuous map $n \mapsto \mathrm{Ad}(n)|_{\mathfrak{t}}$ is therefore constant on the connected set $N(T)_0$, so $\mathrm{Ad}(n)|_{\mathfrak{t}} = \mathrm{id}$ for $n \in N(T)_0$. Thus $N(T)_0 \subset Z_G(T)$ and $\mathrm{Lie}(N(T)_0) \subset \mathfrak{g}^T = \mathfrak{t}$ (Lemma 7.2.3), so $N(T)_0 \subset T$ by Proposition 5.2.4. Hence W is a quotient of $N(T)/N(T)_0$. ☐

Example 7.2.7: For $G = \mathrm{U}(n)$, $W \cong S_n$, acting by permuting the diagonal entries. Indeed, permutation matrices normalize T. Conversely, if $n \in N(T)$ and $t \in T$ has distinct eigenvalues, then n maps the eigenspaces of t (the coordinate axes) to the eigenspaces of $ntn^{-1} \in T$, which are also coordinate axes; so n is a permutation matrix times a diagonal matrix in T.

7.3 The torus theorem

Lemma 7.3.1: Let T be a maximal torus of G. Every $X \in \mathfrak{g}$ is $\mathrm{Ad}(G)$-conjugate to an element of $\mathfrak{t}$.

Proof. Let $t = e^{H_0}$ be a generator of T (Theorem 7.1.1, Proposition 7.1.3). Then $\overline{\exp(\mathbb{R}H_0)} \supset \overline{\{t^m\}} = T$. The continuous function $f(g) := \langle \mathrm{Ad}(g)X, H_0\rangle$ on the compact group G attains its maximum at some g_0. Put $Y := \mathrm{Ad}(g_0)X$. For $Z \in \mathfrak{g}$, by Corollary 5.3.1 and skew-symmetry of ad,

$$0 = \frac{d}{ds}\Big|_{s=0} f(e^{sZ}g_0) = \langle [Z,Y], H_0\rangle = \langle Z, [Y, H_0]\rangle,$$

so $[H_0, Y] = 0$. Hence $\mathrm{Ad}(e^{sH_0})Y = e^{s\,\mathrm{ad}\,H_0}Y = Y$ for all s, so by continuity $\mathrm{Ad}(t')Y = Y$ for all $t' \in \overline{\exp(\mathbb{R}H_0)} = T$, i.e., $Y \in \mathfrak{g}^T = \mathfrak{t}$ (Lemma 7.2.3). ☐

Theorem 7.3.1: Torus theorem (É. Cartan)

Let G be a compact connected matrix Lie group and T a maximal torus. Then $G = \bigcup_{g\in G} gTg^{-1}$. Moreover, any two maximal tori are conjugate.

Proof. We use induction on $\dim G$; for $\dim G = 0$, $G = \{1\} = T$. Let $S := \bigcup_g gTg^{-1}$, the image of the compact space $G \times T$ under $(g,t) \mapsto gtg^{-1}$; so S is closed and nonempty. We show that S is open; then $S = G$ by connectedness. Since S is invariant under conjugation, it suffices to show that every $t_0 \in T$ has a neighborhood contained in S.

Case 1: t_0 is central in G. By Lemma 7.3.1, every $X \in \mathfrak{g}$ can be written as $X = \mathrm{Ad}(g)Y$ with $Y \in \mathfrak{t}$, so $t_0e^X = g(t_0e^Y)g^{-1} \in gTg^{-1}$. Hence S contains $t_0\exp(\mathfrak{g})$, which is a neighborhood of t_0 (Theorem 5.2.2).

Case 2: t_0 is not central. Let $H := Z_G(t_0)_0$, a compact connected matrix Lie group with Lie algebra $\mathfrak{h} = \{X \in \mathfrak{g} \mid \mathrm{Ad}(t_0)X = X\}$. Since T is connected and commutes with t_0, $T \subset H$, and T is a maximal torus of H. If $\dim H = \dim G$, then H would be an open subgroup of G (Theorem 5.2.2), hence closed, hence equal to G, and t_0 would be central. So $\dim H < \dim G$, and by induction $H = \bigcup_{h\in H} hTh^{-1}$. Consider the smooth map $\mu : G\times H \to G$, $\mu(g,h) := ghg^{-1}$, whose image is contained in $\bigcup_g gHg^{-1} \subset S$. For $X \in \mathfrak{g}$, $Y \in \mathfrak{h}$,

$$t_0^{-1}\mu(e^{sX}, t_0e^{sY}) = e^{s\,\mathrm{Ad}(t_0^{-1})X}e^{sY}e^{-sX}, \qquad \frac{d}{ds}\Big|_{s=0}(\dots) = (\mathrm{Ad}(t_0^{-1}) - 1)X + Y.$$

Since $\mathrm{Ad}(t_0^{-1})$ is orthogonal, the image of $\mathrm{Ad}(t_0^{-1}) - 1$ is the orthogonal complement of $\ker(\mathrm{Ad}(t_0^{-1}) - 1) = \mathfrak{h}$. Hence the differential of $(g,h) \mapsto t_0^{-1}\mu(g,h)$ at $(1, t_0)$ is onto $\mathfrak{h}^\perp + \mathfrak{h} = \mathfrak{g}$. By the submersion theorem, $\mu(G\times H)$, and hence S, contains a neighborhood of t_0.

Finally, let T' be another maximal torus and t' a generator of T'. Then $t' = gtg^{-1}$ for some $g \in G$, $t \in T$, so $T' = \overline{\{t'^m\}} \subset gTg^{-1}$, and by maximality $T' = gTg^{-1}$. ☐

Corollary 7.3.2: Let G be compact connected with maximal torus T.

(i) $\exp : \mathfrak{g} \to G$ is surjective, and every element of G lies in a maximal torus.

(ii) Every $X \in \mathfrak{g}$ is $\mathrm{Ad}(G)$-conjugate to an element of $\mathfrak{t}$.

(iii) Two elements of T are conjugate in G if and only if they are conjugate under W.

(iv) A continuous class function on G is determined by its restriction to T, which is W-invariant.

Proof. (i) $g = hth^{-1} \in hTh^{-1}$, a maximal torus, and $t = e^H$ with $H \in \mathfrak{t}$ gives $g = e^{\mathrm{Ad}(h)H}$. (ii) is Lemma 7.3.1. (iv) follows from the theorem and $f(w\cdot t) = f(ntn^{-1}) = f(t)$. (iii) Let $s, t \in T$ with $gsg^{-1} = t$. Then T and gTg^{-1} are maximal tori of G contained in the compact connected group $H := Z_G(t)_0$ (both are connected and commute with t), hence maximal tori of H. By the theorem for H, there is $h \in H$ with $hgTg^{-1}h^{-1} = T$. So $n := hg \in N(T)$ and $nsn^{-1} = hth^{-1} = t$. ☐

7.4 Centralizers of tori

Lemma 7.4.1: Let A be a compact abelian matrix Lie group whose identity component A_0 satisfies: A/A_0 is cyclic. Then A has a generator, i.e., an element a with $\overline{\{a^m\}} = A$.

Proof. A_0 is a torus; let t_0 be a generator of A_0. Let xA_0 generate A/A_0, of order m, so $x^m \in A_0$. Since $\exp : \mathrm{Lie}(A_0) \to A_0$ is a surjective homomorphism, A_0 is divisible: there is $y \in A_0$ with $y^m = x^{-m}t_0$. Put $a := xy$. Then $a^m = t_0$, so $\overline{\{a^k\}} \supset \overline{\{t_0^k\}} = A_0$; and $a^k \in x^k A_0$ meets every coset. Hence $\overline{\{a^k\}} = A$. ☐

Proposition 7.4.2: Let $S \subset G$ be a torus. Then the centralizer $Z_G(S)$ is connected, with Lie algebra $\{X \in \mathfrak{g} \mid \mathrm{Ad}(s)X = X \ \forall s \in S\}$.

Proof. The Lie algebra statement is the definition of $\mathrm{Lie}(Z_G(S))$ together with $e^{tX}s = se^{tX} \iff e^{t\,\mathrm{Ad}(s^{-1})X} = e^{tX}$ for all $t \iff \mathrm{Ad}(s^{-1})X = X$. Let $x \in Z_G(S)$ and let A be the closure of the subgroup generated by x and S, a compact abelian group with $S \subset A_0$. As A/A_0 is finite, $x^m A_0 = A_0$ for some $m \geq 1$, so $\bigcup_k x^k A_0$ is a finite union of closed cosets, a closed subgroup containing x and S; hence it equals A, and A/A_0 is cyclic. By Lemma 7.4.1 A has a generator a, which by Corollary 7.3.2(i) lies in a maximal torus T'. Then $A \subset T'$, so T' is a connected subgroup of $Z_G(S)$ containing 1 and x. Hence $x \in Z_G(S)_0$. ☐

Corollary 7.4.3: (i) $Z_G(T) = T$ for a maximal torus T. Consequently, W acts faithfully on T and on $\mathfrak{t}$.

(ii) If $H \in \mathfrak{t}$ satisfies $\{X \in \mathfrak{g} \mid [H, X] = 0\} = \mathfrak{t}$, then every $g \in G$ with $\mathrm{Ad}(g)H = H$ lies in T.

Proof. (i) $Z_G(T)$ is connected with Lie algebra $\mathfrak{g}^T = \mathfrak{t}$, so it equals T (Proposition 5.2.4). If $n \in N(T)$ acts trivially on T, or on $\mathfrak{t}$ (then also on $T = \exp \mathfrak{t}$), then $n \in Z_G(T) = T$. (ii) $\mathrm{Ad}(g)H = H$ implies that g commutes with e^{sH} for all s, hence with the torus $S := \overline{\exp(\mathbb{R}H)}$. By Proposition 7.4.2, $Z_G(S)$ is connected, and its Lie algebra $\{X \mid \mathrm{Ad}(e^{sH})X = X \ \forall s\} = \{X \mid [H, X] = 0\} = \mathfrak{t}$. Since $T \subset Z_G(S)$, we get $Z_G(S) = T$ and $g \in T$. ☐

8 Roots and weights

Throughout, $T \subset G$ is a fixed maximal torus with Lie algebra $\mathfrak{t}$ and $\Lambda = \ker(\exp|_{\mathfrak{t}})$.

8.1 Weights of a representation

Definition 8.1.1: Let (π, V) be a representation of G. For $\lambda \in P(T)$ let $V_\lambda := \{v \in V \mid \pi(t)v = t^\lambda v \ \forall t \in T\}$. The λ with $V_\lambda \neq 0$ are the *weights* of π, and $m_\lambda := \dim V_\lambda$ their *multiplicities.*

Proposition 8.1.2: (i) $V = \bigoplus_\lambda V_\lambda$, and $d\pi(H)v = i\lambda(H)v$ for $H \in \mathfrak{t}$, $v \in V_\lambda$ (with $d\pi$ extended $\mathbb{C}$-linearly to $\mathfrak{t}_\mathbb{C}$).

(ii) $\chi_\pi(t) = \sum_\lambda m_\lambda t^\lambda$ for $t \in T$, and π is determined up to equivalence by its weights and multiplicities.

(iii) $m_{w\lambda} = m_\lambda$ for all $w \in W$.

Proof. (i) By Corollary 2.2.1, $\pi|_T$ is a sum of irreducible representations of T, which are characters t^λ (Proposition 7.1.5); grouping them gives $V = \bigoplus V_\lambda$. Differentiating $\pi(e^{sH})v = e^{is\lambda(H)}v$ gives the second claim. (ii) The trace formula follows from (i), and χ_π determines π (Corollary 3.2.3(ii), Corollary 7.3.2(iv)). (iii) For $w = nT$, $\chi_\pi(ntn^{-1}) = \chi_\pi(t)$ and $(ntn^{-1})^\lambda = t^{w^{-1}\lambda}$, so $\sum_\lambda m_\lambda t^{w^{-1}\lambda} = \sum_\lambda m_\lambda t^\lambda$. Since the characters of T are linearly independent (orthonormal), $m_{w\lambda} = m_\lambda$. ☐

Example 8.1.3: The weights of the defining representation of $\mathrm{U}(n)$ are $e_1, \dots, e_n$, where $e_j(\operatorname{diag}(i\theta_1, \dots, i\theta_n)) := \theta_j$. For $\mathrm{SU}(2)$ with $\mathfrak{t} = \{\operatorname{diag}(i\theta, -i\theta)\}$ and $\varepsilon(\operatorname{diag}(i\theta, -i\theta)) := \theta$, the weights of π_n are $(n-2k)\varepsilon$, $k = 0, \dots, n$ (proof of Theorem 3.3.1).

8.2 Roots

Let B be the complex-bilinear extension of the invariant inner product to $\mathfrak{g}_\mathbb{C}$ and $X \mapsto \bar{X}$ the complex conjugation of $\mathfrak{g}_\mathbb{C} = \mathfrak{g} \oplus i\mathfrak{g}$. Then $(X, Y)_h := B(X, \bar{Y})$ is a Hermitian inner product on $\mathfrak{g}_\mathbb{C}$, and for $Z \in \mathfrak{g}$, $\operatorname{ad} Z$ is skew-Hermitian for $(\cdot, \cdot)_h$ (Proposition 6.1.1). B is ad-invariant: $B([Z, X], Y) = -B(X, [Z, Y])$ for $X, Y, Z \in \mathfrak{g}_\mathbb{C}$.

For $\lambda \in \mathfrak{t}^*$ let $H_\lambda \in \mathfrak{t}$ be defined by $\langle H_\lambda, H \rangle = \lambda(H)$ for $H \in \mathfrak{t}$, and put $(\lambda, \mu) := \langle H_\lambda, H_\mu \rangle$, an inner product on $\mathfrak{t}^*$. Since W acts on $\mathfrak{t}$ by the orthogonal maps $\operatorname{Ad}(n)$, it acts on $\mathfrak{t}^*$ by orthogonal maps.

Definition 8.2.1: The nonzero weights of the representation Ad of G on $\mathfrak{g}_\mathbb{C}$ are the *roots*; their set is $\Delta \subset P(T)$. For $\alpha \in \Delta$, $\mathfrak{g}_\alpha := \{X \in \mathfrak{g}_\mathbb{C} \mid [H, X] = i\alpha(H)X \ \forall H \in \mathfrak{t}\}$ is the *root space*.

Proposition 8.2.2 (Root space decomposition): (i) $\mathfrak{g}_\mathbb{C} = \mathfrak{t}_\mathbb{C} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha$, orthogonally for $(\cdot, \cdot)_h$.

(ii) $\overline{\mathfrak{g}_\alpha} = \mathfrak{g}_{-\alpha}$; in particular $\Delta = -\Delta$.

(iii) $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$, where $\mathfrak{g}_0 := \mathfrak{t}_\mathbb{C}$ and $\mathfrak{g}_\gamma := 0$ for $\gamma \notin \Delta \cup \{0\}$.

(iv) Δ is W-invariant.

Proof. (i) By Proposition 8.1.2(i) applied to Ad, $\mathfrak{g}_\mathbb{C}$ is the sum of its weight spaces, which for nonzero weights are the $\mathfrak{g}_\alpha$ (as $\operatorname{Ad}(e^H) = e^{\operatorname{ad} H}$). The zero weight space is the complexification of $\mathfrak{g}^T = \mathfrak{t}$ (Lemma 7.2.3). Orthogonality: the $\operatorname{ad} H$, $H \in \mathfrak{t}$, are commuting skew-Hermitian operators, and the weight spaces are their joint eigenspaces for distinct eigenvalue functions. (ii) Conjugating $[H, X] = i\alpha(H)X$ ($H \in \mathfrak{t}$ is real) gives $[H, \bar{X}] = -i\alpha(H)\bar{X}$. (iii) Jacobi identity. (iv) Proposition 8.1.2(iii). □

Example 8.2.3 (Roots of U(n)): Here $\mathfrak{g}_\mathbb{C} = \mathfrak{gl}(n, \mathbb{C})$, and for $H = \operatorname{diag}(i\theta_1, \dots, i\theta_n)$, $[H, E_{jk}] = i(\theta_j - \theta_k)E_{jk}$. Hence $\Delta = \{e_j - e_k \mid j \neq k\}$ and $\mathfrak{g}_{e_j - e_k} = \mathbb{C}E_{jk}$. With $\langle X, Y \rangle = -\operatorname{tr}(XY)$, the e_j are orthonormal in $\mathfrak{t}^*$.

8.3 The SU(2)-subalgebras attached to roots

Definition 8.3.1: For $\alpha \in \Delta$, the *coroot* is $\alpha^\vee := \frac{2}{(\alpha,\alpha)} H_\alpha \in \mathfrak{t}$. Thus $\alpha(\alpha^\vee) = 2$ and $\lambda(\alpha^\vee) = \frac{2(\lambda,\alpha)}{(\alpha,\alpha)}$ for $\lambda \in \mathfrak{t}^*$.

Proposition 8.3.2: Let $\alpha \in \Delta$.

(i) For $X \in \mathfrak{g}_\alpha$ and $Y \in \mathfrak{g}_{-\alpha}$: $[X, Y] = iB(X, Y)H_\alpha$.

(ii) There is $X_\alpha \in \mathfrak{g}_\alpha$ such that, with $H' := -i\alpha^\vee$, $E' := X_\alpha$, $F' := -\bar{X}_\alpha$, the linear map $\varphi_\alpha : \mathfrak{sl}(2, \mathbb{C}) \to \mathfrak{g}_\mathbb{C}$, $H \mapsto H'$, $E \mapsto E'$, $F \mapsto F'$, is an injective Lie algebra homomorphism with $\varphi_\alpha(\mathfrak{su}(2)) \subset \mathfrak{g}$. Explicitly, $\varphi_\alpha(iH) = \alpha^\vee$ and $\varphi_\alpha(E - F) = X_\alpha + \bar{X}_\alpha$.

(iii) $\dim_\mathbb{C} \mathfrak{g}_\alpha = 1$, and $\mathbb{R}\alpha \cap \Delta = \{\pm\alpha\}$.

Proof. (i) $[X, Y] \in \mathfrak{t}_\mathbb{C}$ by Proposition 8.2.2(iii). For $H \in \mathfrak{t}$, by invariance of B,

$$B(H, [X, Y]) = -B([X, H], Y) = B([H, X], Y) = i\alpha(H)B(X, Y) = B\big(H,\ iB(X, Y)H_\alpha\big),$$

and B is nondegenerate on $\mathfrak{t}_{\mathbb{C}}$.

(ii) For $0 \neq X \in \mathfrak{g}_\alpha$, $B(X,\bar{X}) = (X,X)_h > 0$; rescale X to get X_α with $B(X_\alpha,\bar{X}_\alpha) = 2/(\alpha,\alpha)$. Then by (i), using $\bar{X}_\alpha \in \mathfrak{g}_{-\alpha}$,

$$[E',F'] = -[X_\alpha,\bar{X}_\alpha] = -i\tfrac{2}{(\alpha,\alpha)}H_\alpha = H', \qquad [H',E'] = -i[\alpha^\vee,X_\alpha] = -i\cdot i\alpha(\alpha^\vee)X_\alpha = 2E',$$

and, since $[\alpha^\vee,\bar{X}_\alpha] = -i\alpha(\alpha^\vee)\bar{X}_\alpha = -2i\bar{X}_\alpha$, $[H',F'] = i[\alpha^\vee,\bar{X}_\alpha] = 2\bar{X}_\alpha = -2F'$. So φ_α is a homomorphism; it is injective because H',E',F' are linearly independent (they lie in different summands of Proposition 8.2.2(i)). Finally $\varphi_\alpha(iH) = \alpha^\vee$, $\varphi_\alpha(E-F) = X_\alpha + \bar{X}_\alpha$ and $\varphi_\alpha(i(E+F)) = i(X_\alpha - \bar{X}_\alpha)$ are fixed by conjugation, hence lie in $\mathfrak{g}$.

(iii) Let $\mathfrak{s}_\alpha := \varphi_\alpha(\mathfrak{sl}(2,\mathbb{C}))$ and

$$M := \mathbb{C}\alpha^\vee \oplus \bigoplus_{c\in\mathbb{R}\setminus\{0\},\ c\alpha\in\Delta} \mathfrak{g}_{c\alpha}.$$

M is invariant under $\mathrm{ad}(\mathfrak{s}_\alpha)$: $\mathrm{ad}\,H'$ preserves each summand; $\mathrm{ad}\,E'$ maps $\mathfrak{g}_{c\alpha}$ into $\mathfrak{g}_{(c+1)\alpha}$ for $c \neq -1$, maps $\mathfrak{g}_{-\alpha}$ into $\mathbb{C}\alpha^\vee$ by (i), and maps $\alpha^\vee$ into $\mathfrak{g}_\alpha$; similarly for F'. Since $\varphi_\alpha(\mathfrak{su}(2)) \subset \mathfrak{g}$ acts by skew-Hermitian operators, Theorem 5.4.1(ii) applies: M is a direct sum of irreducible representations $V^{(n)}$, and H' acts on $\mathfrak{g}_{c\alpha}$ by $-i\cdot ic\,\alpha(\alpha^\vee) = 2c$ and on $\mathbb{C}\alpha^\vee$ by 0. In particular $2c \in \mathbb{Z}$.

$\mathfrak{s}_\alpha \subset M$ is invariant (it is a subalgebra), with H'-eigenvalues $2,0,-2$. In a decomposition of $\mathfrak{s}_\alpha$ into irreducibles, the summand containing an eigenvector for 2 is some $V^{(n)}$ with $n \geq 2$, of dimension $\geq 3 = \dim\mathfrak{s}_\alpha$; hence $\mathfrak{s}_\alpha \cong V^{(2)}$ is irreducible. Let M' be an invariant complement, $M = \mathfrak{s}_\alpha \oplus M'$ (orthogonal complement). The zero eigenspace of H' on M is $\mathbb{C}\alpha^\vee \subset \mathfrak{s}_\alpha$, so H' has no eigenvalue 0 on M'. Since $V^{(n)}$ has the eigenvalue 0 iff n is even, M' is a sum of $V^{(n)}$ with n odd, and all H'-eigenvalues on M' are odd. Therefore $M' \cap \mathfrak{g}_{c\alpha} = 0$ for $c \in \mathbb{Z}$. Since H'-eigenspaces are compatible with the decomposition $M = \mathfrak{s}_\alpha \oplus M'$, we get $\mathfrak{g}_\alpha = \mathfrak{s}_\alpha \cap \mathfrak{g}_\alpha = \mathbb{C}X_\alpha$ and $\mathfrak{g}_{2\alpha} = \mathfrak{s}_\alpha \cap \mathfrak{g}_{2\alpha} = 0$. As α was arbitrary: *twice a root is never a root.* If $M' \neq 0$, then (odd n) H' has the eigenvalue 1 on M', so $\mathfrak{g}_{\alpha/2} \neq 0$ and $\alpha/2 \in \Delta$; but then $\alpha = 2\cdot\frac{\alpha}{2}$ is twice a root, a contradiction. Hence $M = \mathfrak{s}_\alpha$, and the only nonzero real multiples of α in Δ are $\pm\alpha$. ☐

Corollary 8.3.3: $\dim\mathfrak{g} = \dim\mathfrak{t} + |\Delta|$.

Corollary 8.3.4 (Integrality): For every weight λ of a representation of G and every $\alpha \in \Delta$, $\lambda(\alpha^\vee) = \frac{2(\lambda,\alpha)}{(\alpha,\alpha)} \in \mathbb{Z}$. In particular $\beta(\alpha^\vee) \in \mathbb{Z}$ for $\alpha,\beta \in \Delta$.

Proof. Choose an invariant inner product on V (Theorem 2.2.1); then $d\pi(\mathfrak{g})$ consists of skew-Hermitian operators, so $d\pi\circ\varphi_\alpha$ satisfies the hypothesis of Theorem 5.4.1(ii), and H acts with integer eigenvalues. On V_λ, $d\pi(\varphi_\alpha(H)) = d\pi(-i\alpha^\vee)$ acts by $-i\cdot i\lambda(\alpha^\vee) = \lambda(\alpha^\vee)$. ☐

Lemma 8.3.5: Every $\lambda \in P(T)$ is a weight of some irreducible representation of G. Consequently $\lambda(\alpha^\vee) \in \mathbb{Z}$ for all $\lambda \in P(T)$ and $\alpha \in \Delta$.

Proof. For a representation (π,V) let $P_\lambda := \int_T \overline{t^\lambda}\pi(t)\,dt$. On V_μ it acts by $\int_T t^\mu\overline{t^\lambda}\,dt = \delta_{\lambda\mu}$ (Proposition 7.1.5), so P_λ is the projection onto V_λ along the other weight spaces. Suppose λ is not a weight of any representation. Then $P_\lambda = 0$ always, so for every matrix coefficient $f = f^\pi_{u,v}$ and every $g \in G$,

$$\int_T f(gt)\overline{t^\lambda}\,dt = \langle\pi(g)P_\lambda u, v\rangle = 0.$$

By Theorem 4.2.1(i) this holds for all $f \in C(G)$. By Tietze's extension theorem there is $f \in C(G)$ with $f|_T = t^\lambda$; for $g = e$ we get $\int_T |t^\lambda|^2\,dt = 1 \neq 0$, a contradiction. The last statement now follows from Corollary 8.3.4. ☐

8.4 Reflections and the root system

Definition 8.4.1: For $\alpha \in \Delta$, the *reflection* $s_\alpha : \mathfrak{t}^* \to \mathfrak{t}^*$ is $s_\alpha(\lambda) := \lambda - \lambda(\alpha^\vee)\alpha = \lambda - \frac{2(\lambda,\alpha)}{(\alpha,\alpha)}\alpha$, the orthogonal reflection in the hyperplane $\alpha^\perp$.

Proposition 8.4.2: For $\alpha \in \Delta$ let $n_\alpha := \exp\bigl(\frac{\pi}{2}(X_\alpha + \bar{X}_\alpha)\bigr) \in G$. Then $n_\alpha \in N(T)$, and $w_\alpha := n_\alpha T \in W$ acts on $\mathfrak{t}^*$ by s_α. In particular $s_\alpha \in W$.

Proof. Let $Z := X_\alpha + \bar{X}_\alpha = \varphi_\alpha(E - F) \in \mathfrak{g}$, so $\mathrm{Ad}(n_\alpha) = e^{\frac{\pi}{2}\operatorname{ad} Z}$ (Corollary 5.3.1). For $H \in \ker\alpha \subset \mathfrak{t}$, $[Z,H] = -i\alpha(H)X_\alpha + i\alpha(H)\bar{X}_\alpha = 0$, so $\mathrm{Ad}(n_\alpha)H = H$. The subalgebra $\mathfrak{s}_\alpha$ is $\operatorname{ad} Z$-invariant, and since φ_α is a homomorphism, $\operatorname{ad} Z \circ \varphi_\alpha = \varphi_\alpha \circ \operatorname{ad}(E-F)$. Hence

$$\mathrm{Ad}(n_\alpha)\alpha^\vee = e^{\frac{\pi}{2}\operatorname{ad} Z}\varphi_\alpha(iH) = \varphi_\alpha\bigl(e^{\frac{\pi}{2}\operatorname{ad}(E-F)}iH\bigr) = \varphi_\alpha\bigl(i\,wHw^{-1}\bigr) = -\alpha^\vee,$$

where $w := e^{\frac{\pi}{2}(E-F)} = \bigl(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\bigr)$ (as $(E-F)^2 = -1$) and $e^{\frac{\pi}{2}\operatorname{ad}(E-F)} = \mathrm{Ad}(w)$ (Corollary 5.3.1 for $\mathrm{GL}(2,\mathbb{C})$). So $\mathrm{Ad}(n_\alpha)$ preserves $\mathfrak{t} = \ker\alpha \oplus \mathbb{R}\alpha^\vee$, whence $n_\alpha T n_\alpha^{-1} = \exp(\mathrm{Ad}(n_\alpha)\mathfrak{t}) = T$. On $\mathfrak{t}$, $\mathrm{Ad}(n_\alpha)$ is the reflection r fixing $\ker\alpha$ and negating $\alpha^\vee$; thus $(w_\alpha\lambda)(H) = \lambda(rH)$ equals $\lambda(H)$ on $\ker\alpha$ and $-\lambda(\alpha^\vee)$ for $H = \alpha^\vee$, which are also the values of $s_\alpha\lambda$. ☐

Corollary 8.4.3: Δ is a *reduced root system* in $E := \operatorname{span}_{\mathbb{R}} \Delta \subset \mathfrak{t}^*$, i.e., a finite subset of $E \setminus \{0\}$ such that (i) $s_\alpha\Delta = \Delta$, (ii) $\beta(\alpha^\vee) \in \mathbb{Z}$, and (iii) $\mathbb{R}\alpha \cap \Delta = \{\pm\alpha\}$ for all $\alpha, \beta \in \Delta$.

Proof. Propositions 8.4.2 and 8.2.2(iv), Corollary 8.3.4 and Proposition 8.3.2(iii). ☐

8.5 Positive roots, simple roots and dominant weights

Fix $H_0 \in \mathfrak{t}$ with $\alpha(H_0) \neq 0$ for all $\alpha \in \Delta$, and with $\mu(H_0) \neq \nu(H_0)$ for all distinct $\mu, \nu \in P(T)$ (possible since $P(T)$ is countable and each condition excludes a hyperplane). Put $\Delta^+ := \{\alpha \in \Delta \mid \alpha(H_0) > 0\}$, the *positive roots*; then $\Delta = \Delta^+ \sqcup (-\Delta^+)$, and we write $\alpha > 0$ for $\alpha \in \Delta^+$. A positive root is *simple* if it is not the sum of two positive roots.

Lemma 8.5.1: (i) If $\alpha, \beta \in \Delta$, $\beta \neq \pm\alpha$ and $(\alpha,\beta) > 0$, then $\alpha - \beta \in \Delta$.

(ii) Every positive root is a sum of simple roots (with repetitions).

(iii) If $\alpha \neq \beta$ are simple, then $(\alpha,\beta) \leq 0$.

(iv) The simple roots $\alpha_1, \dots, \alpha_r$ are linearly independent; hence they form a basis of E, and every root is $\sum_j c_j\alpha_j$ with all $c_j \in \mathbb{N}_0$ or all $c_j \in -\mathbb{N}_0$.

Proof. (i) $\alpha(\beta^\vee)\beta(\alpha^\vee) = \frac{4(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)}$ is a product of two positive integers (Corollary 8.3.4) and is < 4 by the strict Cauchy–Schwarz inequality (α, β are not proportional by Proposition 8.3.2(iii)). So one factor is 1. If $\alpha(\beta^\vee) = 1$, then $s_\beta\alpha = \alpha - \beta \in \Delta$; if $\beta(\alpha^\vee) = 1$, then $s_\alpha\beta = \beta - \alpha \in \Delta$, so $\alpha - \beta \in \Delta$. (ii) Induction on the finitely many values $\alpha(H_0) > 0$: a non-simple positive root is $\beta + \gamma$ with $\beta, \gamma > 0$ and $\beta(H_0), \gamma(H_0) < \alpha(H_0)$. (iii) Otherwise $\alpha - \beta \in \Delta$ by (i). If $\alpha - \beta > 0$, then $\alpha = \beta + (\alpha - \beta)$ is not simple; if $\beta - \alpha > 0$, then β is not simple. (iv) Suppose $\sum_j c_j\alpha_j = 0$. Let $v := \sum_{c_j > 0} c_j\alpha_j = \sum_{c_k < 0}(-c_k)\alpha_k$. Then $(v,v) = \sum_{c_j > 0,\, c_k < 0} c_j(-c_k)(\alpha_j, \alpha_k) \leq 0$ by (iii), so $v = 0$; evaluating at H_0 gives $\sum_{c_j > 0} c_j\alpha_j(H_0) = 0$, so no c_j is positive; similarly none is negative. The last statement follows from (ii) and $\Delta = \Delta^+ \sqcup -\Delta^+$. ☐

Definition 8.5.2: (i) The *fundamental Weyl chamber* is $C := \{\lambda \in \mathfrak{t}^* \mid (\lambda,\alpha) > 0\ \forall\alpha \in \Delta^+\}$. The *Weyl chambers* are the connected components of $\mathfrak{t}^*_{\mathrm{reg}} := \{\lambda \mid (\lambda,\alpha) \neq 0\ \forall\alpha \in \Delta\}$; $\lambda \in \mathfrak{t}^*_{\mathrm{reg}}$ is called *regular.*

(ii) $\lambda \in P(T)$ is *dominant* if $(\lambda,\alpha) \geq 0$ for all $\alpha > 0$, and *strictly dominant* if $(\lambda,\alpha) > 0$ for all $\alpha > 0$. P_+ denotes the set of dominant weights.

(iii) $\rho := \frac{1}{2}\sum_{\alpha>0}\alpha$ is the *Weyl vector.*

(iv) $Q_+ := \{\sum_j c_j\alpha_j \mid c_j \geq 0\}$, and $\mu \preceq \lambda$ means $\lambda - \mu \in Q_+$.

By Lemma 8.5.1(ii), λ is dominant iff $(\lambda, \alpha_j) \geq 0$ for all simple roots, and $C = \{\lambda \mid (\lambda, \alpha_j) > 0\ \forall j\}$. Each Weyl chamber is the set of λ with a fixed pattern of signs of (λ, α), $\alpha \in \Delta^+$ (such a set is convex, hence connected, and open, and different sign patterns cannot be connected within $\mathfrak{t}^*_{\mathrm{reg}}$). C is a chamber, as H_0 corresponds to an element of it. Moreover $Q_+ \cap (-Q_+) = \{0\}$ and $\mu(H_0) > 0$ for $0 \neq \mu \in Q_+$, by Lemma 8.5.1(iv).

Lemma 8.5.3: Let α_i be simple. Then s_{α_i} permutes $\Delta^+ \setminus \{\alpha_i\}$ and $s_{\alpha_i}\rho = \rho - \alpha_i$. Hence $\rho(\alpha_i^\vee) = 1$, i.e., $(\rho, \alpha_i) = \frac{1}{2}(\alpha_i, \alpha_i)$, and $\rho \in C$.

Proof. Let $\beta = \sum_j c_j\alpha_j \in \Delta^+ \setminus \{\alpha_i\}$. By Proposition 8.3.2(iii), β is not a multiple of α_i, so $c_j > 0$ for some $j \neq i$. The root $s_{\alpha_i}\beta = \beta - \beta(\alpha_i^\vee)\alpha_i$ has the same coefficient $c_j > 0$, so it is positive by Lemma 8.5.1(iv), and $\neq \alpha_i$. As $s_{\alpha_i}\alpha_i = -\alpha_i$, $s_{\alpha_i}\rho = \rho - \alpha_i$; comparing with $s_{\alpha_i}\rho = \rho - \rho(\alpha_i^\vee)\alpha_i$ gives $\rho(\alpha_i^\vee) = 1$. Finally, $(\rho, \alpha_i) > 0$ for all i means $\rho \in C$. ☐

9 The Weyl group

9.1 W as a reflection group

Let $s_i := s_{\alpha_i}$ for the simple roots $\alpha_1, \ldots, \alpha_r$. Recall that W acts on $\mathfrak{t}^*$ by orthogonal maps permuting Δ (Proposition 8.2.2(iv)); hence it permutes the Weyl chambers.

Theorem 9.1.1: Structure of the Weyl group

(i) W acts simply transitively on the set of Weyl chambers.

(ii) W is generated by $s_1, \ldots, s_r$; in particular it is generated by the reflections s_α, $\alpha \in \Delta$.

(iii) If $\lambda \in \overline{C}$ and $w \in W$, then $\lambda - w\lambda \in Q_+$. If $\lambda \in C$ and $w \neq 1$, then $\lambda - w\lambda \neq 0$, so $(w\lambda)(H_0) < \lambda(H_0)$.

(iv) Every $\lambda \in \mathfrak{t}^*$ is W-conjugate to exactly one element of $\overline{C}$. W acts freely on the regular elements of $\mathfrak{t}^*$.

Proof. Step 1. Let $W' \subset W$ be the subgroup generated by $s_1, \ldots, s_r$ (Proposition 8.4.2). For $\lambda \in \mathfrak{t}^*$ choose $w \in W'$ maximizing $(w\lambda, \rho)$ over the finite set W'. If $(w\lambda, \alpha_i) < 0$ for some i, then by Lemma 8.5.3

$$(s_i w\lambda, \rho) = (w\lambda, s_i\rho) = (w\lambda, \rho) - (w\lambda, \alpha_i) > (w\lambda, \rho),$$

a contradiction. So $w\lambda \in \overline{C}$; if λ is regular, so is $w\lambda$, and $w\lambda \in C$. Hence W' acts transitively on chambers.

Step 2: if $w \in W$ and $wC = C$, then $w = 1$. For $\mu \in C$ and $\alpha > 0$, $(\mu, w\alpha) = (w^{-1}\mu, \alpha) > 0$, so $w\Delta^+ = \Delta^+$ and $w\rho = \rho$. Write $w = nT$; then $\mathrm{Ad}(n)H_\rho = H_\rho$. Since $\alpha(H_\rho) = (\alpha, \rho) \neq 0$ for all $\alpha \in \Delta$, Proposition 8.2.2(i) shows that $\{X \in \mathfrak{g} \mid [H_\rho, X] = 0\} = \mathfrak{t}$. By Corollary 7.4.3(ii), $n \in T$, i.e., $w = 1$.

(i) Transitivity is Step 1. If $wC' = C'$ for a chamber $C' = uC$, $u \in W$, then $u^{-1}wuC = C$, so $w = 1$ by Step 2. (ii) For $w \in W$ choose $w' \in W'$ with $w'wC = C$ (Step 1); then $w'w = 1$ by Step 2, so $w \in W'$.

(iii) For $w \in W$ let $N(w) := |\{\beta \in \Delta^+ \mid w\beta < 0\}|$. If $w \neq 1$, there is a simple root with $w\alpha_i < 0$: otherwise w maps every positive root $\sum c_j\alpha_j$ to a root in Q_+, i.e., to a positive root, so $wC = C$ and $w = 1$. Since s_i permutes $\Delta^+ \setminus \{\alpha_i\}$ and $ws_i\alpha_i = -w\alpha_i > 0$, we get

$N(ws_i) = N(w) - 1$. We prove the claim by induction on $N(w)$; $N(w) = 0$ forces $w = 1$. For $w \neq 1$ choose i as above. Since $s_i\lambda = \lambda - \lambda(\alpha_i^\vee)\alpha_i$,

$$\lambda - w\lambda = (\lambda - ws_i\lambda) - \lambda(\alpha_i^\vee)\, w\alpha_i.$$

The first term is in Q_+ by induction, and $-w\alpha_i \in \Delta^+$ and $\lambda(\alpha_i^\vee) \geq 0$, so $\lambda - w\lambda \in Q_+$. If $\lambda \in C$, then $\lambda(\alpha_i^\vee) > 0$, and a sum $a + b$ with $a \in Q_+$, $0 \neq b \in Q_+$ is nonzero (since $Q_+ \cap -Q_+ = 0$). Finally, $\mu(H_0) > 0$ for $0 \neq \mu \in Q_+$.

(iv) Existence is Step 1. If $\lambda, \mu \in \overline{C}$ and $\mu = w\lambda$, then $\lambda - \mu \in Q_+$ and $\mu - \lambda = \mu - w^{-1}\mu \in Q_+$ by (iii), so $\lambda = \mu$. If λ is regular and $w\lambda = \lambda$, then w fixes the chamber containing λ, so $w = 1$ by (i). ☐

Definition 9.1.1: The *sign* of $w \in W$ is $\varepsilon(w) := \det(w|_{\mathfrak{t}^*}) \in \{\pm 1\}$. It is a homomorphism with $\varepsilon(s_\alpha) = -1$.

Lemma 9.1.2: For $w \in W$, $\rho - w\rho = \sum_{\beta>0,\ w^{-1}\beta<0} \beta$. In particular $w\rho - \rho \in P(T)$, so $\rho + P(T)$ is W-invariant.

Proof. $w\rho = \frac{1}{2}\sum_{\alpha>0} w\alpha$. As α runs through Δ^+, $w\alpha$ runs through the positive roots β with $w^{-1}\beta > 0$ and the negatives of the positive roots β with $w^{-1}\beta < 0$. Hence $w\rho = \rho - \sum_{\beta>0,\, w^{-1}\beta<0} \beta$. Roots lie in $P(T)$, and W preserves $P(T)$. ☐

9.2 Alternating sums and the Weyl denominator

For $\mu \in \mathfrak{t}^*$ let $e^\mu : \mathfrak{t} \to \mathbb{C}$, $e^\mu(H) := e^{i\mu(H)}$. For distinct $\mu_1, \ldots, \mu_m$ these functions are linearly independent: choose H with distinct values $\mu_j(H)$; then $s \mapsto e^{is\mu_j(H)}$ are linearly independent functions of $s \in \mathbb{R}$ (consider derivatives at $s = 0$ and a Vandermonde determinant). W acts on functions on $\mathfrak{t}$ by $(w \cdot f)(H) := f(\mathrm{Ad}(n)^{-1}H)$, so that $w \cdot e^\mu = e^{w\mu}$. For $\mu \in \mathfrak{t}^*$ define the *alternating sum*

$$A_\mu := \sum_{w \in W} \varepsilon(w)\, e^{w\mu}.$$

If $\mu \in P(T)$, then $e^\mu(H) = (e^H)^\mu$ is a function on T.

Lemma 9.2.1: Let $L \subset \mathfrak{t}^*$ be a W-invariant subset and $f = \sum_{\mu \in L} c_\mu e^\mu$ (finite sum) with $w \cdot f = \varepsilon(w) f$ for all $w \in W$. Then $f = \sum_{\nu \in L \cap C} c_\nu A_\nu$.

Proof. By linear independence, $w \cdot f = \varepsilon(w) f$ means $c_{w\mu} = \varepsilon(w) c_\mu$. If μ is not regular, then $(\mu, \alpha) = 0$ for some α, so $s_\alpha \mu = \mu$ and $c_\mu = -c_\mu = 0$. By Theorem 9.1.1(iv), the regular $\mu \in L$ are partitioned into orbits $\{w\nu\}_{w \in W}$ with unique $\nu \in C \cap L$ and $|W|$ distinct elements; the corresponding part of f is $c_\nu A_\nu$. ☐

Lemma 9.2.2: If $\nu, \nu' \in C$ and $\nu - \nu' \in P(T)$ and $w\nu - \nu \in P(T)$ for all w, then $|A_\nu|^2$ and $A_\nu \overline{A_{\nu'}}$ are functions on T, and $\int_T A_\nu \overline{A_{\nu'}}\, dt = |W| \delta_{\nu\nu'}$.

Proof. $A_\nu \overline{A_{\nu'}} = \sum_{w,w'} \varepsilon(w)\varepsilon(w') e^{w\nu - w'\nu'}$ with $w\nu - w'\nu' = (w\nu - \nu) + (\nu - \nu') + (\nu' - w'\nu') \in P(T)$, so each term is a character of T. By Proposition 7.1.5, the integral counts the pairs with $w\nu = w'\nu'$. If $\nu \neq \nu'$ there are none (Theorem 9.1.1(iv)); if $\nu = \nu'$, exactly the pairs $w = w'$ (free action). ☐

Proposition 9.2.3 (Weyl denominator formula): As functions on $\mathfrak{t}$,

$$A_\rho = \sum_{w \in W} \varepsilon(w)\, e^{w\rho} = e^\rho \prod_{\alpha>0} (1 - e^{-\alpha}) = \prod_{\alpha>0} (e^{\alpha/2} - e^{-\alpha/2}).$$

Consequently, for $t = e^H \in T$, $|A_\rho(H)|^2 = \prod_{\alpha>0} |1 - t^\alpha|^2 =: |\Delta(t)|^2$ depends only on t, and $\int_T |\Delta(t)|^2\, dt = |W|$.

Proof. Let $D := \prod_{\alpha>0}(e^{\alpha/2} - e^{-\alpha/2}) = e^{\rho}\prod_{\alpha>0}(1 - e^{-\alpha})$. By Lemma 8.5.3, $s_i \cdot D$ is the same product with the factor for α_i replaced by $e^{-\alpha_i/2} - e^{\alpha_i/2}$; so $s_i \cdot D = -D$. By Theorem 9.1.1(ii), $w \cdot D = \varepsilon(w)D$ for all w. Expanding, $D = \sum_S c_S e^{\rho - \sum_{\alpha\in S}\alpha}$ over subsets $S \subset \Delta^+$, with exponents in the W-invariant set $L := \rho + P(T)$ (Lemma 9.1.2); the exponent ρ occurs only for $S = \emptyset$ (as $\sum_{\alpha\in S}\alpha$ is positive on H_0 for $S \neq \emptyset$), with coefficient 1. By Lemma 9.2.1, $D = \sum_{\nu\in C\cap L} c_\nu A_\nu$, where only ν of the form $\rho - \sum_{\alpha\in S}\alpha$ occur. For such ν, $\nu(\alpha_i^\vee) = 1 - \sum_{\alpha\in S}\alpha(\alpha_i^\vee) \in \mathbb{Z}$ (Corollary 8.3.4) and > 0, hence $\geq 1 = \rho(\alpha_i^\vee)$. So $\eta := \nu - \rho$ satisfies $(\eta, \alpha_i) \geq 0$ for all i, while $-\eta = \sum_{\alpha\in S}\alpha = \sum_i c_i\alpha_i$ with $c_i \geq 0$. Then $(\eta,\eta) = -\sum_i c_i(\eta,\alpha_i) \leq 0$, so $\eta = 0$. Hence $D = A_\rho$. Since $|e^{\alpha/2} - e^{-\alpha/2}| = |1 - e^{\alpha}|$ on $\mathfrak{t}$, $|A_\rho(H)|^2 = \prod_{\alpha>0}|1 - t^{\alpha}|^2$. Finally apply Lemma 9.2.2 with $\nu = \nu' = \rho$. □

9.3 Regular elements

Definition 9.3.1: $t \in T$ is *regular* if $t^\alpha \neq 1$ for all $\alpha \in \Delta$. Let T^{reg} be the set of regular elements of T and $G^{\mathrm{reg}} := \bigcup_g gT^{\mathrm{reg}}g^{-1}$.

Lemma 9.3.2: (i) $\dim Z_G(t) = \dim T + |\{\alpha \in \Delta \mid t^\alpha = 1\}|$ for $t \in T$. Hence $t \in T$ is regular iff $\dim Z_G(t) = \dim T$, and then $Z_G(t)_0 = T$. In particular, $G^{\mathrm{reg}} \cap T = T^{\mathrm{reg}}$.

(ii) $T \setminus T^{\mathrm{reg}}$ is a null set in T, and $G \setminus G^{\mathrm{reg}}$ is a null set in G.

(iii) If $s, t \in T^{\mathrm{reg}}$ and $nsn^{-1} = t$ with $n \in G$, then $n \in N(T)$.

Proof. (i) $\mathrm{Lie}(Z_G(t)) = \{X \in \mathfrak{g} \mid \mathrm{Ad}(t)X = X\}$, whose complexification is $\mathfrak{t}_{\mathbb{C}} \oplus \bigoplus_{t^\alpha=1}\mathfrak{g}_\alpha$ by Proposition 8.2.2, and $\dim \mathfrak{g}_\alpha = 1$. If t is regular, $T \subset Z_G(t)_0$ have the same dimension, so T is open in the connected group $Z_G(t)_0$, hence equal to it. The dimension of $Z_G(t)$ is invariant under conjugation, which gives the last claim. (ii) $T \setminus T^{\mathrm{reg}} = \bigcup_{\alpha>0} K_\alpha$ with $K_\alpha := \{t \mid t^\alpha = 1\}$, a closed subgroup with Lie algebra $\ker\alpha$, hence a finite union of submanifolds of dimension $\dim T - 1$; such sets are null for Haar measure of T (which is a smooth density in charts). By the torus theorem, $G \setminus G^{\mathrm{reg}}$ is the image of the compact manifold-pieces $G/T \times K_\alpha$, of dimension $\dim G - 1$, under the smooth map $q(gT, t) := gtg^{-1}$; so it is null. (iii) $nZ_G(s)_0n^{-1} = Z_G(t)_0$, i.e., $nTn^{-1} = T$ by (i). □

9.4 The Weyl integration formula

We use the manifold G/T (see the prerequisites) and the smooth map $q : G/T \times T \to G$, $q(gT, t) := gtg^{-1}$, which is well defined as T is abelian. Let $\mathfrak{p} := \mathfrak{t}^\perp \subset \mathfrak{g}$; then $\mathrm{Ad}(t)\mathfrak{p} = \mathfrak{p}$ for $t \in T$, and $\mathfrak{p}_{\mathbb{C}} = \bigoplus_{\alpha\in\Delta}\mathfrak{g}_\alpha$. The differential of $G \to G/T$ at e restricts to an isomorphism $\mathfrak{p} \to T_{eT}(G/T)$ (its kernel is $\mathfrak{t}$).

Lemma 9.4.1 (Invariant measures): Fix Lebesgue measures on $\mathfrak{p}$, $\mathfrak{t}$ and the product measure on $\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{t}$.

(i) There are unique smooth positive measures (densities) ω_G on G and ω_T on T, invariant under left translations, which at e are the given Lebesgue measures on $\mathfrak{g}$ and $\mathfrak{t}$. They are positive multiples of the Haar measures.

(ii) There is a unique smooth positive density $\omega_{G/T}$ on G/T, invariant under the action of G, which at eT is the given Lebesgue measure on $\mathfrak{p} \cong T_{eT}(G/T)$.

(iii) There is $c > 0$ with $\int_{G/T} h\, d\omega_{G/T} = c\int_G h(gT)\, dg$ for all $h \in C(G/T)$.

Proof. (i) Transport the density at e by left translations; the resulting measure is left invariant, so its normalization is Haar measure by uniqueness (Theorem 2.1.1). (ii) For $g \in G$ let $\ell_g(xT) :=$

gxT. Define $\omega_{G/T}$ at gT as the image of the density at eT under $d\ell_g$. This is independent of the representative: for $s \in T$, $d\ell_s$ at eT corresponds to $\mathrm{Ad}(s)|_{\mathfrak{p}}$ (differentiate $se^{uX}T = e^{u\,\mathrm{Ad}(s)X}sT = e^{u\,\mathrm{Ad}(s)X}T$), which has $|\det| = 1$ as $\mathrm{Ad}(s)$ is orthogonal. Smoothness follows from the existence of smooth local sections of $G \to G/T$, and invariance is clear. (iii) For $f \in C(G)$, $g \mapsto \int_T f(gs)\,ds$ is continuous and right T-invariant, so it defines $\tilde{f} \in C(G/T)$. The functional $f \mapsto \int_{G/T} \tilde{f}\,d\omega_{G/T}$ is positive, left invariant and nonzero; by the uniqueness part of Theorem 2.1.1 it is $c\int_G f\,dg$ with $c = \omega_{G/T}(G/T)$. For $f(g) := h(gT)$, $\tilde{f} = h$. $\square$

Lemma 9.4.2 (Jacobian of q): $q^*\omega_G = J(t)\,\omega_{G/T} \otimes \omega_T$ with

$$J(t) = \left|\det\!\big((\mathrm{Ad}(t^{-1}) - 1)|_{\mathfrak{p}}\big)\right| = \prod_{\alpha\in\Delta}(t^{-\alpha} - 1) = |\Delta(t)|^2.$$

In particular q is a local diffeomorphism near every point of $G/T \times T^{\mathrm{reg}}$.

Proof. Fix (g_0T, t_0). The map $\psi(X,Y) := (g_0e^XT, t_0e^Y)$, $(X,Y) \in \mathfrak{p} \times \mathfrak{t}$ near 0, is a local diffeomorphism whose differential at 0 maps the Lebesgue measure on $\mathfrak{p} \times \mathfrak{t}$ to $\omega_{G/T} \otimes \omega_T$ (by construction). The map $Z \mapsto q(g_0T, t_0)e^Z$, $Z \in \mathfrak{g}$ near 0, has the analogous property for ω_G. Now

$$\begin{aligned} q(g_0T, t_0)^{-1}\,q(\psi(sX, sY)) &= g_0t_0^{-1}e^{sX}t_0e^{sY}e^{-sX}g_0^{-1} \\ &= \exp\Big(s\,\mathrm{Ad}(g_0)\big((\mathrm{Ad}(t_0^{-1}) - 1)X + Y\big) + O(s^2)\Big). \end{aligned}$$

So in these coordinates the differential of q is $(X,Y) \mapsto \mathrm{Ad}(g_0)\big((\mathrm{Ad}(t_0^{-1}) - 1)X + Y\big)$. Its absolute determinant is $|\det \mathrm{Ad}(g_0)| \cdot |\det((\mathrm{Ad}(t_0^{-1}) - 1)|_{\mathfrak{p}})| = J(t_0)$, as $\mathrm{Ad}(g_0)$ is orthogonal and $(\mathrm{Ad}(t_0^{-1}) - 1)$ maps $\mathfrak{p}$ to $\mathfrak{p}$. On $\mathfrak{p}_{\mathbb{C}} = \bigoplus_\alpha \mathfrak{g}_\alpha$, $\mathrm{Ad}(t_0^{-1})$ acts on $\mathfrak{g}_\alpha$ by $t_0^{-\alpha}$, so $\det((\mathrm{Ad}(t_0^{-1}) - 1)|_{\mathfrak{p}}) = \prod_{\alpha\in\Delta}(t_0^{-\alpha} - 1) = \prod_{\alpha>0}(\overline{t_0^\alpha} - 1)(t_0^\alpha - 1) = \prod_{\alpha>0}|1 - t_0^\alpha|^2 \geq 0$, which equals $|\Delta(t_0)|^2$ by Proposition 9.2.3. It is nonzero iff t_0 is regular, and then q is a local diffeomorphism by the inverse function theorem. $\square$

Lemma 9.4.3: Let $M := G/T \times T^{\mathrm{reg}}$ and $N := G^{\mathrm{reg}}$. Then $q^{-1}(N) = M$, every fibre $q^{-1}(y)$, $y \in N$, has exactly $|W|$ points, and every $y \in N$ has an open neighborhood V such that $q^{-1}(V)$ is a disjoint union of $|W|$ open sets, each mapped diffeomorphically onto V by q.

Proof. If $q(gT, t) \in N$, then t is conjugate to an element of T^{reg}, hence regular by Lemma 9.3.2(i); so $q^{-1}(N) = M$. Fix $(g_1T, t_1) \in q^{-1}(y)$. If $q(gT, t) = y$, then $n := g_1^{-1}g$ satisfies $ntn^{-1} = t_1$, so $n \in N(T)$ by Lemma 9.3.2(iii), and $(gT, t) = (g_1nT, n^{-1}t_1n)$. Conversely these points lie in the fibre, and they are distinct for distinct $nT \in W$. So $|q^{-1}(y)| = |W|$.

Let $q^{-1}(y) = \{x_1, \ldots, x_k\}$. By Lemma 9.4.2 choose disjoint open $U_j \ni x_j$ mapped diffeomorphically onto open sets $q(U_j)$. We claim that there is an open $V \ni y$ with $V \subset \bigcap_j q(U_j)$ and $q^{-1}(V) \subset \bigcup_j U_j$. Otherwise there are $z_m \notin \bigcup_j U_j$ with $q(z_m) \to y$; by compactness of $G/T \times T$ a subsequence converges to some z with $q(z) = y$, i.e., $z = x_j$ for some j, contradicting $z_m \notin U_j$ with U_j open. Then the sets $U_j \cap q^{-1}(V)$ have the required properties. $\square$

Theorem 9.4.1: Weyl integration formula

Let G be compact connected with maximal torus T, and let dg, dt be the normalized Haar measures. For every $f \in C(G)$, and more generally for every Borel function $f \geq 0$ and every $f \in L^1(G)$,

$$\int_G f(g)\,dg = \frac{1}{|W|}\int_G\int_T f(gtg^{-1})\,|\Delta(t)|^2\,dt\,dg.$$

In particular, for a class function f (continuous, or Borel and ≥ 0, or integrable),

$$\int_G f(g)\,dg = \frac{1}{|W|}\int_T f(t)\,|\Delta(t)|^2\,dt.$$

Proof. Let $F \in C(G)$ with compact support in $N = G^{\mathrm{reg}}$. If the support lies in a set V as in Lemma 9.4.3, then by the change of variables formula on each of the $|W|$ sheets and Lemma 9.4.2,

$$\int_M (F\circ q)\,J\,d(\omega_{G/T}\otimes\omega_T) = |W|\int_N F\,d\omega_G.$$

By a partition of unity this holds for all $F \in C_c(N)$. For $F \in C(G)$ choose $\varphi_k \in C_c(N)$ with $0 \leq \varphi_k \uparrow 1_N$ pointwise (possible as N is open in the compact metric space G), apply the formula to $F\varphi_k$ and use dominated convergence; since $G \setminus N$ and $(G/T)\times(T\setminus T^{\mathrm{reg}})$ are null sets (Lemma 9.3.2(ii)), we obtain

$$\int_{G/T}\int_T F(gtg^{-1})J(t)\,d\omega_T(t)\,d\omega_{G/T}(gT) = |W|\int_G F\,d\omega_G.$$

The inner integral is a continuous function of gT (since $(gs)t(gs)^{-1} = gtg^{-1}$ for $s \in T$). By Lemma 9.4.1 and $J = |\Delta|^2$, there is a constant $\kappa > 0$ with

$$\int_G F(g)\,dg = \kappa\int_G\int_T F(gtg^{-1})|\Delta(t)|^2\,dt\,dg \qquad (F \in C(G)).$$

Taking $F = 1$ and using $\int_T |\Delta|^2\,dt = |W|$ (Proposition 9.2.3) gives $\kappa = 1/|W|$.

The right-hand side, as a functional of F, is integration against the image of the finite Borel measure $\frac{1}{|W|}|\Delta(t)|^2\,dt\,dg$ on $G\times T$ under $(g,t)\mapsto gtg^{-1}$. This is a finite Borel measure on the compact metrizable space G, hence regular, and it agrees with Haar measure on $C(G)$; by the uniqueness in the Riesz–Markov theorem the two measures coincide. This gives the formula for Borel $f \geq 0$ and integrable f. For class functions, $f(gtg^{-1}) = f(t)$. ☐

Example 9.4.4: For SU(2): $W \cong \mathbb{Z}/2$, $\Delta^+ = \{\alpha\}$ with $t_\theta^\alpha = e^{2i\theta}$, $|\Delta(t_\theta)|^2 = |1-e^{2i\theta}|^2 = 4\sin^2\theta$, and Theorem 9.4.1 reproduces Lemma 3.3.2: for a class function f,

$$\frac{1}{2}\cdot\frac{1}{2\pi}\int_0^{2\pi} f(t_\theta)\,4\sin^2\theta\,d\theta = \frac{2}{\pi}\int_0^\pi f(t_\theta)\sin^2\theta\,d\theta,$$

where we substituted $\theta\mapsto 2\pi-\theta$ on $[\pi,2\pi]$ and used $f(t_{-\theta}) = f(t_\theta)$.

Example 9.4.5 (Random unitary matrices): For $G = \mathrm{U}(n)$, $W = S_n$ (Example 7.2.7) and $|\Delta|^2 = \prod_{j<k}|e^{i\theta_j}-e^{i\theta_k}|^2$. So the eigenvalue angles of a Haar-random unitary matrix have joint density $\frac{1}{n!(2\pi)^n}\prod_{j<k}|e^{i\theta_j}-e^{i\theta_k}|^2$ on $[0,2\pi)^n$ (circular unitary ensemble).

10 The Weyl character formula

Definition 10.0.1: The *highest weight* of a representation π of G is the weight λ of π with maximal value $\lambda(H_0)$ (unique, since $\mu\mapsto\mu(H_0)$ is injective on $P(T)$).

For a representation π we regard $\chi_\pi\circ\exp$ as a function on $\mathfrak{t}$; by Proposition 8.1.2(ii), $\chi_\pi\circ\exp = \sum_\mu m_\mu e^\mu$, and it is W-invariant by Corollary 7.3.2(iv). Recall that $A_\rho(H)\neq 0$ iff e^H is regular (Proposition 9.2.3).

Theorem 10.0.1: Weyl character formula and theorem of the highest weight

Let G be a compact connected Lie group with maximal torus T and positive roots Δ^+.

(i) The highest weight λ of an irreducible representation π is dominant and has multiplicity 1, and for all $H \in \mathfrak{t}$

$$\chi_\pi(e^H)\, A_\rho(H) = A_{\lambda+\rho}(H), \quad \text{i.e.} \quad \chi_\pi(e^H) = \frac{\sum_{w\in W} \varepsilon(w) e^{i(w(\lambda+\rho))(H)}}{\sum_{w\in W} \varepsilon(w) e^{i(w\rho)(H)}}$$

whenever $e^H \in T^{\mathrm{reg}}$.

(ii) Two irreducible representations are equivalent if and only if they have the same highest weight.

(iii) For every $\lambda \in P_+$ there is an irreducible representation with highest weight λ.

Hence $\lambda \mapsto \pi_\lambda$ is a bijection from P_+ onto $\widehat{G}$.

Proof. Let $L := \rho + P(T)$, a W-invariant set (Lemma 9.1.2).

(i) The function $f := (\chi_\pi \circ \exp) \cdot A_\rho = \sum_{\mu,w} m_\mu \varepsilon(w) e^{\mu + w\rho}$ has exponents in L and integer coefficients, and is anti-invariant. By Lemma 9.2.1, $f = \sum_{\nu \in C\cap L} c_\nu A_\nu$ with $c_\nu \in \mathbb{Z}$. Now $|f(H)|^2 = |\chi_\pi(t)|^2 |\Delta(t)|^2$ for $t = e^H$, so by Theorems 3.2.1 and 9.4.1 and Lemma 9.2.2,

$$1 = \int_G |\chi_\pi|^2 \, dg = \frac{1}{|W|} \int_T |f|^2 \, dt = \sum_\nu c_\nu^2.$$

Hence $f = \pm A_\nu$ for a single $\nu \in C \cap L$. Compare the exponents with maximal value at H_0. In $\pm A_\nu$ it is ν, uniquely, with coefficient ± 1 (Theorem 9.1.1(iii)). In f, $(\mu + w\rho)(H_0) \le \lambda(H_0) + \rho(H_0)$, with equality only if $\mu(H_0) = \lambda(H_0)$ and $(w\rho)(H_0) = \rho(H_0)$, i.e., $\mu = \lambda$ and $w = 1$ (Theorem 9.1.1(iii), $\rho \in C$). So the top exponent of f is $\lambda + \rho$ with coefficient $m_\lambda \ge 1$. Therefore $\nu = \lambda + \rho$, the sign is $+$, and $m_\lambda = 1$. As $\lambda + \rho \in C$, $(\lambda+\rho)(\alpha_i^\vee) \ge 1$ is an integer (Lemma 8.3.5) and $\rho(\alpha_i^\vee) = 1$, so $\lambda(\alpha_i^\vee) \ge 0$ for all i: λ is dominant.

(ii) By (i), the highest weight determines χ_π on T^{reg}, which is dense in T (its complement is a finite union of closed subgroups of lower dimension, Lemma 9.3.2), hence on T by continuity, hence on G (Corollary 7.3.2(iv)), hence π (Corollary 3.2.3(ii)).

(iii) Suppose $\lambda \in P_+$ is not the highest weight of any irreducible representation. For $L' \in \Lambda$ and $\nu \in L$, $A_\nu(H + L') = e^{i\nu(L')} A_\nu(H)$, since $w\nu - \nu \in P(T)$. Hence $A_{\lambda+\rho}/A_\rho$ is Λ-periodic on $\{H \mid e^H \in T^{\mathrm{reg}}\}$ (the factors $e^{i(\lambda+\rho)(L')}$ and $e^{i\rho(L')}$ differ by $e^{i\lambda(L')} = 1$), and W-invariant (quotient of two anti-invariant functions). Define $f : G \to \mathbb{C}$ by $f(gtg^{-1}) := A_{\lambda+\rho}(H)/A_\rho(H)$ for $t = e^H \in T^{\mathrm{reg}}$, and $f := 0$ on $G \setminus G^{\mathrm{reg}}$. This is well defined by Corollary 7.3.2(iii), it is a class function, and it is continuous on the open set G^{reg} since locally $f = \phi \circ \mathrm{pr}_2 \circ (q|_{U_j})^{-1}$ with ϕ continuous (Lemma 9.4.3). By Theorem 9.4.1 and Lemma 9.2.2 (note $\lambda + \rho \in C \cap L$),

$$\|f\|_2^2 = \frac{1}{|W|} \int_T |A_{\lambda+\rho}|^2 \, dt = 1.$$

If π is irreducible with highest weight μ, then $\mu \ne \lambda$ by assumption, and by (i), Theorem 9.4.1 (for the integrable class function $f\overline{\chi_\pi}$) and Lemma 9.2.2,

$$\langle f, \chi_\pi \rangle_{L^2} = \frac{1}{|W|} \int_T A_{\lambda+\rho} \overline{A_{\mu+\rho}} \, dt = 0.$$

This contradicts Theorem 4.2.1(iv). □

Remark 10.0.2: Part (iii) is Weyl's analytic proof via Peter–Weyl, as in Simon's book. Algebraic (Verma modules) and geometric (Borel–Weil) constructions of π_λ also exist; we do not need them.

Proposition 10.0.3: Let π be irreducible with highest weight λ. Every weight μ of π satisfies $\mu \preceq \lambda$.

Proof. For $\beta \in \Delta^+$ fix $0 \neq X_\beta \in \mathfrak{g}_\beta$ and $Y_\beta := \bar{X}_\beta \in \mathfrak{g}_{-\beta}$. By Proposition 8.2.2(iii), $d\pi(\mathfrak{g}_\gamma)V_\mu \subset V_{\mu+\gamma}$. Let $0 \neq v \in V_\lambda$. Then $d\pi(X_\beta)v = 0$ for $\beta > 0$, since $(\lambda+\beta)(H_0) > \lambda(H_0)$. Let U be the span of all vectors $d\pi(Y_{\beta_1})\cdots d\pi(Y_{\beta_k})v$, $k \geq 0$, $\beta_j > 0$ ("words of length k"). U is invariant under $\mathfrak{t}_\mathbb{C}$ (it is spanned by weight vectors) and under all $d\pi(Y_\beta)$. We show by induction on k that $d\pi(X_\beta)u \in U$ for all $\beta > 0$ and all words u of length k. For $k = 0$ this was shown. For $u = d\pi(Y_{\beta_1})u'$ with u' of length $k-1$,

$$d\pi(X_\beta)u = d\pi(Y_{\beta_1})d\pi(X_\beta)u' + d\pi([X_\beta, Y_{\beta_1}])u'.$$

The first term lies in U by induction. Since $\dim \mathfrak{g}_\gamma = 1$, $[X_\beta, Y_{\beta_1}] \in \mathfrak{g}_{\beta-\beta_1}$ is a multiple of $X_{\beta-\beta_1}$ if $\beta - \beta_1 > 0$ (induction applies), of $Y_{\beta_1-\beta}$ if $\beta - \beta_1 < 0$ (in U by definition), lies in $\mathfrak{t}_\mathbb{C}$ if $\beta = \beta_1$, and is 0 otherwise. So U is invariant under $\mathfrak{g}_\mathbb{C}$, hence under G (Proposition 5.3.3), and $U = V$. The weights of the spanning vectors are $\lambda - \beta_1 - \cdots - \beta_k \preceq \lambda$. ☐

Theorem 10.0.2: Weyl dimension formula

The irreducible representation with highest weight λ has dimension $d_\lambda = \displaystyle\prod_{\alpha>0} \frac{(\lambda+\rho,\alpha)}{(\rho,\alpha)}$.

Proof. Recall $(w\mu)(H_\nu) = (w\mu,\nu) = (\mu, w^{-1}\nu)$ (W acts orthogonally) and $\varepsilon(w^{-1}) = \varepsilon(w)$. Hence, using Proposition 9.2.3 and $e^{\alpha/2}(H) - e^{-\alpha/2}(H) = 2i\sin(\alpha(H)/2)$,

$$A_{\lambda+\rho}(sH_\rho) = \sum_w \varepsilon(w)e^{is(\lambda+\rho,w^{-1}\rho)} = A_\rho(sH_{\lambda+\rho}) = \prod_{\alpha>0} 2i\sin\bigl(\tfrac{s}{2}(\alpha,\lambda+\rho)\bigr),$$

and likewise $A_\rho(sH_\rho) = \prod_{\alpha>0} 2i\sin(\frac{s}{2}(\alpha,\rho))$, which is nonzero for small $s \neq 0$ as $(\alpha,\rho) > 0$. By Theorem 10.0.1(i) and continuity of χ_λ,

$$d_\lambda = \chi_\lambda(e) = \lim_{s\to 0} \frac{A_{\lambda+\rho}(sH_\rho)}{A_\rho(sH_\rho)} = \prod_{\alpha>0} \frac{(\alpha,\lambda+\rho)}{(\alpha,\rho)}.$$

☐

Example 10.0.4 (SU(2) revisited): For SU(2), $\Delta^+ = \{\alpha\}$ with $\alpha = 2\varepsilon$, $\rho = \varepsilon$, $P(T) = \mathbb{Z}\varepsilon$, $P_+ = \mathbb{N}_0\varepsilon$, $W = \{1, s_\alpha\}$ with $s_\alpha\varepsilon = -\varepsilon$. With $x = e^{i\theta}$, the character formula reads $\chi_{n\varepsilon}(t_\theta) = \frac{x^{n+1}-x^{-n-1}}{x-x^{-1}}$ and $d_{n\varepsilon} = n+1$, in accordance with Theorem 3.3.1.

11 Representations of the classical groups

11.1 Root data of the classical groups

Lemma 11.1.1: $\mathrm{Sp}(n)$ is connected, and every element of $\mathrm{Sp}(n)$ is conjugate in $\mathrm{Sp}(n)$ to an element of the torus T of Example 7.2.4(iv).

Proof. Let $\beta(x,y) := x^TJy$, an antisymmetric bilinear form on $\mathbb{C}^{2n}$, and $j(x) := J\bar{x}$, an antilinear map. Since J is real with $J^T = -J = J^{-1}$, we have $\langle x, j(y)\rangle = (J\bar{y})^*x = y^TJ^Tx = -\beta(y,x) = \beta(x,y)$, $\langle j(x), j(y)\rangle = \langle y, x\rangle$ and $j^2 = -1$. In particular $\langle j(x), x\rangle = \overline{\beta(x,x)} = 0$.

Let $g \in \mathrm{Sp}(n)$. From $g^TJg = J$ and $(g^T)^{-1} = \bar{g}$ (unitarity) we get $Jg = \bar{g}J$, hence $gJ = J\bar{g}$ and $g\,j(x) = j(gx)$. Let v be a unit eigenvector, $gv = \lambda v$; then $g\,j(v) = j(\lambda v) = \bar{\lambda}\,j(v)$, and $v \perp j(v)$. The subspace $E := \mathrm{span}\{v, j(v)\}$ is invariant under g and j, and so is $E^\perp$ (for g by unitarity; for j because for $x \in E^\perp$, $\langle j(x), j(v)\rangle = \langle v, x\rangle = 0$ and $\langle j(x), v\rangle = -\langle j(x), j(j(v))\rangle = -\langle j(v), x\rangle = 0$). By induction we obtain an orthonormal basis $v_1, \dots, v_n, j(v_1), \dots, j(v_n)$ of eigenvectors of g, with $gv_a = \lambda_a v_a$. Let k be the unitary matrix with columns $v_1, \dots, v_n, -j(v_1), \dots, -j(v_n)$. Its Gram matrix for β is J: $\beta(v_a, v_b) = \langle v_a, j(v_b)\rangle = 0$, $\beta(j(v_a), j(v_b)) = \langle j(v_a), j^2(v_b)\rangle = -\langle j(v_a), v_b\rangle = 0$, and $\beta(v_a, -j(v_b)) = -\langle v_a, j^2(v_b)\rangle = \langle v_a, v_b\rangle = \delta_{ab}$. Hence $k^T J k = J$, $k \in \mathrm{Sp}(n)$, and $k^{-1}gk = \mathrm{diag}(\lambda_1, \dots, \lambda_n, \bar\lambda_1, \dots, \bar\lambda_n) \in T$. As T is connected, $\mathrm{Sp}(n) = \bigcup_k kTk^{-1}$ is connected. ☐

We equip the Lie algebras with the invariant inner products $\langle X, Y\rangle = -\operatorname{tr}(XY)$ for $\mathrm{U}(n)$ and $-\frac{1}{2}\operatorname{tr}(XY)$ for $\mathrm{SO}(m)$ and $\mathrm{Sp}(n)$. With the maximal tori of Example 7.2.4, $\mathfrak{t} \cong \mathbb{R}^n$ via the angles $(\theta_1, \dots, \theta_n)$, and the functionals $e_j(\theta) := \theta_j$ form an orthonormal basis of $\mathfrak{t}^*$.

Theorem 11.1.1: Root data

For the classical groups and their maximal tori from Example 7.2.4, $P(T) = \mathbb{Z}^n$ (with respect to $e_1, \dots, e_n$), and the roots, positive roots (for a suitable H_0), simple roots, Weyl groups and Weyl vectors are as follows:

G	Δ^+	simple roots	W, $\lvert W\rvert$	ρ
$\mathrm{U}(n)$	$e_j - e_k$, $j < k$	$e_j - e_{j+1}$	S_n, $n!$	$\sum_j \frac{n+1-2j}{2} e_j$
$\mathrm{SO}(2n+1)$	$e_j \pm e_k$ $(j<k)$, e_j	$e_j - e_{j+1}$, e_n	$S_n \ltimes (\mathbb{Z}/2)^n$, $2^n n!$	$\sum_j (n - j + \frac{1}{2}) e_j$
$\mathrm{Sp}(n)$	$e_j \pm e_k$ $(j<k)$, $2e_j$	$e_j - e_{j+1}$, $2e_n$	$S_n \ltimes (\mathbb{Z}/2)^n$, $2^n n!$	$\sum_j (n - j + 1) e_j$
$\mathrm{SO}(2n)$	$e_j \pm e_k$ $(j<k)$	$e_j - e_{j+1}$, $e_{n-1} + e_n$	$S_n \ltimes (\mathbb{Z}/2)^{n-1}$, $2^{n-1} n!$	$\sum_j (n - j) e_j$

Here $\Delta = \Delta^+ \cup -\Delta^+$, S_n permutes the e_j, $(\mathbb{Z}/2)^n$ acts by sign changes $e_j \mapsto -e_j$, and for $\mathrm{SO}(2n)$ only even numbers of sign changes occur ($n \geq 2$ for $\mathrm{SO}(2n)$). Consequently, P_+ consists of the $\lambda \in \mathbb{Z}^n$ with

$$\begin{array}{ll} \mathrm{U}(n): & \lambda_1 \geq \dots \geq \lambda_n; \\ \mathrm{SO}(2n+1),\ \mathrm{Sp}(n): & \lambda_1 \geq \dots \geq \lambda_n \geq 0; \\ \mathrm{SO}(2n): & \lambda_1 \geq \dots \geq \lambda_{n-1} \geq |\lambda_n|. \end{array}$$

Proof. Weight lattice. In all cases $e^H = 1$ for $H \leftrightarrow \theta$ iff all $\theta_j \in 2\pi\mathbb{Z}$, so $\Lambda = 2\pi\mathbb{Z}^n$ and $P(T) = \mathbb{Z}^n$.

Roots. $\mathrm{U}(n)$: Example 8.2.3. $\mathrm{SO}(m)$: on $\mathbb{C}^m$, the block $R(\theta_j)$ has eigenvectors $e_{2j-1} \mp ie_{2j}$ with eigenvalues $e^{\pm i\theta_j}$, so the weights of $\mathbb{C}^m$ are $\pm e_j$, together with 0 if m is odd, each with multiplicity one. The map $\Lambda^2\mathbb{C}^m \to \mathfrak{so}(m)_\mathbb{C} = \{X \in M_m(\mathbb{C}) \mid X^T = -X\}$, $u \wedge v \mapsto uv^T - vu^T$, is a linear isomorphism, and it is $\mathrm{SO}(m)$-equivariant since $g(uv^T - vu^T)g^{-1} = (gu)(gv)^T - (gv)(gu)^T$ for $g^{-1} = g^T$. The weights of $\Lambda^2\mathbb{C}^m$ are the sums of two weights of $\mathbb{C}^m$ belonging to distinct basis vectors. The nonzero ones are $\pm e_j \pm e_k$ $(j < k)$, and for odd m also $\pm e_j$. $\mathrm{Sp}(n)$: the weights of $\mathbb{C}^{2n}$ are $\pm e_j$. Let $\mathfrak{sp}(2n, \mathbb{C}) := \{X \in M_{2n}(\mathbb{C}) \mid X^T J + JX = 0\}$. By the block computation in Example 5.2.3, its complex dimension is $n^2 + n(n+1)$, the real dimension of $\mathfrak{sp}(n) \subset \mathfrak{sp}(2n, \mathbb{C})$; as $\mathfrak{sp}(n) \cap i\mathfrak{sp}(n) = 0$ (skew-Hermitian versus Hermitian), $\mathfrak{sp}(2n, \mathbb{C}) = \mathfrak{sp}(n)_\mathbb{C}$. The map $X \mapsto JX$ is a linear isomorphism of $\mathfrak{sp}(2n, \mathbb{C})$ onto the symmetric matrices, and $J(tXt^{-1}) = (t^{-1})^T(JX)t^{-1}$ for $t \in \mathrm{Sp}(n)$ (using $Jt = (t^T)^{-1}J$). So $\mathfrak{sp}(n)_\mathbb{C}$ is the space of symmetric bilinear forms on $\mathbb{C}^{2n}$, i.e., Sym^2 of the dual of $\mathbb{C}^{2n}$, whose weights are $\pm e_j \pm e_k$ ($j < k$, from distinct basis vectors), $\pm 2e_j$ and 0.

Positive and simple roots. Take $H_0 \leftrightarrow (n, n-1, \dots, 1) +$ a small generic perturbation; then

Δ^+ is as stated. Let S be the proposed set of simple roots. It is linearly independent, has $\dim \operatorname{span} \Delta$ elements, and every positive root is a nonnegative integer combination of S:

$$e_j - e_k = \sum_{l=j}^{k-1} (e_l - e_{l+1}), \quad e_j = (e_j - e_n) + e_n, \quad 2e_j = 2(e_j - e_n) + 2e_n,$$

$$e_j + e_k = (e_j - e_n) + (e_k - e_n) + 2e_n$$

in types B_n ($\mathrm{SO}(2n+1)$, write $2e_n = e_n + e_n$) and C_n ($\mathrm{Sp}(n)$), and $e_j + e_k = (e_j - e_{n-1}) + (e_k - e_n) + (e_{n-1} + e_n)$ for $j < k \le n-1$, $e_j + e_n = (e_j - e_{n-1}) + (e_{n-1} + e_n)$ in type D_n ($\mathrm{SO}(2n)$). Then S is the set of simple roots: each $\alpha \in S$ is simple, since writing $\alpha = \beta + \gamma$ with $\beta, \gamma > 0$ and expanding β, γ in S with nonnegative coefficients would, by linear independence, force $\beta = 0$ or $\gamma = 0$; and the simple roots form a basis of $\operatorname{span} \Delta$ (Lemma 8.5.1(iv)), so they cannot be more than $|S|$.

Weyl groups. By Theorem 9.1.1(ii), W is generated by the reflections s_α. Here $s_{e_j - e_k}$ swaps e_j and e_k; s_{e_j} and s_{2e_j} change the sign of e_j; $s_{e_j + e_k}$ maps $e_j \mapsto -e_k$, $e_k \mapsto -e_j$. In types B_n, C_n these generate all signed permutations. In type D_n, every generator changes an even number of signs, and $s_{e_j - e_k} s_{e_j + e_k}$ changes the signs of e_j and e_k; together with the transpositions these generate all signed permutations with an even number of sign changes.

Weyl vectors and dominant weights. ρ is obtained by summing: e.g., for B_n, $\sum_{j<k} \big((e_j - e_k) + (e_j + e_k) \big) + \sum_j e_j = \sum_j (2(n-j)+1) e_j$. Dominance means $(\lambda, \alpha) \ge 0$ for the simple roots: $\lambda_j \ge \lambda_{j+1}$, and additionally $\lambda_n \ge 0$ (types B, C), resp. $\lambda_{n-1} + \lambda_n \ge 0$ (type D), which together with $\lambda_{n-1} \ge \lambda_n$ means $\lambda_{n-1} \ge |\lambda_n|$. ☐

11.2 The unitary groups

For $\mathrm{U}(n)$ let $\delta := (n-1, n-2, \dots, 0) = \rho + \frac{n-1}{2}(1, \dots, 1)$. Since $(1, \dots, 1)$ is W-invariant, $A_{\mu+\delta} = e^{\frac{n-1}{2}(1,\dots,1)} A_{\mu+\rho}$ for every μ, so $A_{\lambda+\rho}/A_\rho = A_{\lambda+\delta}/A_\delta$. With $x_j := e^{i\theta_j}$ and $\varepsilon(w) = \operatorname{sgn}(w)$ for $w \in S_n$,

$$A_{\lambda+\delta} = \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \prod_j x_{\sigma(j)}^{\lambda_j + n - j} = \det\big(x_k^{\lambda_j + n - j}\big)_{j,k}, \qquad A_\delta = \det\big(x_k^{n-j}\big)_{j,k} = \prod_{j<k} (x_j - x_k)$$

(Vandermonde).

Theorem 11.2.1: Irreducible representations of U(n)

The irreducible representations π_λ of $\mathrm{U}(n)$ are labelled by $\lambda \in \mathbb{Z}^n$ with $\lambda_1 \ge \dots \ge \lambda_n$. The character at a matrix with eigenvalues $x_1, \dots, x_n$ is the *Schur function*

$$s_\lambda(x_1, \dots, x_n) = \frac{\det\big(x_k^{\lambda_j + n - j}\big)_{j,k}}{\prod_{j<k} (x_j - x_k)}$$

(for distinct x_j; by continuity otherwise), and $d_\lambda = \prod_{j<k} \frac{\lambda_j - \lambda_k + k - j}{k - j}$.

Proof. Theorems 10.0.1, 10.0.2, 11.1.1 and the formulas above; regularity of $\operatorname{diag}(x_j)$ means $x_j \ne x_k$ for $j \ne k$, and $(\lambda + \rho, e_j - e_k) = \lambda_j - \lambda_k + k - j$, $(\rho, e_j - e_k) = k - j$. ☐

Example 11.2.1: (i) $\Lambda^k \mathbb{C}^n \cong \pi_{(1^k, 0^{n-k})}$, with character the elementary symmetric polynomial $e_k(x)$.

(ii) $\mathrm{Sym}^k \mathbb{C}^n \cong \pi_{(k,0,\dots,0)}$, with character the complete symmetric polynomial $h_k(x)$.

(iii) $\det^m = \pi_{(m,\dots,m)}$, and $\pi_\lambda \otimes \det^m \cong \pi_{\lambda+(m,\dots,m)}$.

(iv) $\pi_\lambda^* \cong \pi_{\lambda^*}$ with $\lambda^* = (-\lambda_n, \dots, -\lambda_1)$.

Proof. (i) The weights of $\Lambda^k\mathbb{C}^n$ are $e_{j_1}+\cdots+e_{j_k}$, $j_1 < \cdots < j_k$, each with multiplicity one, forming one S_n-orbit, the orbit of $(1^k, 0^{n-k})$. The irreducible summand containing the weight vector for $(1^k, 0^{n-k})$ has all weights of that orbit (Proposition 8.1.2(iii)), so it has dimension $\geq \binom{n}{k}$ and is everything. Its highest weight is the dominant element $(1^k, 0^{n-k})$ of the orbit (Theorem 9.1.1(iii)). (ii) $e_1^k \in \mathrm{Sym}^k\mathbb{C}^n$ has weight $(k, 0, \dots, 0)$, which is the highest weight of Sym^k (the weights are $\mu \in \mathbb{N}_0^n$ with $|\mu| = k$, and $(k, 0, \dots, 0)$ maximizes $\mu(H_0)$). So Sym^k contains $\pi_{(k,0,\dots,0)}$ (the summand containing the one-dimensional top weight space). By Theorem 11.2.1, its dimension is $\prod_{l=2}^{n} \frac{k+l-1}{l-1} = \binom{n+k-1}{k} = \dim \mathrm{Sym}^k\mathbb{C}^n$; so they are equal. (iii) det is one-dimensional with weight $(1, \dots, 1)$, and tensoring with it shifts all weights. (iv) The weights of π^* are the negatives of those of π (Lemma 3.2.2(iv)). The lowest weight of π_λ is $w_0\lambda = (\lambda_n, \dots, \lambda_1)$, where w_0 reverses the order: $w_0\lambda$ is a weight, and for any weight μ, $w_0\mu$ is a weight, so $\lambda - w_0\mu \in Q_+$ (Proposition 10.0.3); as w_0 maps Δ^+ to $-\Delta^+$, $w_0\lambda - \mu \in -Q_+$, i.e., $\mu(H_0) \geq (w_0\lambda)(H_0)$. Hence the highest weight of π^* is $-w_0\lambda = \lambda^*$. ☐

Remark 11.2.2 (Young diagrams): Highest weights with $\lambda_n \geq 0$ are partitions with at most n parts; one depicts λ by a *Young diagram* with λ_j boxes in row j. Its *columns* have lengths $\lambda'_1 \geq \lambda'_2 \geq \cdots$, where $\lambda'_i := |\{j \mid \lambda_j \geq i\}|$.

Lemma 11.2.3: Let λ be a partition with at most n parts and column lengths $\lambda'_1, \dots, \lambda'_m$. Then π_λ is a subrepresentation of $\Lambda^{\lambda'_1}\mathbb{C}^n \otimes \cdots \otimes \Lambda^{\lambda'_m}\mathbb{C}^n \subset (\mathbb{C}^n)^{\otimes|\lambda|}$.

Proof. The vector $v := \bigotimes_i (e_1 \wedge \cdots \wedge e_{\lambda'_i})$ has weight $\sum_i (1^{\lambda'_i}, 0, \dots) = \lambda$. In each factor, $e_1 \wedge \cdots \wedge e_{\lambda'_i}$ spans the weight space with maximal value at H_0; hence λ is the weight of the tensor product with maximal value at H_0, and its weight space is $\mathbb{C}v$. The irreducible summand containing v then has highest weight λ. Finally, $\Lambda^c\mathbb{C}^n$ embeds into $(\mathbb{C}^n)^{\otimes c}$ as antisymmetric tensors. ☐

Theorem 11.2.2: Schur–Weyl duality

Let $\mathrm{U}(n)$ act on $V := (\mathbb{C}^n)^{\otimes k}$ by $g \mapsto g^{\otimes k}$ and S_k by permuting the factors. These actions commute, and

$$V \cong \bigoplus_\lambda \pi_\lambda \otimes M_\lambda \qquad \text{as a representation of } \mathrm{U}(n) \times S_k,$$

where λ runs over the partitions of k with at most n parts, $M_\lambda := \mathrm{Hom}_{\mathrm{U}(n)}(\pi_\lambda, V)$ with S_k acting by composition, and the M_λ are irreducible and pairwise inequivalent representations of S_k.

Proof. Step 1: The commutant of S_k *in* $\mathrm{End}(V)$ *is* $\mathcal{A} := \mathrm{span}\{g^{\otimes k} \mid g \in \mathrm{U}(n)\}$. Clearly $\mathcal{A}$ commutes with S_k. Conversely, $\mathrm{End}(V) = \mathrm{End}(\mathbb{C}^n)^{\otimes k}$, and the commutant of S_k is the space of symmetric tensors. It is spanned by the $X^{\otimes k}$, $X \in \mathrm{End}(\mathbb{C}^n)$: the coefficient of $s_1 \cdots s_k$ in the polynomial $(s_1X_1 + \cdots + s_kX_k)^{\otimes k}$ is $\sum_{\sigma \in S_k} X_{\sigma(1)} \otimes \cdots \otimes X_{\sigma(k)}$, and coefficients of a polynomial with values in a subspace lie in that subspace. Let ℓ be a linear functional vanishing on $\mathcal{A}$. Then $Z \mapsto \ell((e^Z)^{\otimes k})$ is holomorphic on $M_n(\mathbb{C}) = \mathfrak{u}(n) \oplus i\mathfrak{u}(n) \cong \mathbb{C}^{n^2}$ and vanishes on $\mathfrak{u}(n) \cong \mathbb{R}^{n^2}$, hence vanishes identically. Since $\exp : M_n(\mathbb{C}) \to \mathrm{GL}(n, \mathbb{C})$ is surjective (by the Jordan normal form it suffices to treat a block $\mu(1 + N)$ with N nilpotent; choose c with $e^c = \mu$ and put $L := \sum_{k \geq 1} (-1)^{k+1} N^k/k$, a finite sum; then $e^{c+L} = \mu e^L = \mu(1 + N)$ by the formal identity $\exp(\log(1 + x)) = 1 + x$) and $\mathrm{GL}(n, \mathbb{C})$ is dense in $M_n(\mathbb{C})$, $\ell(X^{\otimes k}) = 0$ for all X. Hence $\mathcal{A}$ contains all symmetric tensors.

Step 2: decomposition. The map $\bigoplus_\lambda \pi_\lambda \otimes M_\lambda \to V$, $u \otimes A \mapsto Au$, is an isomorphism (isotypic decomposition, Proposition 2.2.3) intertwining $\mathrm{U}(n) \times S_k$. Every $g^{\otimes k}$ acts on $\pi_\lambda \otimes M_\lambda$ as $\pi_\lambda(g) \otimes 1$, hence so does every element of $\mathcal{A}$ (as $T_\lambda \otimes 1$ for some T_λ).

Irreducibility. If $M_\lambda = M' \oplus M''$ with S_k-invariant $M' \neq 0, M_\lambda$, let $p : M_\lambda \to M'$ be an S_k-equivariant projection (complete reducibility for the finite group S_k). The operator P equal to $1 \otimes p$ on $\pi_\lambda \otimes M_\lambda$ and 0 on the other summands commutes with S_k, so $P \in \mathcal{A}$ and $1 \otimes p = T_\lambda \otimes 1$, which forces p to be scalar, a contradiction.

Inequivalence. If $\phi : M_\lambda \to M_\mu$ is an S_k-isomorphism with $\lambda \neq \mu$, choose any linear $\psi \neq 0$ from π_λ to π_μ and let Q be $\psi \otimes \phi$ on $\pi_\lambda \otimes M_\lambda$ (with values in $\pi_\mu \otimes M_\mu$) and 0 elsewhere. Q commutes with S_k, so $Q \in \mathcal{A}$; but elements of $\mathcal{A}$ preserve each summand, while $Q \neq 0$ maps the λ-summand into the μ-summand. Contradiction.

Which λ occur. The weights of V are $e_{j_1} + \cdots + e_{j_k}$, i.e., $\mu \in \mathbb{N}_0^n$ with $\sum \mu_j = k$, so highest weights are partitions of k with at most n parts. Conversely each such λ occurs by Lemma 11.2.3. ☐

Remark 11.2.4: In the representation theory of symmetric groups, M_λ is identified with the Specht module $\mathcal{S}^\lambda$; we do not use this. For $k = 2$: $\mathbb{C}^n \otimes \mathbb{C}^n = \mathrm{Sym}^2\mathbb{C}^n \oplus \Lambda^2\mathbb{C}^n$, with S_2 acting trivially, resp. by the sign.

11.3 The special unitary groups

Proposition 11.3.1: The restriction of π_λ from $\mathrm{U}(n)$ to $\mathrm{SU}(n)$ is irreducible; $\pi_\lambda|_{\mathrm{SU}(n)} \cong \pi_\mu|_{\mathrm{SU}(n)}$ iff $\lambda - \mu \in \mathbb{Z}(1, \ldots, 1)$; and every irreducible representation of $\mathrm{SU}(n)$ arises in this way. Hence $\widehat{\mathrm{SU}(n)}$ is labelled by $\lambda_1 \geq \cdots \geq \lambda_{n-1} \geq \lambda_n = 0$, equivalently by the *Dynkin labels* $a_j := \lambda_j - \lambda_{j+1} \in \mathbb{N}_0$, $j = 1, \ldots, n-1$.

Proof. Every $g \in \mathrm{U}(n)$ is $g = c \cdot g'$ with $c \in \mathbb{T}$ (an n-th root of $\det g$) and $g' \in \mathrm{SU}(n)$. Scalars act by scalars on π_λ (Schur), so $\mathrm{SU}(n)$-invariant subspaces are $\mathrm{U}(n)$-invariant, and the restriction is irreducible. The maximal torus of $\mathrm{SU}(n)$ is $T' := T \cap \mathrm{SU}(n)$, with $\mathfrak{t}' = \{\theta \mid \sum \theta_j = 0\}$; restriction $\mathfrak{t}^* \to \mathfrak{t}'^*$ has kernel $\mathbb{R}(1, \ldots, 1)$, maps roots to roots and $P(T) = \mathbb{Z}^n$ onto $P(T')$ (a character of $T' \cong \mathbb{T}^{n-1}$, $(z_1, \ldots, z_{n-1}) \mapsto \mathrm{diag}(z_1, \ldots, z_{n-1}, (z_1 \cdots z_{n-1})^{-1})$, is $\prod_{j<n} z_j^{m_j}$, the restriction of $\prod_{j<n} x_j^{m_j}$). The highest weight of $\pi_\lambda|_{\mathrm{SU}(n)}$ is the restriction of λ; restrictions of λ, μ agree iff $\lambda - \mu \in \mathbb{R}(1, \ldots, 1) \cap \mathbb{Z}^n = \mathbb{Z}(1, \ldots, 1)$. Every dominant weight of $\mathrm{SU}(n)$ is the restriction of a dominant $\lambda \in \mathbb{Z}^n$ (choose a preimage and use that dominance only involves the differences $\lambda_j - \lambda_{j+1}$). Now apply Theorem 10.0.1 to $\mathrm{SU}(n)$. ☐

Example 11.3.2 (SU(3)): For Dynkin labels (a, b), i.e., $\lambda = (a + b, b, 0)$, Theorem 11.2.1 gives

$$d_{(a,b)} = \frac{(a+1)(b+1)(a+b+2)}{2}.$$

In physics notation: $\mathbf{1} = (0,0)$, $\mathbf{3} = (1,0) = \mathbb{C}^3$, $\bar{\mathbf{3}} = (0,1) = \Lambda^2\mathbb{C}^3 \cong (\mathbb{C}^3)^*$, $\mathbf{6} = (2,0) = \mathrm{Sym}^2\mathbb{C}^3$, $\mathbf{8} = (1,1)$, $\mathbf{10} = (3,0) = \mathrm{Sym}^3\mathbb{C}^3$. Then

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}, \qquad \mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}, \qquad \mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1},$$

and $\mathbf{8}$ is the complexified adjoint representation.

Proof. The identifications of $\mathbf{3}, \mathbf{6}, \mathbf{10}$ and $\Lambda^2\mathbb{C}^3 = \pi_{(1,1,0)}$, of Dynkin labels $(0, 1)$, follow from Example 11.2.1 and Proposition 11.3.1. Also $(\mathbb{C}^3)^* \cong \pi_{(0,0,-1)}$ by Example 11.2.1(iv), and $(0, 0, -1) + (1, 1, 1) = (1, 1, 0)$, so $(\mathbb{C}^3)^* \cong \Lambda^2\mathbb{C}^3$ as SU(3)-representations.

First, $\mathbb{C}^3 \otimes \mathbb{C}^3 = \mathrm{Sym}^2\mathbb{C}^3 \oplus \Lambda^2\mathbb{C}^3 = \mathbf{6} \oplus \bar{\mathbf{3}}$. Next, $\mathbf{3} \otimes \bar{\mathbf{3}} \cong \mathbb{C}^3 \otimes (\mathbb{C}^3)^* \cong \mathrm{End}(\mathbb{C}^3) = \mathfrak{sl}(3, \mathbb{C}) \oplus \mathbb{C} \cdot 1$, with SU(3) acting by conjugation. The weights of $\mathfrak{sl}(3, \mathbb{C})$ are 0 and the roots $e_j - e_k$, and $e_1 - e_3$

(weight vector E_{13}) has maximal value at H_0; so $\mathfrak{sl}(3,\mathbb{C})$ contains $\pi_{(1,0,-1)} = \pi_{(2,1,0)}|_{\mathrm{SU}(3)}$, of Dynkin labels $(1,1)$ and dimension $8 = \dim\mathfrak{sl}(3,\mathbb{C})$. Hence $\mathfrak{sl}(3,\mathbb{C}) \cong \mathbf{8}$ and $\mathbf{3}\otimes\bar{\mathbf{3}} = \mathbf{8}\oplus\mathbf{1}$.

Finally, $\mathbf{3}^{\otimes 3} = (\mathbf{6}\otimes\mathbf{3})\oplus(\bar{\mathbf{3}}\otimes\mathbf{3}) = (\mathbf{6}\otimes\mathbf{3})\oplus\mathbf{8}\oplus\mathbf{1}$. The multiplication map $\mathrm{Sym}^2\mathbb{C}^3\otimes\mathbb{C}^3 \to \mathrm{Sym}^3\mathbb{C}^3$ is equivariant and surjective; by complete reducibility $\mathbf{6}\otimes\mathbf{3} \cong \mathbf{10}\oplus K$ with K its kernel, $\dim K = 8$. We compare weight multiplicities (for U(3)). In $\mathrm{Sym}^2\mathbb{C}^3\otimes\mathbb{C}^3$, the weight $(3,0,0)$ and its permutations have multiplicity 1, the permutations of $(2,1,0)$ have multiplicity 2 (e.g. $e_1^2\otimes e_2$ and $e_1e_2\otimes e_1$), and $(1,1,1)$ has multiplicity 3; in $\mathrm{Sym}^3\mathbb{C}^3$ all these have multiplicity 1. So the weights of K are the six permutations of $(2,1,0)$ (multiplicity 1) and $(1,1,1)$ (multiplicity 2), and the highest weight of K is $(2,1,0)$. Hence $K \supset \pi_{(2,1,0)} = \mathbf{8}$, and $K\cong\mathbf{8}$ by comparing dimensions. □

11.4 The orthogonal groups and spherical harmonics

Theorem 11.4.1: Irreducible representations of SO(m)

The irreducible representations of $\mathrm{SO}(2n+1)$ are labelled by $\lambda\in\mathbb{Z}^n$ with $\lambda_1\geq\cdots\geq\lambda_n\geq 0$, those of $\mathrm{SO}(2n)$ $(n\geq 2)$ by $\lambda\in\mathbb{Z}^n$ with $\lambda_1\geq\cdots\geq\lambda_{n-1}\geq|\lambda_n|$; characters and dimensions are given by Theorems 10.0.1 and 10.0.2 with the data of Theorem 11.1.1.

Proof. $\mathrm{SO}(m)$ is connected (Lemma 5.4.1); apply Theorems 10.0.1 and 11.1.1. □

Remark 11.4.1: For SO(3) $(n=1)$: $\Delta^+ = \{e_1\}$, $\rho = \frac{1}{2}e_1$, and $\lambda = \ell e_1$ gives $d = (\ell+\frac{1}{2})/\frac{1}{2} = 2\ell+1$, as in Proposition 5.4.2. Dominant "weights" with $\lambda_j\in\frac{1}{2}+\mathbb{Z}$ satisfy the inequalities but are not in $P(T)=\mathbb{Z}^n$; for $n=1$ they are the highest weights of the representations of SU(2) not factoring through SO(3).

Theorem 11.4.2: Spherical harmonics

Let $\mathcal{P}_\ell$ be the space of complex homogeneous polynomials of degree ℓ on $\mathbb{R}^3$ and $\mathcal{H}_\ell := \{P\in\mathcal{P}_\ell \mid \Delta P = 0\}$, with SO(3) acting by $(gP)(x) := P(g^{-1}x)$. Then $\mathcal{H}_\ell\cong D^{(\ell)}$ is irreducible of dimension $2\ell+1$, and

$$L^2(S^2) = \widehat{\bigoplus_{\ell\in\mathbb{N}_0}}\mathcal{H}_\ell|_{S^2}$$

(orthogonal Hilbert sum), each irreducible representation of SO(3) occurring exactly once.

Proof. Dimension. Define the inner product on $\mathcal{P} := \bigoplus_\ell \mathcal{P}_\ell$ by $\langle x^a, x^b\rangle := a!\,\delta_{ab}$ for multi-indices a,b. Then $\langle x_jP,Q\rangle = \langle P,\partial_jQ\rangle$ (check on monomials), so multiplication by $|x|^2$ is adjoint to Δ. Hence $\mathcal{P}_\ell = \mathcal{H}_\ell\oplus|x|^2\mathcal{P}_{\ell-2}$ (orthogonally), and as multiplication by $|x|^2$ is injective, $\dim\mathcal{H}_\ell = \binom{\ell+2}{2}-\binom{\ell}{2} = 2\ell+1$. Since Δ commutes with rotations, $\mathcal{H}_\ell$ is invariant.

Irreducibility. Let $T=\mathrm{SO}(2)$ be the rotations about the x_3-axis by angle θ. On $z:=x_1+ix_2$, $\bar z$, x_3, the element g^{-1} acts by $z\mapsto e^{-i\theta}z$, $\bar z\mapsto e^{i\theta}\bar z$, $x_3\mapsto x_3$; so the weights of $\mathcal{P}_\ell$ lie in $\{-\ell,\dots,\ell\}$ (in units of e_1), and $\bar z^\ell = (x_1-ix_2)^\ell$, which is harmonic, has weight ℓ. Hence $\mathcal{H}_\ell$ contains $D^{(\ell)}$ (the summand containing $\bar z^\ell$ has highest weight ℓ), and as $\dim\mathcal{H}_\ell = 2\ell+1$, $\mathcal{H}_\ell\cong D^{(\ell)}$.

Decomposition. SO(3) acts transitively on S^2, and the stabilizer of e_3 is T; so $g\mapsto ge_3$ identifies G/T with S^2, and the image of Haar measure is an SO(3)-invariant probability measure on S^2, which by the uniqueness argument of Lemma 9.4.1(iii) is the normalized surface measure. Thus $f\mapsto f(\cdot\, e_3)$ is a unitary SO(3)-isomorphism from $L^2(S^2)$ onto $L^2(G/T)$ in the sense of Proposition 4.3.2. By that proposition, $D^{(\ell)}$ occurs with multiplicity $\dim(D^{(\ell)})^T$, the multiplicity of the weight 0, which is 1. Restriction to S^2 is injective on $\mathcal{H}_\ell$ (by homogeneity), so $\mathcal{H}_\ell|_{S^2}$ is the (unique) $D^{(\ell)}$-isotypic component of $L^2(S^2)$. Isotypic components for inequivalent irreducibles are orthogonal (the orthogonal projection onto one restricted to the other is an intertwiner, hence 0 by Schur), and their sum is dense by Theorem 4.3.1. □

11.5 The compact symplectic groups

$\mathrm{Sp}(n)$ is connected (Lemma 11.1.1), so by Theorems 10.0.1 and 11.1.1 its irreducible representations are labelled by $\lambda \in \mathbb{Z}^n$ with $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$. The defining representation $\mathbb{C}^{2n}$ (weights $\pm e_j$) has highest weight e_1, and the complexified adjoint representation, which is $\mathrm{Sym}^2((\mathbb{C}^{2n})^*)$ by the proof of Theorem 11.1.1, has highest weight $2e_1$. For $n = 1$, $\mathrm{Sp}(1) = \mathrm{SU}(2)$: for $g \in M_2(\mathbb{C})$ one has $g^T J g = (\det g) J$, so $\mathrm{Sp}(1) = \{g \in \mathrm{U}(2) \mid \det g = 1\}$.

12 Applications to physics

In this section, statements labelled *physical input* are taken from physics and are not proved here; all mathematical consequences drawn from them are proved.

12.1 Symmetries in quantum mechanics

Physical input. Pure states of a quantum system are rays in a Hilbert space $\mathcal{H}$, and a symmetry group G acts on $\mathcal{H}$ by unitary operators, possibly only up to phases (Wigner's theorem). For rotations, the operators can be chosen to form a unitary representation of $\mathrm{SU}(2)$ (rather than of $\mathrm{SO}(3)$). If the representation is finite-dimensional, Corollary 2.2.1 and Theorem 3.3.1 decompose it into the $D^{(j)}$, $j \in \frac{1}{2}\mathbb{N}_0$; the representations with half-integral j do not factor through $\mathrm{SO}(3)$ (Proposition 5.4.2) and describe *spin*.

12.2 Angular momentum

Let (π, V) be a finite-dimensional unitary representation of $\mathrm{SU}(2)$, $X_a = -\frac{i}{2}\sigma_a$ as in Proposition 5.4.2, and define the self-adjoint *angular momentum operators* $J_a := i\, d\pi(X_a)$. From $[X_1, X_2] = X_3$ (and cyclically) we get $[J_1, J_2] = iJ_3$ (and cyclically). Since $X_3 = -\frac{i}{2}H$ and $X_1 \pm iX_2$ correspond to $-iF$ and $-iE$, we have $d\pi(H) = 2J_3$, $d\pi(E) = J_+ := J_1 + iJ_2$, $d\pi(F) = J_- := J_1 - iJ_2$. Let $J^2 := J_1^2 + J_2^2 + J_3^2$.

Proposition 12.2.1: J^2 commutes with $\pi(g)$ for all g and acts on $D^{(j)}$ by the scalar $j(j+1)$. $D^{(j)}$ has an orthonormal basis $|j, m\rangle$, $m = -j, -j+1, \ldots, j$, with $J_3|j,m\rangle = m|j,m\rangle$ and $J_\pm|j,m\rangle = \sqrt{j(j+1) - m(m\pm 1)}\,|j, m\pm 1\rangle$.

Proof. $[J^2, J_a] = 0$ follows from the commutation relations (e.g. $[J_1^2, J_3] = J_1[J_1,J_3] + [J_1,J_3]J_1 = -i(J_1J_2 + J_2J_1)$ and $[J_2^2, J_3] = i(J_2J_1 + J_1J_2)$). So J^2 commutes with $d\pi(\mathfrak{g})$, hence with $\pi(G)$ (Proposition 5.3.3), and is a scalar on the irreducible $D^{(j)}$ by Schur. With $J_-J_+ = J_1^2 + J_2^2 + i[J_1, J_2] = J^2 - J_3^2 - J_3$, on the highest weight vector ($J_+v = 0$, $J_3 v = jv$, as $Hv = 2jv$ by Theorem 5.4.1) we get $J^2 v = (j^2 + j)v$. The weights of $D^{(j)} = \pi_{2j}$ under $H = 2J_3$ are $2j, 2j-2, \ldots, -2j$, each with multiplicity one; normalized eigenvectors $|j,m\rangle$ of J_3 are orthonormal (J_3 self-adjoint). Since $J_+^* = J_-$ and $J_+|j,m\rangle$ is an eigenvector of J_3 for $m+1$, $J_+|j,m\rangle = c_m|j, m+1\rangle$ with $|c_m|^2 = \langle J_-J_+|j,m\rangle, |j,m\rangle\rangle = j(j+1) - m^2 - m$; the phases can be normalized to make all $c_m \geq 0$, and then $J_-|j,m+1\rangle = J_+^*|j,m+1\rangle = c_m|j,m\rangle$. □

Corollary 3.3.3 decomposes $D^{(j_1)} \otimes D^{(j_2)}$ into $D^{(J)}$, $|j_1 - j_2| \leq J \leq j_1 + j_2$; the coefficients $\langle j_1 m_1\, j_2 m_2 | JM\rangle$ of $|J, M\rangle$ in the basis $|j_1, m_1\rangle \otimes |j_2, m_2\rangle$ (nonzero only for $m_1 + m_2 = M$, by comparing J_3-eigenvalues) are the *Clebsch–Gordan coefficients*. For example, $D^{(1/2)} \otimes D^{(1/2)} = D^{(0)} \oplus D^{(1)}$, with the singlet spanning $\Lambda^2\mathbb{C}^2$ and the triplet spanning $\mathrm{Sym}^2\mathbb{C}^2$.

Theorem 12.2.1: Wigner–Eckart

Let π be a unitary representation of $\mathrm{SU}(2)$ on a finite-dimensional space $\mathcal{H}$, and let T_q, $q = -k, \ldots, k$, be operators on $\mathcal{H}$ such that $\pi(g)T_q\pi(g)^{-1} = \sum_{q'} D^{(k)}_{q'q}(g) T_{q'}$, where $D^{(k)}_{q'q}(g) :=$

$\langle k,q'|D^{(k)}(g)|k,q\rangle$. Let $W_1, W_2 \subset \mathcal{H}$ be irreducible invariant subspaces with $W_1 \cong D^{(j)}$, $W_2 \cong D^{(j')}$, and orthonormal bases $|j,m\rangle_1$, $|j',m'\rangle_2$ corresponding to the standard bases. Then there is a constant c (the reduced matrix element) with

$$_1\langle j,m|T_q|j',m'\rangle_2 = c\,\langle j'm'\,kq|jm\rangle \qquad \text{for all } m,m',q.$$

In particular these matrix elements vanish unless $m = m'+q$ and $|j'-k| \le j \le j'+k$.

Proof. Let P be the orthogonal projection onto W_1, an intertwiner. Define $\Phi : D^{(k)} \otimes D^{(j')} \to D^{(j)}$ by $\Phi(|k,q\rangle \otimes |j',m'\rangle) := PT_q|j',m'\rangle_2$ (identifying W_1 with $D^{(j)}$). The transformation law of T_q shows that Φ is an intertwiner: $\Phi(D^{(k)}(g)|k,q\rangle \otimes \pi(g)v) = \sum_{q'} D^{(k)}_{q'q}(g)PT_{q'}\pi(g)v = P\pi(g)T_qv = \pi(g)PT_qv$. By Corollary 3.3.3 and Proposition 2.2.3, $D^{(j)}$ occurs at most once in $D^{(k)}\otimes D^{(j')}$, so $\mathrm{Hom}_G(D^{(k)}\otimes D^{(j')}, D^{(j)})$ has dimension ≤ 1, and if nonzero it is spanned by $\Phi_0 :=$ the orthogonal projection onto the $D^{(j)}$-component followed by the isometry onto $D^{(j)}$ given by the Clebsch–Gordan basis. Hence $\Phi = c\Phi_0$, and $\langle j,m|\Phi_0(|k,q\rangle \otimes |j',m'\rangle) = \langle j'm'\,kq|jm\rangle$ (taking the factors in the order $D^{(j')} \otimes D^{(k)} \cong D^{(k)} \otimes D^{(j')}$). ☐

For electric dipole transitions ($k=1$) this gives the *selection rules* $\Delta j \in \{0,\pm1\}$ and $\Delta m \in \{0,\pm1\}$.

12.3 The hydrogen atom and SO(4)

Physical input (Pauli 1926). For the hydrogen atom (atomic units), the angular momentum L and the Runge–Lenz vector A commute with the Hamiltonian H. On an eigenspace $\mathcal{E}$ of H with eigenvalue $E<0$, assumed finite-dimensional, the operators L_a and $K_a := A_a/\sqrt{-2E}$ are self-adjoint and satisfy

$$[L_a,L_b] = i\epsilon_{abc}L_c, \quad [L_a,K_b] = i\epsilon_{abc}K_c, \quad [K_a,K_b] = i\epsilon_{abc}L_c,$$

$$L\cdot K = 0, \qquad E = -\frac{1}{2(L^2+K^2+1)}.$$

Proposition 12.3.1: Under these assumptions, $E = -\frac{1}{2n^2}$ for some $n \in \mathbb{N}$, and $\mathcal{E} \cong (D^{(j)} \otimes D^{(j)})^{\oplus m}$ with $n = 2j+1$ as a representation of $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. In particular $\dim\mathcal{E}$ is a multiple of n^2, and under the rotations L, $\mathcal{E} \cong (D^{(0)} \oplus D^{(1)} \oplus \cdots \oplus D^{(n-1)})^{\oplus m}$.

Proof. $J^\pm := \frac{1}{2}(L \pm K)$ satisfy $[J^\pm_a, J^\pm_b] = i\epsilon_{abc}J^\pm_c$ and $[J^+_a, J^-_b] = 0$ (direct computation from the relations). So $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ acts on $\mathcal{E}$ by $X_a \mapsto -iJ^+_a$ on the first and $-iJ^-_a$ on the second summand, by skew-Hermitian operators. By Theorem 5.4.1(ii) applied to the first summand, $\mathcal{E} = \bigoplus_j \mathcal{E}_j$ with $\mathcal{E}_j$ the $D^{(j)}$-isotypic component for J^+; as J^- commutes with J^+, each $\mathcal{E}_j$ is J^--invariant, and $\mathcal{E}_j \cong D^{(j)} \otimes M_j$ with J^- acting on $M_j := \mathrm{Hom}_{J^+}(D^{(j)}, \mathcal{E})$; decomposing M_j under J^- gives $\mathcal{E} \cong \bigoplus D^{(j)} \otimes D^{(j')}$. Now $L\cdot K = 0$ gives $(J^+)^2 - (J^-)^2 = L\cdot K = 0$ (the cross terms cancel since $L\cdot K = K\cdot L$, which follows from $L\cdot K = 0$ and $K\cdot L = L\cdot K + \sum_a [K_a, L_a] = L\cdot K$), so $j = j'$ by Proposition 12.2.1. Further $L^2 + K^2 = 2((J^+)^2 + (J^-)^2) = 4j(j+1)$, so $E = -\frac{1}{2(2j+1)^2}$, which determines j. Finally, $L = J^+ + J^-$ acts on $D^{(j)} \otimes D^{(j)}$ as the tensor product representation, which decomposes by Corollary 3.3.3. ☐

(Fock showed that in fact $m=1$, i.e., the degeneracy is exactly n^2.)

12.4 Isospin and the eightfold way

Physical input. Proton and neutron form a doublet $D^{(1/2)}$ of an approximate internal *isospin* symmetry SU(2), and the pions a triplet $D^{(1)}$. Gell-Mann and Ne'eman (1961) extended this to an approximate *flavour* symmetry SU(3); in the quark model, the quarks u, d, s span $\mathbf{3}$, the antiquarks $\bar{\mathbf{3}}$, mesons are $q\bar{q}$ states and baryons qqq states. Isospin I_3 and hypercharge Y are coordinates on $\mathfrak{t}^*$ for the maximal torus of SU(3).

By Example 11.3.2, mesons occur in $\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$ (e.g. the pseudoscalar octet π, K, η, and the singlet η'), and baryons in $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$ (the octet containing p, n, Λ, Σ, Ξ; the decuplet containing Δ, Σ^*, Ξ^*, Ω^-). The Ω^-, corresponding to the weight $3e_3$ of $\mathrm{Sym}^3\mathbb{C}^3$ (a corner of the weight diagram of $\mathbf{10}$), was predicted and then discovered in 1964.

A second, exact SU(3) symmetry, *colour*, is a gauge symmetry of quantum chromodynamics, and observed hadrons are colour singlets. Representation theory explains why quarks bind in pairs $q\bar{q}$ and triples qqq: by Example 11.3.2, $\mathbf{3} \otimes \bar{\mathbf{3}}$ and $\mathbf{3}^{\otimes 3}$ contain the trivial representation exactly once (for $\mathbf{3}^{\otimes 3}$, it is $\Lambda^3\mathbb{C}^3$), whereas $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ does not contain it.

Further reading: B. Simon, *Representations of finite and compact groups*, AMS, 1996 (main reference); J. F. Adams, *Lectures on Lie groups*, University of Chicago Press, 1969; T. Bröcker and T. tom Dieck, *Representations of compact Lie groups*, Springer, 1985; D. Bump, *Lie groups*, 2nd ed., Springer, 2013; W. Fulton and J. Harris, *Representation theory: A first course*, Springer, 1991; B. C. Hall, *Lie groups, Lie algebras, and representations*, 2nd ed., Springer, 2015; J. E. Humphreys, *Introduction to Lie algebras and representation theory*, Springer, 1972; A. W. Knapp, *Lie groups beyond an introduction*, 2nd ed., Birkhäuser, 2002; M. R. Sepanski, *Compact Lie groups*, Springer, 2007; S. Sternberg, *Group theory and physics*, Cambridge University Press, 1994.